Chapter-2 Boundary-Value Problems in Electrostatics: I Problem Solutions

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Problem 2.1

1.1 Problem Statement

A point charge q is brought to a position a distance d away from an infinite plane conductor held at zero potential. Using the method of images, find:

(a) the surface-charge density induced on the plane, and plot it;

(b) the force between the plane and the charge by using Coulomb's law for the force between the charge and its image;

(c) the total force acting on the plane by integrating $\sigma^2/2\varepsilon_0$ over the whole plane;

(d) the work necessary to remove the charge q from its position to infinity;

(e) the potential energy between the charge q and its image [compare the answer to part **d** and discuss];

(f) Find the answer to part **d** in electron volts for an electron originally one angstrom from the surface.

1.2 Solution

Let the grounded conducting plane be the plane

$$z = 0,$$

and let the real charge q be located at

$$(0, 0, d), \qquad d > 0.$$

The physical region is

$$z > 0.$$

By the method of images, the potential in the physical region is the same as that produced by

- the real charge q at $(0, 0, d)$,
- an image charge $-q$ at $(0, 0, -d)$,

with no conductor present.
Thus the potential for $z > 0$ is

$$\Phi(x, y, z) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}}\right].$$

We first check that this satisfies the boundary condition on the plane $z = 0$:

$$\Phi(x, y, 0) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{x^2 + y^2 + d^2}} - \frac{1}{\sqrt{x^2 + y^2 + d^2}}\right] = 0.$$

So the plane is indeed an equipotential at zero potential.
It is convenient to introduce the cylindrical radial coordinate on the plane:

$$\rho = \sqrt{x^2 + y^2}.$$

(a) Induced surface-charge density on the plane

The induced surface-charge density on a conductor is related to the normal electric field just outside the surface by

$$\sigma = \varepsilon_0 E_n.$$

Since the outward normal from the conductor into the region $z > 0$ is $\hat{\mathbf{z}}$,

$$E_n = E_z(0^+).$$

Also

$$E_z = -\frac{\partial \Phi}{\partial z}.$$

Therefore

$$\sigma(\rho) = -\varepsilon_0 \left.\frac{\partial \Phi}{\partial z}\right|_{z=0^+}.$$

Now compute the derivative. Write

$$\Phi(\rho, z) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{\rho^2+(z-d)^2}} - \frac{1}{\sqrt{\rho^2+(z+d)^2}}\right].$$

Use

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\rho^2+(z\mp d)^2\right)^{-1/2} = -\frac{z\mp d}{[\rho^2+(z\mp d)^2]^{3/2}}.$$

Hence

$$\frac{\partial \Phi}{\partial z} = \frac{q}{4\pi\varepsilon_0}\left[-\frac{z-d}{\left(\rho^2+(z-d)^2\right)^{3/2}} + \frac{z+d}{\left(\rho^2+(z+d)^2\right)^{3/2}}\right].$$

Now set $z=0$:

$$\left.\frac{\partial \Phi}{\partial z}\right|_{z=0} = \frac{q}{4\pi\varepsilon_0}\left[-\frac{-d}{(\rho^2+d^2)^{3/2}} + \frac{d}{(\rho^2+d^2)^{3/2}}\right].$$

So

$$\left.\frac{\partial \Phi}{\partial z}\right|_{z=0} = \frac{q}{4\pi\varepsilon_0}\frac{2d}{(\rho^2+d^2)^{3/2}}.$$

Therefore

$$E_z(0^+) = -\left.\frac{\partial \Phi}{\partial z}\right|_{z=0} = -\frac{q}{4\pi\varepsilon_0}\frac{2d}{(\rho^2+d^2)^{3/2}}.$$

Thus

$$\sigma(\rho) = \varepsilon_0 E_z(0^+) = -\frac{q}{4\pi}\frac{2d}{(\rho^2+d^2)^{3/2}}.$$

Hence

$$\boxed{\sigma(\rho) = -\frac{qd}{2\pi(\rho^2+d^2)^{3/2}}.}$$

This is negative everywhere if $q>0$, as expected: a positive charge induces negative charge on a grounded conductor.

Behavior of $\sigma(\rho)$. At the point on the plane directly below the charge, $\rho=0$,

$$\sigma(0) = -\frac{q}{2\pi d^2}.$$

For large ρ,

$$\sigma(\rho) \sim -\frac{qd}{2\pi\rho^3}, \qquad \rho \gg d.$$

So the charge density is most negative directly below the point charge and falls off as $1/\rho^3$.

Check of the total induced charge. Integrate over the whole plane:

$$Q_{\text{ind}} = \int \sigma\, da = \int_0^{2\pi}\int_0^{\infty} \sigma(\rho)\,\rho\,d\rho\,d\phi.$$

Substitute $\sigma(\rho)$:

$$Q_{\text{ind}} = \int_0^{2\pi}\int_0^{\infty}\left(-\frac{qd}{2\pi(\rho^2+d^2)^{3/2}}\right)\rho\,d\rho\,d\phi.$$

The ϕ-integral gives 2π, so

$$Q_{\text{ind}} = -qd\int_0^{\infty}\frac{\rho\,d\rho}{(\rho^2+d^2)^{3/2}}.$$

Let

$$u = \rho^2+d^2, \qquad du = 2\rho\,d\rho.$$

Then

$$Q_{\text{ind}} = -\frac{qd}{2}\int_{d^2}^{\infty} u^{-3/2}\,du.$$

Since

$$\int u^{-3/2}\,du = -2u^{-1/2},$$

we get

$$Q_{\text{ind}} = -\frac{qd}{2}\left[-2u^{-1/2}\right]_{d^2}^{\infty} = qd\left[u^{-1/2}\right]_{d^2}^{\infty} = qd\left(0-\frac{1}{d}\right) = -q.$$

So

$$\boxed{Q_{\text{ind}} = -q.}$$

That is exactly what we expect for a grounded infinite plane.

(b) Force between the plane and the charge from the image charge

In the image problem, the force on the real charge q is exactly the force due to the image charge $-q$ located a distance $2d$ away.

By Coulomb's law, the magnitude of the force is

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2d)^2} = \frac{q^2}{16\pi\varepsilon_0 d^2}.$$

The force is attractive, directed toward the plane, that is, in the $-\hat{\mathbf{z}}$ direction. Therefore

$$\boxed{\mathbf{F} = -\hat{\mathbf{z}}\,\frac{q^2}{16\pi\varepsilon_0 d^2}.}$$

(c) Total force on the plane from integrating $\sigma^2/2\varepsilon_0$

The electrostatic pressure on a conductor surface is

$$p = \frac{\sigma^2}{2\varepsilon_0}.$$

This pressure acts normal to the surface, pushing outward from the conductor into the field region. For the plane, that means the field pulls the plane upward toward the charge. The total force on the plane is therefore

$$F = \int p\,da = \int \frac{\sigma^2}{2\varepsilon_0}\,da.$$

Using cylindrical coordinates,

$$da = 2\pi\rho\,d\rho.$$

Also

$$\sigma(\rho) = -\frac{qd}{2\pi(\rho^2+d^2)^{3/2}},$$

so

$$\sigma^2(\rho) = \frac{q^2d^2}{4\pi^2(\rho^2+d^2)^3}.$$

Therefore

$$F = \int_0^{\infty} \frac{1}{2\varepsilon_0}\frac{q^2d^2}{4\pi^2(\rho^2+d^2)^3}(2\pi\rho\,d\rho).$$

Simplify:

$$F = \frac{q^2d^2}{4\pi\varepsilon_0}\int_0^{\infty}\frac{\rho\,d\rho}{(\rho^2+d^2)^3}.$$

Now let

$$u = \rho^2+d^2, \qquad du = 2\rho\,d\rho, \qquad \rho\,d\rho = \frac{du}{2}.$$

Then

$$F = \frac{q^2d^2}{4\pi\varepsilon_0}\cdot\frac{1}{2}\int_{d^2}^{\infty} u^{-3}\,du.$$

Since

$$\int u^{-3}\,du = -\frac{1}{2}u^{-2},$$

we get

$$F = \frac{q^2 d^2}{8\pi\varepsilon_0}\left[-\frac{1}{2}u^{-2}\right]_{d^2}^{\infty} = \frac{q^2 d^2}{8\pi\varepsilon_0}\left(\frac{1}{2d^4}\right).$$

Therefore

$$F = \frac{q^2}{16\pi\varepsilon_0 d^2}.$$

So the total force on the plane has magnitude

$$\boxed{F = \frac{q^2}{16\pi\varepsilon_0 d^2},}$$

in agreement with part **(b)**. Its direction is toward the charge.

(d) Work necessary to remove the charge to infinity

The force on the charge is attractive:

$$F_z(d) = -\frac{q^2}{16\pi\varepsilon_0 d^2}.$$

To remove the charge slowly to infinity, an external agent must apply an equal and opposite force:

$$F_{\text{ext}}(d) = +\frac{q^2}{16\pi\varepsilon_0 d^2}.$$

The required work is

$$W_{\text{ext}} = \int_d^{\infty} F_{\text{ext}}(z)\,dz = \int_d^{\infty} \frac{q^2}{16\pi\varepsilon_0 z^2}\,dz.$$

Since

$$\int_d^{\infty} z^{-2}\,dz = \left[-\frac{1}{z}\right]_d^{\infty} = \frac{1}{d},$$

we obtain

$$W_{\text{ext}} = \frac{q^2}{16\pi\varepsilon_0 d}.$$

Hence the work necessary to remove the charge to infinity is

$$\boxed{W_{\text{remove}} = \frac{q^2}{16\pi\varepsilon_0 d}.}$$

This is positive, as it should be, because one must do work against the attractive force.

(e) Potential energy between the charge and its image

If we naively treat the real charge and the image charge as two actual point charges separated by $2d$, their mutual Coulomb potential energy is

$$U_{\text{pair}} = \frac{1}{4\pi\varepsilon_0}\frac{q(-q)}{2d} = -\frac{q^2}{8\pi\varepsilon_0 d}.$$

So

$$\boxed{U_{\text{pair}} = -\frac{q^2}{8\pi\varepsilon_0 d}.}$$

But this is *not* the actual electrostatic energy of the physical system consisting of one real charge plus the conductor.

From part **(d)**, the actual energy change relative to infinity is

$$U_{\text{phys}} = -W_{\text{remove}} = -\frac{q^2}{16\pi\varepsilon_0 d}.$$

Thus

$$\boxed{U_{\text{phys}} = -\frac{q^2}{16\pi\varepsilon_0 d}.}$$

This is exactly one-half of the naive charge-image interaction energy:

$$U_{\text{phys}} = \frac{1}{2}U_{\text{pair}}.$$

Why the factor 1/2? The image charge is not a real independent charge distribution; it is only a mathematical device for reproducing the field in the physical region. If we computed the full pair energy of the real charge and the image charge, we would be counting twice what belongs to the physical system. The correct physical energy is one-half the naive interaction energy with the image.

Equivalently, the physical energy can be written as

$$U = \frac{1}{2} q\, \Phi_{\text{image}}(\text{at real charge}),$$

not simply $q\Phi_{\text{image}}$.

Indeed, the image potential at the location of the real charge is

$$\Phi_{\text{image}} = \frac{1}{4\pi\varepsilon_0} \frac{-q}{2d} = -\frac{q}{8\pi\varepsilon_0 d}.$$

So

$$U = \frac{1}{2} q\Phi_{\text{image}} = \frac{1}{2} q \left(-\frac{q}{8\pi\varepsilon_0 d} \right) = -\frac{q^2}{16\pi\varepsilon_0 d},$$

which agrees with part **(d)**.

(f) Numerical answer for an electron one angstrom from the surface

For an electron,

$$q = e, \qquad d = 1\ \text{Å} = 10^{-10}\ \text{m}.$$

From part **(d)**,

$$W_{\text{remove}} = \frac{e^2}{16\pi\varepsilon_0 d}.$$

Using the standard constant

$$\frac{e^2}{4\pi\varepsilon_0} = 14.4\ \text{eV} \cdot \text{Å},$$

we have

$$\frac{e^2}{16\pi\varepsilon_0 d} = \frac{1}{4} \frac{e^2}{4\pi\varepsilon_0 d}.$$

At $d = 1$ Å,

$$\frac{e^2}{4\pi\varepsilon_0 d} = 14.4\ \text{eV}.$$

Therefore

$$W_{\text{remove}} = \frac{1}{4}(14.4\ \text{eV}) = 3.6\ \text{eV}.$$

Hence

$$\boxed{W_{\text{remove}} = 3.6\ \text{eV}}$$

for an electron initially 1 Å from the surface.

The corresponding physical potential energy is

$$\boxed{U_{\text{phys}} = -3.6\ \text{eV}.}$$

Final results

$$\boxed{\sigma(\rho) = -\frac{qd}{2\pi(\rho^2 + d^2)^{3/2}}}$$

$$\boxed{\mathbf{F}_{\text{on charge}} = -\hat{\mathbf{z}} \frac{q^2}{16\pi\varepsilon_0 d^2}}$$

$$\boxed{F_{\text{on plane}} = \frac{q^2}{16\pi\varepsilon_0 d^2}}$$

$$\boxed{W_{\text{remove}} = \frac{q^2}{16\pi\varepsilon_0 d}}$$

$$\boxed{U_{\text{pair}} = -\frac{q^2}{8\pi\varepsilon_0 d}, \qquad U_{\text{phys}} = -\frac{q^2}{16\pi\varepsilon_0 d}}$$

$$\boxed{W_{\text{remove}} = 3.6 \text{ eV} \quad \text{for an electron at } d = 1 \text{ Å}.}$$

Problem 2.29

2.1 Problem Statement

For the Galerkin method on a two-dimensional square lattice with lattice spacing h, verify the relations (2.81) for the localized "pyramid" basis functions,

$$\phi_{ij}(x,y) = \left(1 - \frac{|x|}{h}\right)\left(1 - \frac{|y|}{h}\right), \qquad |x| < h, \quad |y| < h,$$

where x and y are measured from the site (i,j). In particular, show that

$$\int dx \int dy\, \phi_{ij}(x,y) = h^2,$$

$$\int dx \int dy\, \nabla\phi_{ij} \cdot \nabla\phi_{ij} = \frac{8}{3},$$

$$\int dx \int dy\, \nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = -\frac{1}{3},$$

$$\int dx \int dy\, \nabla\phi_{i,j+1} \cdot \nabla\phi_{ij} = -\frac{1}{3},$$

$$\int dx \int dy\, \nabla\phi_{i+1,j+1} \cdot \nabla\phi_{ij} = -\frac{1}{3}.$$

2.2 Solution

The localized pyramid basis function centered at the site (i,j) is

$$\phi_{ij}(x,y) = \left(1 - \frac{|x|}{h}\right)\left(1 - \frac{|y|}{h}\right), \qquad |x| < h, \quad |y| < h,$$

and it vanishes outside the square

$$-h < x < h, \qquad -h < y < h.$$

It is convenient to introduce the one-dimensional hat function

$$f(t) = 1 - \frac{|t|}{h}, \qquad |t| < h,$$

and

$$f(t) = 0, \qquad |t| \geq h.$$

Then

$$\phi_{ij}(x,y) = f(x)f(y).$$

Its derivative is

$$f'(t) = \begin{cases} \dfrac{1}{h}, & -h < t < 0, \\ -\dfrac{1}{h}, & 0 < t < h, \end{cases}$$

and $f'(t) = 0$ outside the support. The derivative is undefined exactly at $t = 0$, but that single point has measure zero and does not affect any integral.

We will repeatedly use the two basic one-dimensional integrals

$$\int_{-h}^{h} f(t)\, dt = 2\int_0^h \left(1 - \frac{t}{h}\right) dt = 2\left[t - \frac{t^2}{2h}\right]_0^h = h,$$

$$\int_{-h}^{h} f(t)^2\,dt = 2\int_0^h \left(1-\frac{t}{h}\right)^2 dt.$$

Expand the square:

$$\left(1-\frac{t}{h}\right)^2 = 1-\frac{2t}{h}+\frac{t^2}{h^2}.$$

Therefore

$$\int_{-h}^{h} f(t)^2\,dt = 2\left[t-\frac{t^2}{h}+\frac{t^3}{3h^2}\right]_0^h = 2\left(h-h+\frac{h}{3}\right) = \frac{2h}{3}.$$

Also,

$$\int_{-h}^{h} (f'(t))^2\,dt = \int_{-h}^{0} \frac{1}{h^2}\,dt + \int_0^h \frac{1}{h^2}\,dt = \frac{h}{h^2}+\frac{h}{h^2} = \frac{2}{h}.$$

1. Verification of $\int \phi_{ij}\,dx\,dy = h^2$

Since

$$\phi_{ij}(x,y) = f(x)f(y),$$

the integral factorizes:

$$\int dx \int dy\, \phi_{ij}(x,y) = \left(\int_{-h}^{h} f(x)\,dx\right)\left(\int_{-h}^{h} f(y)\,dy\right).$$

Using

$$\int_{-h}^{h} f(t)\,dt = h,$$

we get

$$\boxed{\int dx \int dy\, \phi_{ij}(x,y) = h^2.}$$

2. Verification of $\int \nabla\phi_{ij}\cdot\nabla\phi_{ij}\,dx\,dy = \frac{8}{3}$

Because

$$\phi_{ij}(x,y) = f(x)f(y),$$

its partial derivatives are

$$\frac{\partial\phi_{ij}}{\partial x} = f'(x)f(y), \qquad \frac{\partial\phi_{ij}}{\partial y} = f(x)f'(y).$$

Hence

$$\nabla\phi_{ij}\cdot\nabla\phi_{ij} = \left(\frac{\partial\phi_{ij}}{\partial x}\right)^2 + \left(\frac{\partial\phi_{ij}}{\partial y}\right)^2 = (f'(x))^2 f(y)^2 + f(x)^2 (f'(y))^2.$$

Integrating over the support,

$$\int dx \int dy\, \nabla\phi_{ij}\cdot\nabla\phi_{ij} = \left(\int (f'(x))^2 dx\right)\left(\int f(y)^2 dy\right) + \left(\int f(x)^2 dx\right)\left(\int (f'(y))^2 dy\right).$$

By symmetry the two terms are equal, so

$$= 2\left(\int_{-h}^{h} (f'(t))^2 dt\right)\left(\int_{-h}^{h} f(t)^2 dt\right).$$

Substitute the one-dimensional values:

$$= 2\left(\frac{2}{h}\right)\left(\frac{2h}{3}\right) = \frac{8}{3}.$$

Therefore

$$\boxed{\int dx \int dy\, \nabla\phi_{ij}\cdot\nabla\phi_{ij} = \frac{8}{3}.}$$

3. Verification of $\int \nabla\phi_{i+1,j} \cdot \nabla\phi_{ij}\, dx\, dy = -\frac{1}{3}$

Now consider the basis function centered one lattice spacing to the right. In the local coordinates measured from the site (i,j), the neighboring basis function is

$$\phi_{i+1,j}(x,y) = f(x-h)\, f(y).$$

Its support overlaps that of ϕ_{ij} only for

$$0 < x < h, \qquad -h < y < h.$$

On this overlap region,

$$f(x) = 1 - \frac{x}{h}, \qquad f(x-h) = 1 - \frac{|x-h|}{h} = \frac{x}{h},$$

because $0 < x < h$ implies $|x-h| = h - x$.

Thus in the overlap region,

$$\phi_{ij}(x,y) = \left(1 - \frac{x}{h}\right) f(y), \qquad \phi_{i+1,j}(x,y) = \frac{x}{h} f(y).$$

Their derivatives are

$$\frac{\partial\phi_{ij}}{\partial x} = -\frac{1}{h} f(y), \qquad \frac{\partial\phi_{i+1,j}}{\partial x} = +\frac{1}{h} f(y),$$

and

$$\frac{\partial\phi_{ij}}{\partial y} = \left(1 - \frac{x}{h}\right) f'(y), \qquad \frac{\partial\phi_{i+1,j}}{\partial y} = \frac{x}{h} f'(y).$$

Therefore

$$\nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = \left(+\frac{1}{h} f(y)\right)\left(-\frac{1}{h} f(y)\right) + \left(\frac{x}{h} f'(y)\right)\left(\left(1 - \frac{x}{h}\right) f'(y)\right).$$

So

$$\nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = -\frac{1}{h^2} f(y)^2 + \frac{x}{h}\left(1 - \frac{x}{h}\right)(f'(y))^2.$$

Now integrate over the overlap region:

$$\int dx \int dy\, \nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = \int_0^h dx \int_{-h}^h dy \left[-\frac{1}{h^2} f(y)^2 + \frac{x}{h}\left(1 - \frac{x}{h}\right)(f'(y))^2\right].$$

Separate the two terms.

For the first term,

$$-\frac{1}{h^2}\left(\int_0^h dx\right)\left(\int_{-h}^h f(y)^2 dy\right) = -\frac{1}{h^2}(h)\left(\frac{2h}{3}\right) = -\frac{2}{3}.$$

For the second term,

$$\left(\int_0^h \frac{x}{h}\left(1 - \frac{x}{h}\right) dx\right)\left(\int_{-h}^h (f'(y))^2 dy\right).$$

Compute the x-integral:

$$\int_0^h \left(\frac{x}{h} - \frac{x^2}{h^2}\right) dx = \left[\frac{x^2}{2h} - \frac{x^3}{3h^2}\right]_0^h = \frac{h}{2} - \frac{h}{3} = \frac{h}{6}.$$

Then

$$\frac{h}{6} \cdot \frac{2}{h} = \frac{1}{3}.$$

Hence the total is

$$-\frac{2}{3} + \frac{1}{3} = -\frac{1}{3}.$$

Therefore

$$\boxed{\int dx \int dy\, \nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = -\frac{1}{3}.}$$

4. Verification of $\int \nabla\phi_{i,j+1}\cdot\nabla\phi_{ij}\,dx\,dy = -\frac{1}{3}$

This case is completely analogous by symmetry under interchange of x and y. The neighboring basis function is

$$\phi_{i,j+1}(x,y) = f(x)\,f(y-h),$$

and the overlap region is

$$-h < x < h, \qquad 0 < y < h.$$

Repeating exactly the same steps as in the previous part, but with x and y interchanged, gives

$$\boxed{\int dx \int dy\, \nabla\phi_{i,j+1}\cdot\nabla\phi_{ij} = -\frac{1}{3}.}$$

5. Verification of $\int \nabla\phi_{i+1,j+1}\cdot\nabla\phi_{ij}\,dx\,dy = -\frac{1}{3}$

Now consider the diagonal neighbor. In local coordinates,

$$\phi_{i+1,j+1}(x,y) = f(x-h)\,f(y-h).$$

Its support overlaps that of ϕ_{ij} only on the square

$$0 < x < h, \qquad 0 < y < h.$$

On this overlap region,

$$\phi_{ij}(x,y) = \left(1-\frac{x}{h}\right)\left(1-\frac{y}{h}\right),$$

$$\phi_{i+1,j+1}(x,y) = \frac{x}{h}\frac{y}{h}.$$

Therefore

$$\frac{\partial\phi_{ij}}{\partial x} = -\frac{1}{h}\left(1-\frac{y}{h}\right), \qquad \frac{\partial\phi_{i+1,j+1}}{\partial x} = \frac{1}{h}\frac{y}{h},$$

and

$$\frac{\partial\phi_{ij}}{\partial y} = -\frac{1}{h}\left(1-\frac{x}{h}\right), \qquad \frac{\partial\phi_{i+1,j+1}}{\partial y} = \frac{1}{h}\frac{x}{h}.$$

Thus

$$\nabla\phi_{i+1,j+1}\cdot\nabla\phi_{ij} = \left(\frac{y}{h^2}\right)\left(-\frac{1}{h}\left(1-\frac{y}{h}\right)\right) + \left(\frac{x}{h^2}\right)\left(-\frac{1}{h}\left(1-\frac{x}{h}\right)\right).$$

So

$$\nabla\phi_{i+1,j+1}\cdot\nabla\phi_{ij} = -\frac{1}{h^2}\frac{y}{h}\left(1-\frac{y}{h}\right) - \frac{1}{h^2}\frac{x}{h}\left(1-\frac{x}{h}\right).$$

Integrate over $0<x<h$, $0<y<h$:

$$\int_0^h dx \int_0^h dy\, \nabla\phi_{i+1,j+1}\cdot\nabla\phi_{ij}.$$

For the first term,

$$-\frac{1}{h^2}\left(\int_0^h dx\right)\left(\int_0^h \frac{y}{h}\left(1-\frac{y}{h}\right)dy\right) = -\frac{1}{h^2}(h)\left(\frac{h}{6}\right) = -\frac{1}{6}.$$

For the second term, by symmetry,

$$-\frac{1}{6}.$$

Therefore

$$-\frac{1}{6}-\frac{1}{6} = -\frac{1}{3}.$$

Hence

$$\boxed{\int dx \int dy\, \nabla\phi_{i+1,j+1}\cdot\nabla\phi_{ij} = -\frac{1}{3}.}$$

Final results

We have verified all the required relations:

$$\boxed{\int dx \int dy\, \phi_{ij}(x,y) = h^2}$$

$$\boxed{\int dx \int dy\, \nabla\phi_{ij} \cdot \nabla\phi_{ij} = \frac{8}{3}}$$

$$\boxed{\int dx \int dy\, \nabla\phi_{i+1,j} \cdot \nabla\phi_{ij} = -\frac{1}{3}}$$

$$\boxed{\int dx \int dy\, \nabla\phi_{i,j+1} \cdot \nabla\phi_{ij} = -\frac{1}{3}}$$

$$\boxed{\int dx \int dy\, \nabla\phi_{i+1,j+1} \cdot \nabla\phi_{ij} = -\frac{1}{3}}$$

These are the overlap and stiffness-matrix elements for the localized two-dimensional pyramid basis functions used in the Galerkin method.

Problem 2.30

3.1 Problem Statement

Using the results of Problem 2.29, apply the Galerkin method to the integral equivalent of the Poisson equation with zero potential on the boundary,

$$\int_V dx\, dy\, [\nabla\phi_{ij} \cdot \nabla\psi - 4\pi\phi_{ij}] = 0 \qquad \text{with} \qquad \psi(x,y) = \sum_{i',j'=1}^{N} \psi_{i'j'}\, \phi_{i'j'}(x,y)$$

for the lattice of Problem 1.24, with its three independent interior lattice sites. Show that you get three coupled equations for the ψ_{ij} values (ψ_1, ψ_2, ψ_3) and solve them to find the Galerkin approximations for the potential at these sites. Compare with the exact values and the results of the various iterations of Problem 1.24c. Comment.

$$[\psi = 4\pi\varepsilon_0\Phi]$$

3.2 Solution

We use the same square domain as in Problem 1.24:

$$0 \le x \le 1, \qquad 0 \le y \le 1,$$

with zero potential on the boundary and uniform source density of unit strength in the interior. As in the problem statement, define

$$\psi(x,y) = 4\pi\varepsilon_0\Phi(x,y).$$

Then Poisson's equation becomes

$$\nabla^2\psi = -4\pi$$

inside the square, with

$$\psi = 0 \qquad \text{on the boundary.}$$

We now apply the Galerkin method using the localized pyramid basis functions of Problem 2.29 on the square lattice with spacing

$$h = 0.25.$$

1. Independent lattice sites and symmetry

There are $3 \times 3 = 9$ interior lattice points:

$$(0.25, 0.25),\ (0.50, 0.25),\ (0.75, 0.25),$$
$$(0.25, 0.50),\ (0.50, 0.50),\ (0.75, 0.50),$$
$$(0.25, 0.75),\ (0.50, 0.75),\ (0.75, 0.75).$$

By symmetry, these reduce to three independent values:

$$\psi_1 = \psi(0.25, 0.25), \qquad \psi_2 = \psi(0.50, 0.25), \qquad \psi_3 = \psi(0.50, 0.50).$$

Thus the four corner-interior nodes have value ψ_1, the four edge-center interior nodes have value ψ_2, and the central node has value ψ_3.

The Galerkin approximation is therefore

$$\psi(x, y) = \sum_{i,j} \psi_{ij} \phi_{ij}(x, y),$$

with only these three independent coefficients.

2. Galerkin equations

The Galerkin condition is

$$\int_V dx\, dy\, [\nabla \phi_{ij} \cdot \nabla \psi - 4\pi \phi_{ij}] = 0$$

for each interior test function ϕ_{ij}. Substituting

$$\psi = \sum_{i',j'} \psi_{i'j'} \phi_{i'j'},$$

gives

$$\sum_{i',j'} \psi_{i'j'} \int_V dx\, dy\, \nabla \phi_{ij} \cdot \nabla \phi_{i'j'} = 4\pi \int_V dx\, dy\, \phi_{ij}.$$

From Problem 2.29 we use the matrix elements

$$\int dx\, dy\, \phi_{ij} = h^2,$$

$$\int dx\, dy\, \nabla \phi_{ij} \cdot \nabla \phi_{ij} = \frac{8}{3},$$

$$\int dx\, dy\, \nabla \phi_{i+1,j} \cdot \nabla \phi_{ij} = \int dx\, dy\, \nabla \phi_{i,j+1} \cdot \nabla \phi_{ij} = \int dx\, dy\, \nabla \phi_{i+1,j+1} \cdot \nabla \phi_{ij} = -\frac{1}{3}.$$

All other overlaps vanish because the supports do not overlap.

Also,

$$h = 0.25 = \frac{1}{4}, \qquad h^2 = \frac{1}{16},$$

so the right-hand side is

$$4\pi h^2 = 4\pi \left(\frac{1}{16}\right) = \frac{\pi}{4}.$$

We now write the three independent equations.

3. Equation for the corner-interior site ψ_1

Take the test function centered at the corner-interior node $(0.25, 0.25)$. Its overlaps are:

- with itself: coefficient $8/3$,
- with the two adjacent edge-center nodes: two contributions of $-1/3$,
- with the central node: one diagonal contribution of $-1/3$.

There are no other contributions because outside the domain the basis functions are absent, and boundary values are fixed to zero.

Therefore

$$\frac{8}{3}\psi_1 - \frac{1}{3}\psi_2 - \frac{1}{3}\psi_2 - \frac{1}{3}\psi_3 = \frac{\pi}{4}.$$

So

$$\frac{8}{3}\psi_1 - \frac{2}{3}\psi_2 - \frac{1}{3}\psi_3 = \frac{\pi}{4}.$$

Multiplying by 3,

$$\boxed{8\psi_1 - 2\psi_2 - \psi_3 = \frac{3\pi}{4}.}$$

4. Equation for the edge-center site ψ_2

Now use the test function centered at $(0.50, 0.25)$. Its overlaps are:

- with itself: coefficient $8/3$,
- with the two neighboring corner-interior nodes left and right: two contributions of $-1/3$,
- with the central node above: one orthogonal contribution of $-1/3$,
- with the two corner-interior nodes diagonally above left and above right: these are both of type ψ_2? No — by symmetry the diagonal neighbors of $(0.50, 0.25)$ at $(0.25, 0.50)$ and $(0.75, 0.50)$ are edge-center-type interior nodes, so both contribute $-1/3\psi_2$.

Thus the equation is

$$\frac{8}{3}\psi_2 - \frac{1}{3}\psi_1 - \frac{1}{3}\psi_1 - \frac{1}{3}\psi_3 - \frac{1}{3}\psi_2 - \frac{1}{3}\psi_2 = \frac{\pi}{4}.$$

So

$$\left(\frac{8}{3} - \frac{2}{3}\right)\psi_2 - \frac{2}{3}\psi_1 - \frac{1}{3}\psi_3 = \frac{\pi}{4},$$

that is,

$$2\psi_2 - \frac{2}{3}\psi_1 - \frac{1}{3}\psi_3 = \frac{\pi}{4}.$$

Multiplying by 3,

$$\boxed{-2\psi_1 + 6\psi_2 - \psi_3 = \frac{3\pi}{4}.}$$

5. Equation for the center site ψ_3

Finally use the test function centered at $(0.50, 0.50)$. Its overlaps are:

- with itself: coefficient $8/3$,
- with the four edge-center nodes above, below, left, right: four contributions of $-1/3\psi_2$,
- with the four corner-interior nodes on the diagonals: four contributions of $-1/3\psi_1$.

Therefore

$$\frac{8}{3}\psi_3 - \frac{4}{3}\psi_2 - \frac{4}{3}\psi_1 = \frac{\pi}{4}.$$

Multiplying by 3,

$$\boxed{-4\psi_1 - 4\psi_2 + 8\psi_3 = \frac{3\pi}{4}.}$$

6. Solve the three coupled equations

We must solve

$$8\psi_1 - 2\psi_2 - \psi_3 = \frac{3\pi}{4},$$

$$-2\psi_1 + 6\psi_2 - \psi_3 = \frac{3\pi}{4},$$

$$-4\psi_1 - 4\psi_2 + 8\psi_3 = \frac{3\pi}{4}.$$

We solve step by step.

Step 1: eliminate ψ_3 from the first two equations. Subtract the second equation from the first:

$$(8\psi_1 - 2\psi_2 - \psi_3) - (-2\psi_1 + 6\psi_2 - \psi_3) = 0.$$

So

$$10\psi_1 - 8\psi_2 = 0,$$

hence

$$\boxed{5\psi_1 = 4\psi_2 \qquad \Rightarrow \qquad \psi_2 = \frac{5}{4}\psi_1.}$$

Step 2: use the first equation to express ψ_3. From

$$8\psi_1 - 2\psi_2 - \psi_3 = \frac{3\pi}{4},$$

we get

$$\psi_3 = 8\psi_1 - 2\psi_2 - \frac{3\pi}{4}.$$

Substitute $\psi_2 = \frac{5}{4}\psi_1$:

$$\psi_3 = 8\psi_1 - \frac{10}{4}\psi_1 - \frac{3\pi}{4} = \frac{22}{4}\psi_1 - \frac{3\pi}{4} = \frac{11}{2}\psi_1 - \frac{3\pi}{4}.$$

So

$$\boxed{\psi_3 = \frac{11}{2}\psi_1 - \frac{3\pi}{4}.}$$

Step 3: substitute into the third equation. The third equation is

$$-4\psi_1 - 4\psi_2 + 8\psi_3 = \frac{3\pi}{4}.$$

Substitute $\psi_2 = \frac{5}{4}\psi_1$:

$$-4\psi_1 - 5\psi_1 + 8\psi_3 = \frac{3\pi}{4},$$

so

$$-9\psi_1 + 8\psi_3 = \frac{3\pi}{4}.$$

Now substitute $\psi_3 = \frac{11}{2}\psi_1 - \frac{3\pi}{4}$:

$$-9\psi_1 + 8\left(\frac{11}{2}\psi_1 - \frac{3\pi}{4}\right) = \frac{3\pi}{4}.$$

Thus

$$-9\psi_1 + 44\psi_1 - 6\pi = \frac{3\pi}{4}.$$

So

$$35\psi_1 = 6\pi + \frac{3\pi}{4} = \frac{24\pi + 3\pi}{4} = \frac{27\pi}{4}.$$

Hence

$$\boxed{\psi_1 = \frac{27\pi}{140}.}$$

Then

$$\psi_2 = \frac{5}{4}\psi_1 = \frac{5}{4} \cdot \frac{27\pi}{140} = \frac{27\pi}{112},$$

so

$$\boxed{\psi_2 = \frac{27\pi}{112}.}$$

Finally,

$$\psi_3 = \frac{11}{2}\frac{27\pi}{140} - \frac{3\pi}{4} = \frac{297\pi}{280} - \frac{210\pi}{280} = \frac{87\pi}{280}.$$

Thus

$$\boxed{\psi_3 = \frac{87\pi}{280}.}$$

7. Numerical values

Numerically,

$$\begin{aligned}
\psi_1 &= \frac{27\pi}{140} \approx 0.60588, \\
\psi_2 &= \frac{27\pi}{112} \approx 0.75735, \\
\psi_3 &= \frac{87\pi}{280} \approx 0.97614.
\end{aligned}$$

Therefore the Galerkin approximations are

$$\boxed{\psi_1 \approx 0.60588, \qquad \psi_2 \approx 0.75735, \qquad \psi_3 \approx 0.97614.}$$

Since

$$\psi = 4\pi\varepsilon_0\Phi,$$

these are directly the approximations for the dimensionless potentials requested.

8. Comparison with the exact values

From Problem 1.24, the exact continuum values are

$$\begin{aligned}
\psi_{\text{exact}}(0.25, 0.25) &= 0.5691, \\
\psi_{\text{exact}}(0.50, 0.25) &= 0.7205, \\
\psi_{\text{exact}}(0.50, 0.50) &= 0.9258.
\end{aligned}$$

So the Galerkin errors are

$$\begin{aligned}
\psi_1 - \psi_{1,\text{exact}} &\approx 0.60588 - 0.5691 = 0.03678, \\
\psi_2 - \psi_{2,\text{exact}} &\approx 0.75735 - 0.7205 = 0.03685, \\
\psi_3 - \psi_{3,\text{exact}} &\approx 0.97614 - 0.9258 = 0.05034.
\end{aligned}$$

Thus

$$\boxed{\text{Errors} \approx (0.0368,\ 0.0368,\ 0.0503).}$$

The Galerkin approximation overestimates the exact solution at all three points.

9. Comparison with Problem 1.24 relaxation results

From Problem 1.24, the exact-continuum comparison values quoted there were

$$(0.5691,\ 0.7205,\ 0.9258).$$

The coarse-grid improved-grid relaxation converged to approximately

$$(0.57415,\ 0.72709,\ 0.93311),$$

while the Jacobi sequence approached nearby values more slowly.

Compared with those, the present Galerkin values

$$(0.60588,\ 0.75735,\ 0.97614)$$

are noticeably larger.

So on this coarse $h = 0.25$ mesh:

- the Galerkin approximation captures the qualitative shape correctly,
- but it is less accurate pointwise than the improved-grid relaxation result of Problem 1.24,
- and it systematically overshoots the exact continuum solution at the three independent sites.

10. Comment

This behavior is understandable. The Galerkin method here uses piecewise pyramid basis functions, so the approximation is globally constrained by a small finite-dimensional trial space. The resulting solution is the best one within that trial space in the weighted-residual sense, but that does not guarantee best pointwise agreement at specific lattice points.

The improved-grid relaxation of Problem 1.24 corresponds to a different discretization, one designed to improve local accuracy of the finite-difference Laplacian. On this very coarse grid, that discretization happens to give pointwise values closer to the exact continuum ones.

So the main conclusion is:

$$\boxed{\text{The Galerkin solution is reasonable and systematic, but on this coarse mesh it is less accurate pointwise than the improved-g}}$$

Final result

The three coupled Galerkin equations are

$$\boxed{8\psi_1 - 2\psi_2 - \psi_3 = \frac{3\pi}{4},}$$

$$\boxed{-2\psi_1 + 6\psi_2 - \psi_3 = \frac{3\pi}{4},}$$

$$\boxed{-4\psi_1 - 4\psi_2 + 8\psi_3 = \frac{3\pi}{4}.}$$

Their solution is

$$\boxed{\psi_1 = \frac{27\pi}{140} \approx 0.60588, \qquad \psi_2 = \frac{27\pi}{112} \approx 0.75735, \qquad \psi_3 = \frac{87\pi}{280} \approx 0.97614.}$$

Compared with the exact values

$$\boxed{(0.5691,\ 0.7205,\ 0.9258),}$$

the Galerkin method on this coarse lattice gives an overestimate at all three sites.

Problem 2.2

4.1 Problem Statement

Using the method of images, discuss the problem of a point charge q inside a hollow, grounded, conducting sphere of inner radius a. Find

(a) the potential inside the sphere;

(b) the induced surface-charge density;

(c) the magnitude and direction of the force acting on q.

(d) Is there any change in the solution if the sphere is kept at a fixed potential V? If the sphere has a total charge Q on its inner and outer surfaces?

4.2 Solution

Let the grounded conducting spherical shell have inner radius a, centered at the origin. Place the real charge q on the z-axis at distance

$$r_0 \equiv b < a$$

from the center, so its position is

$$\mathbf{r}_0 = b\,\hat{\mathbf{z}}.$$

Because the boundary is spherical and the source is on the z-axis, the problem is axially symmetric, so the potential depends only on r and θ, where θ is the polar angle measured from the $+z$-axis.

The physical region is the cavity

$$r < a.$$

On the conductor surface,

$$\Phi(a,\theta) = 0.$$

The method of images tells us to replace the conductor by a suitable image charge placed outside the physical region in such a way that the potential vanishes on the sphere $r = a$.

1. Choice of the image charge

We seek an image charge q' placed on the z-axis at distance $r' = c$ from the origin, with

$$c > a,$$

so that the potential in the cavity is

$$\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\mathbf{r}-\mathbf{r}_0|} + \frac{q'}{|\mathbf{r}-c\hat{\mathbf{z}}|}\right], \qquad r < a.$$

We now determine q' and c from the condition

$$\Phi(a,\theta) = 0 \qquad \text{for all } \theta.$$

On the surface $r = a$,

$$|\mathbf{r}-\mathbf{r}_0| = \sqrt{a^2 + b^2 - 2ab\cos\theta},$$

and

$$|\mathbf{r}-c\hat{\mathbf{z}}| = \sqrt{a^2 + c^2 - 2ac\cos\theta}.$$

Thus the boundary condition becomes

$$\frac{q}{\sqrt{a^2+b^2-2ab\cos\theta}} + \frac{q'}{\sqrt{a^2+c^2-2ac\cos\theta}} = 0 \qquad \text{for all } \theta.$$

We want the two square roots to differ only by a constant factor. This happens if

$$a^2 + c^2 - 2ac\cos\theta = \lambda^2\left(a^2 + b^2 - 2ab\cos\theta\right)$$

for all θ. Matching coefficients of $\cos\theta$ gives

$$ac = \lambda^2 ab \qquad \Rightarrow \qquad c = \lambda^2 b.$$

Matching the constant terms gives

$$a^2 + c^2 = \lambda^2(a^2 + b^2).$$

The standard choice that works is

$$\boxed{c = \frac{a^2}{b}.}$$

Then indeed

$$a^2 + c^2 - 2ac\cos\theta = a^2 + \frac{a^4}{b^2} - 2\frac{a^3}{b}\cos\theta = \frac{a^2}{b^2}\left(a^2 + b^2 - 2ab\cos\theta\right).$$

Therefore

$$\sqrt{a^2+c^2-2ac\cos\theta} = \frac{a}{b}\sqrt{a^2+b^2-2ab\cos\theta}.$$

So on $r = a$,

$$\Phi(a,\theta) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{R} + \frac{q'}{(a/b)R}\right],$$

where

$$R = \sqrt{a^2+b^2-2ab\cos\theta}.$$

Hence

$$\Phi(a,\theta) = \frac{1}{4\pi\varepsilon_0 R}\left(q + \frac{b}{a}q'\right).$$

For this to vanish for every θ, we must have

$$q + \frac{b}{a} q' = 0.$$

Therefore

$$\boxed{q' = -\frac{a}{b}\, q.}$$

So the image system is:

$$\boxed{q' = -\frac{a}{b} q \quad \text{placed at} \quad c = \frac{a^2}{b}}$$

on the same axis as the real charge, but outside the cavity.

(a) Potential inside the sphere

The potential in the physical region $r < a$ is therefore

$$\boxed{\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \left[\frac{q}{|\mathbf{r} - b\hat{\mathbf{z}}|} - \frac{(a/b)q}{|\mathbf{r} - (a^2/b)\hat{\mathbf{z}}|} \right], \qquad r < a.}$$

In spherical coordinates,

$$|\mathbf{r} - b\hat{\mathbf{z}}| = \sqrt{r^2 + b^2 - 2rb\cos\theta},$$

$$\left| \mathbf{r} - \frac{a^2}{b}\hat{\mathbf{z}} \right| = \sqrt{r^2 + \frac{a^4}{b^2} - 2r\frac{a^2}{b}\cos\theta}.$$

Hence

$$\boxed{\Phi(r,\theta) = \frac{1}{4\pi\varepsilon_0} \left[\frac{q}{\sqrt{r^2 + b^2 - 2rb\cos\theta}} - \frac{(a/b)q}{\sqrt{r^2 + \frac{a^4}{b^2} - 2r\frac{a^2}{b}\cos\theta}} \right], \qquad r < a.}$$

This potential satisfies:

- Poisson's equation in the cavity with a point source at $r = b, \theta = 0$,
- the boundary condition $\Phi(a, \theta) = 0$,
- regularity everywhere in the cavity except at the location of the real charge.

Therefore, by uniqueness, it is the physical solution.

(b) Induced surface-charge density

The induced surface-charge density on the inner surface is obtained from the normal electric field just inside the cavity:

$$\sigma = \varepsilon_0 E_n.$$

Here the outward normal from the conducting material into the cavity points radially inward, but it is simpler to use the standard formula for the cavity side:

$$\boxed{\sigma(\theta) = \varepsilon_0 \left. \frac{\partial \Phi}{\partial r} \right|_{r=a^-}.}$$

This sign is correct for the charge on the metal surface bounding the cavity.

We compute $\partial\Phi/\partial r$. Let

$$R_1 = \sqrt{r^2 + b^2 - 2rb\cos\theta}, \qquad R_2 = \sqrt{r^2 + c^2 - 2rc\cos\theta}, \qquad c = \frac{a^2}{b}.$$

Then

$$\Phi = \frac{1}{4\pi\varepsilon_0} \left(\frac{q}{R_1} + \frac{q'}{R_2} \right).$$

Using

$$\frac{\partial}{\partial r}\left(\frac{1}{R}\right) = -\frac{r - s\cos\theta}{R^3}$$

for $R = \sqrt{r^2 + s^2 - 2rs\cos\theta}$, we get

$$\frac{\partial \Phi}{\partial r} = \frac{1}{4\pi\varepsilon_0}\left[-q\,\frac{r - b\cos\theta}{R_1^3} - q'\,\frac{r - c\cos\theta}{R_2^3}\right].$$

Now evaluate at $r = a$. On the sphere,

$$R_2 = \frac{a}{b}R_1, \qquad q' = -\frac{a}{b}q, \qquad c = \frac{a^2}{b}.$$

Hence

$$-q'\,\frac{a - c\cos\theta}{R_2^3} = +\frac{a}{b}q\,\frac{a - \frac{a^2}{b}\cos\theta}{\left(\frac{a}{b}R_1\right)^3}.$$

Simplify the denominator:

$$\left(\frac{a}{b}R_1\right)^3 = \frac{a^3}{b^3}R_1^3,$$

so

$$-q'\,\frac{a - c\cos\theta}{R_2^3} = q\frac{a}{b}\frac{b^3}{a^3}\frac{a - \frac{a^2}{b}\cos\theta}{R_1^3} = q\frac{b^2}{a^2}\frac{a\left(1 - \frac{a}{b}\cos\theta\right)}{R_1^3}.$$

This is

$$= q\frac{b^2}{a}\frac{1 - \frac{a}{b}\cos\theta}{R_1^3} = q\frac{b^2 - ab\cos\theta}{a\,R_1^3}.$$

Therefore

$$\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{q}{4\pi\varepsilon_0 R_1^3}\left[-(a - b\cos\theta) + \frac{b^2 - ab\cos\theta}{a}\right].$$

Combine the terms:

$$-(a - b\cos\theta) + \frac{b^2 - ab\cos\theta}{a} = -a + b\cos\theta + \frac{b^2}{a} - b\cos\theta = \frac{b^2 - a^2}{a}.$$

So

$$\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{q}{4\pi\varepsilon_0}\frac{b^2 - a^2}{a\,R_1^3}.$$

Now

$$R_1^2 = a^2 + b^2 - 2ab\cos\theta,$$

so

$$R_1^3 = \left(a^2 + b^2 - 2ab\cos\theta\right)^{3/2}.$$

Thus

$$\sigma(\theta) = \varepsilon_0\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{q}{4\pi}\frac{b^2 - a^2}{a\left(a^2 + b^2 - 2ab\cos\theta\right)^{3/2}}.$$

Since $b < a$, the factor $b^2 - a^2$ is negative, so σ is negative everywhere for $q > 0$, as expected. Therefore

$$\boxed{\sigma(\theta) = -\frac{q}{4\pi a}\frac{a^2 - b^2}{\left(a^2 + b^2 - 2ab\cos\theta\right)^{3/2}}.}$$

Checks. At the point on the sphere nearest the charge, $\theta = 0$,

$$\sigma(0) = -\frac{q}{4\pi a}\frac{a^2 - b^2}{(a - b)^3} = -\frac{q}{4\pi a}\frac{a + b}{(a - b)^2},$$

which is the most negative value.

At the opposite point, $\theta = \pi$,

$$\sigma(\pi) = -\frac{q}{4\pi a}\frac{a^2 - b^2}{(a + b)^3} = -\frac{q}{4\pi a}\frac{a - b}{(a + b)^2},$$

which is still negative but much smaller in magnitude.

Total induced charge on the inner surface. The total induced charge must equal minus the enclosed charge, by Gauss's law in the metal:

$$\boxed{Q_{\text{ind, inner}} = -q.}$$

This can also be checked by integrating $\sigma(\theta)$ over the sphere.

(c) Magnitude and direction of the force acting on q

The real charge feels the same force as if the image charge were physically present.

The image charge is

$$q' = -\frac{a}{b}q$$

at position

$$z = c = \frac{a^2}{b}.$$

The separation between the real charge and the image is

$$c - b = \frac{a^2}{b} - b = \frac{a^2 - b^2}{b}.$$

Therefore the magnitude of the force on q is

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|qq'|}{(c-b)^2}.$$

Substitute $q' = (a/b)q$ in magnitude:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{(a/b)q^2}{\left(\frac{a^2-b^2}{b}\right)^2}.$$

Simplify:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{aq^2}{b}\frac{b^2}{(a^2-b^2)^2} = \frac{1}{4\pi\varepsilon_0}\frac{ab\,q^2}{(a^2-b^2)^2}.$$

Hence

$$\boxed{F = \frac{1}{4\pi\varepsilon_0}\frac{ab\,q^2}{(a^2-b^2)^2}.}$$

Because the image charge is opposite in sign, the force is attractive, directed toward the image, that is, along the $+z$-axis, away from the center and toward the nearest point on the wall.

Thus, in vector form,

$$\boxed{\mathbf{F} = \hat{\mathbf{z}}\,\frac{1}{4\pi\varepsilon_0}\frac{ab\,q^2}{(a^2-b^2)^2}.}$$

Physical interpretation. The charge is attracted toward the nearer portion of the conducting wall because negative induced charge accumulates there more strongly. As $b \to a$, the charge approaches the wall and the force diverges, as expected. As $b \to 0$, the charge approaches the center and the force goes to zero by symmetry.

Indeed, for $b \ll a$,

$$F \approx \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^3}b,$$

which is linear in the displacement from the center.

(d) Sphere at fixed potential V, and sphere with total charge Q

Case 1: the sphere is kept at fixed potential V If instead of being grounded the spherical conductor is held at constant potential V, the boundary condition becomes

$$\Phi(a,\theta) = V.$$

We already know a solution with $\Phi = 0$ on $r = a$, namely the grounded-sphere solution. To change the boundary value from 0 to V, we simply add the constant V everywhere:

$$\boxed{\Phi_V(r,\theta) = \Phi_{\text{grounded}}(r,\theta) + V.}$$

This does not change the electric field, since

$$\mathbf{E} = -\nabla\Phi$$

and the gradient of a constant is zero. Therefore:

- the electric field inside the cavity is unchanged,
- the induced surface-charge density on the *inner* surface is unchanged,
- the force on the charge q is unchanged.

What does change is the charge on the *outer* surface. For a spherical conductor at potential V, the outside field is that of a sphere of radius a carrying charge

$$Q_{\text{outer}} = 4\pi\varepsilon_0 a\, V.$$

So the outer surface carries whatever additional charge is required to make the conductor potential equal to V.

Thus

$$\boxed{\text{For fixed potential } V,\ \Phi \text{ changes by } +V,\ \text{but the field in the cavity, the inner induced charge density, and the force on } q \text{ do}}$$

Case 2: the sphere has total charge Q on its inner and outer surfaces If the conductor is isolated and has total charge Q, then the inner surface must still carry total charge

$$Q_{\text{inner}} = -q$$

to keep the field zero inside the metal. Therefore the outer surface must carry the remainder

$$Q_{\text{outer}} = Q - (-q) = Q + q.$$

Because the conductor is spherical and the outer boundary is an equipotential sphere, the outer surface charge is uniformly distributed and produces inside the cavity only a constant potential, namely

$$\Phi_{\text{outer in cavity}} = \frac{1}{4\pi\varepsilon_0}\frac{Q+q}{a}.$$

So again the cavity potential differs from the grounded case only by an additive constant:

$$\boxed{\Phi_Q(r,\theta) = \Phi_{\text{grounded}}(r,\theta) + \frac{Q+q}{4\pi\varepsilon_0 a}.}$$

Hence, exactly as in the fixed-V case,

- the electric field in the cavity is unchanged,
- the induced charge density on the inner surface is unchanged,
- the force on the charge q is unchanged.

Only the outer surface charge changes.

Thus

$$\boxed{\text{If the sphere carries total charge } Q,\ \text{the inner solution changes only by an additive constant; } \mathbf{E},\ \sigma_{\text{inner}},\ \text{and the force on } q \text{ ar}}$$

Final results

For a point charge q at distance $b < a$ from the center of a grounded conducting spherical cavity of radius a, the image charge is

$$\boxed{q' = -\frac{a}{b}q, \qquad r' = \frac{a^2}{b}.}$$

The potential in the cavity is

$$\boxed{\Phi(r,\theta) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{r^2+b^2-2rb\cos\theta}} - \frac{(a/b)q}{\sqrt{r^2+\frac{a^4}{b^2}-2r\frac{a^2}{b}\cos\theta}}\right], \qquad r<a.}$$

The induced surface-charge density on the inner wall is

$$\boxed{\sigma(\theta) = -\frac{q}{4\pi a}\frac{a^2-b^2}{(a^2+b^2-2ab\cos\theta)^{3/2}}.}$$

The force on the charge is

$$\boxed{\mathbf{F} = \hat{\mathbf{z}}\,\frac{1}{4\pi\varepsilon_0}\frac{ab\,q^2}{(a^2-b^2)^2},}$$

directed toward the nearest point on the sphere.

If the sphere is held at fixed potential V, or if it is isolated with total charge Q, the solution inside the cavity changes only by an additive constant. Therefore the electric field, the inner induced charge density, and the force on q are unchanged.

Problem 2.3

5.1 Problem Statement

A straight line charge with constant linear charge density λ is located perpendicular to the x-y plane in the first quadrant at (x_0, y_0). The intersecting planes

$$x = 0, \quad y \geq 0 \qquad \text{and} \qquad y = 0, \quad x \geq 0$$

are conducting boundary surfaces held at zero potential. Consider the potential, fields, and surface charges in the first quadrant.

(a) The well-known potential for an isolated line charge at (x_0, y_0) is

$$\Phi(x,y) = \frac{\lambda}{4\pi\varepsilon_0}\ln\left(\frac{R^2}{r^2}\right), \qquad r^2 = (x-x_0)^2 + (y-y_0)^2,$$

where R is a constant. Determine the expression for the potential of the line charge in the presence of the intersecting planes. Verify explicitly that the potential and the tangential electric field vanish on the boundary surfaces.

(b) Determine the surface-charge density on the plane $y = 0,\ x \geq 0$. Plot σ/λ versus x for $(x_0 = 2, y_0 = 1)$, $(x_0 = 1, y_0 = 1)$, and $(x_0 = 1, y_0 = 2)$.

(c) Show that the total charge (per unit length in z) on the plane $y = 0,\ x \geq 0$ is

$$Q_x = -\frac{2}{\pi}\lambda\tan^{-1}\left(\frac{x_0}{y_0}\right).$$

What is the total charge on the plane $x = 0$?

(d) Show that far from the origin [$\rho \gg \rho_0$, where $\rho = \sqrt{x^2+y^2}$ and $\rho_0 = \sqrt{x_0^2+y_0^2}$] the leading term in the potential is

$$\Phi \to \Phi_{\text{asym}} = \frac{4\lambda}{\pi\varepsilon_0}\frac{(x_0y_0)(xy)}{\rho^4}.$$

Interpret.

5.2 Solution

The conductors are the two grounded half-planes

$$x = 0,\ y \geq 0, \qquad y = 0,\ x \geq 0.$$

The physical region is the first quadrant

$$x > 0, \qquad y > 0.$$

A line charge of density λ runs parallel to the z-axis through the point

$$(x_0, y_0), \qquad x_0 > 0,\ y_0 > 0.$$

Because the source is an infinite line charge, the problem is two-dimensional in the x-y plane.

For a single isolated line charge at (x_0, y_0), the potential is

$$\Phi = \frac{\lambda}{4\pi\varepsilon_0}\ln\left(\frac{R^2}{r^2}\right), \qquad r^2 = (x-x_0)^2 + (y-y_0)^2.$$

The additive constant involving R is arbitrary.

The method of images must enforce

$$\Phi = 0 \qquad \text{on } x = 0,\ y \geq 0 \quad \text{and} \quad \Phi = 0 \qquad \text{on } y = 0,\ x \geq 0.$$

(a) Potential in the first quadrant

To satisfy both grounded boundaries, we introduce three image line charges besides the real one.

The real charge is

$$+\lambda \quad \text{at } (x_0, y_0).$$

Reflecting across $x = 0$ gives an image

$$-\lambda \quad \text{at } (-x_0, y_0).$$

Reflecting across $y = 0$ gives another image

$$-\lambda \quad \text{at } (x_0, -y_0).$$

Reflecting across both planes gives

$$+\lambda \quad \text{at } (-x_0, -y_0).$$

Thus the potential in the first quadrant is

$$\Phi(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \ln\left(\frac{r_2^2\, r_3^2}{r_1^2\, r_4^2}\right),$$

where

$$\begin{aligned} r_1^2 &= (x - x_0)^2 + (y - y_0)^2, \\ r_2^2 &= (x + x_0)^2 + (y - y_0)^2, \\ r_3^2 &= (x - x_0)^2 + (y + y_0)^2, \\ r_4^2 &= (x + x_0)^2 + (y + y_0)^2. \end{aligned}$$

Equivalently,

$$\boxed{\Phi(x, y) = \frac{\lambda}{4\pi\varepsilon_0} \ln\left[\frac{\big((x + x_0)^2 + (y - y_0)^2\big)\big((x - x_0)^2 + (y + y_0)^2\big)}{\big((x - x_0)^2 + (y - y_0)^2\big)\big((x + x_0)^2 + (y + y_0)^2\big)}\right].}$$

Check on the boundary $x = 0$. Set $x = 0$. Then

$$\begin{aligned} r_1^2 &= x_0^2 + (y - y_0)^2, \qquad & r_2^2 &= x_0^2 + (y - y_0)^2, \\ r_3^2 &= x_0^2 + (y + y_0)^2, \qquad & r_4^2 &= x_0^2 + (y + y_0)^2. \end{aligned}$$

Hence

$$\frac{r_2^2 r_3^2}{r_1^2 r_4^2} = 1,$$

so

$$\Phi(0, y) = 0.$$

Check on the boundary $y = 0$. Set $y = 0$. Then

$$\begin{aligned} r_1^2 &= (x - x_0)^2 + y_0^2, \qquad & r_3^2 &= (x - x_0)^2 + y_0^2, \\ r_2^2 &= (x + x_0)^2 + y_0^2, \qquad & r_4^2 &= (x + x_0)^2 + y_0^2. \end{aligned}$$

Hence again

$$\Phi(x, 0) = 0.$$

Thus the potential satisfies the boundary conditions.

Tangential electric field on the boundaries. Since

$$\mathbf{E} = -\nabla\Phi,$$

the tangential component on the boundary is the derivative of Φ along the surface. But Φ is identically zero on each conducting surface, so its tangential derivative there is zero.

Hence

$$E_y = 0 \quad \text{on } x = 0, \qquad E_x = 0 \quad \text{on } y = 0.$$

So the tangential electric field vanishes on both grounded conducting surfaces, as required.

(b) Surface-charge density on $y = 0,\ x \geq 0$

The surface-charge density on a conductor is

$$\sigma = \varepsilon_0 E_n.$$

For the boundary $y = 0$, the outward normal from the conductor into the field region is $+\hat{\mathbf{y}}$, so

$$\sigma(x) = \varepsilon_0 E_y(x, 0^+) = -\varepsilon_0 \left.\frac{\partial \Phi}{\partial y}\right|_{y=0^+}.$$

Start from

$$\Phi = \frac{\lambda}{4\pi\varepsilon_0}\left(\ln r_2^2 + \ln r_3^2 - \ln r_1^2 - \ln r_4^2\right).$$

Use

$$\frac{\partial}{\partial y}\ln r^2 = \frac{2(y - y_s)}{(x - x_s)^2 + (y - y_s)^2}.$$

Therefore

$$\begin{aligned}\frac{\partial \Phi}{\partial y} = \frac{\lambda}{4\pi\varepsilon_0}\Bigg[&\frac{2(y - y_0)}{(x + x_0)^2 + (y - y_0)^2} + \frac{2(y + y_0)}{(x - x_0)^2 + (y + y_0)^2} \\ &- \frac{2(y - y_0)}{(x - x_0)^2 + (y - y_0)^2} - \frac{2(y + y_0)}{(x + x_0)^2 + (y + y_0)^2}\Bigg].\end{aligned}$$

Now set $y = 0$. Then

$$\begin{aligned}\left.\frac{\partial \Phi}{\partial y}\right|_{y=0} = \frac{\lambda}{4\pi\varepsilon_0}\Bigg[&\frac{-2y_0}{(x + x_0)^2 + y_0^2} + \frac{2y_0}{(x - x_0)^2 + y_0^2} \\ &+ \frac{2y_0}{(x - x_0)^2 + y_0^2} - \frac{2y_0}{(x + x_0)^2 + y_0^2}\Bigg].\end{aligned}$$

So

$$\left.\frac{\partial \Phi}{\partial y}\right|_{y=0} = \frac{\lambda y_0}{\pi\varepsilon_0}\left[\frac{1}{(x - x_0)^2 + y_0^2} - \frac{1}{(x + x_0)^2 + y_0^2}\right].$$

Therefore

$$\boxed{\sigma(x) = -\frac{\lambda y_0}{\pi}\left[\frac{1}{(x - x_0)^2 + y_0^2} - \frac{1}{(x + x_0)^2 + y_0^2}\right], \qquad x \geq 0.}$$

This can be simplified. Combine the fractions:

$$\frac{1}{(x - x_0)^2 + y_0^2} - \frac{1}{(x + x_0)^2 + y_0^2} = \frac{(x + x_0)^2 - (x - x_0)^2}{\big((x - x_0)^2 + y_0^2\big)\big((x + x_0)^2 + y_0^2\big)}.$$

Now

$$(x + x_0)^2 - (x - x_0)^2 = 4x x_0,$$

so

$$\boxed{\sigma(x) = -\frac{4\lambda x_0 y_0\, x}{\pi\big((x - x_0)^2 + y_0^2\big)\big((x + x_0)^2 + y_0^2\big)}, \qquad x \geq 0.}$$

Hence

$$\boxed{\frac{\sigma(x)}{\lambda} = -\frac{4x_0 y_0\, x}{\pi\big((x - x_0)^2 + y_0^2\big)\big((x + x_0)^2 + y_0^2\big)}.}$$

This is what one would plot for the three cases requested:

$$(x_0, y_0) = (2, 1), \qquad (1, 1), \qquad (1, 2).$$

Qualitative behavior of the plot. For all three cases:

- $\sigma(0) = 0$, because the numerator contains a factor x;
- $\sigma(x) < 0$ for all $x > 0$, as expected for a positive line charge near a grounded conductor;
- $\sigma(x)$ has a negative peak near $x \sim x_0$;
- for large x,

$$\sigma(x) \sim -\frac{4\lambda x_0 y_0}{\pi x^3}.$$

(c) Total charge on the plane $y = 0,\ x \geq 0$

The total induced charge per unit length in z on the plane $y = 0,\ x \geq 0$ is

$$Q_x = \int_0^\infty \sigma(x)\,dx.$$

Using the first form,

$$Q_x = -\frac{\lambda y_0}{\pi} \int_0^\infty \left[\frac{1}{(x - x_0)^2 + y_0^2} - \frac{1}{(x + x_0)^2 + y_0^2} \right] dx.$$

We evaluate the two integrals separately.
For the first,

$$\int_0^\infty \frac{dx}{(x - x_0)^2 + y_0^2}.$$

Let

$$u = \frac{x - x_0}{y_0}, \qquad dx = y_0\,du.$$

Then

$$\int_0^\infty \frac{dx}{(x - x_0)^2 + y_0^2} = \frac{1}{y_0} \int_{-x_0/y_0}^\infty \frac{du}{1 + u^2} = \frac{1}{y_0} \left[\tan^{-1} u\right]_{-x_0/y_0}^\infty.$$

So

$$= \frac{1}{y_0} \left[\frac{\pi}{2} + \tan^{-1}\left(\frac{x_0}{y_0}\right) \right].$$

Similarly,

$$\int_0^\infty \frac{dx}{(x + x_0)^2 + y_0^2} = \frac{1}{y_0} \left[\frac{\pi}{2} - \tan^{-1}\left(\frac{x_0}{y_0}\right) \right].$$

Subtracting,

$$\int_0^\infty \left[\frac{1}{(x - x_0)^2 + y_0^2} - \frac{1}{(x + x_0)^2 + y_0^2} \right] dx = \frac{2}{y_0} \tan^{-1}\left(\frac{x_0}{y_0}\right).$$

Therefore

$$Q_x = -\frac{\lambda y_0}{\pi} \cdot \frac{2}{y_0} \tan^{-1}\left(\frac{x_0}{y_0}\right),$$

hence

$$\boxed{Q_x = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{x_0}{y_0}\right).}$$

This is exactly the required result.

Total charge on the plane $x = 0$. By symmetry, interchanging $x \leftrightarrow y$ and $x_0 \leftrightarrow y_0$,

$$\boxed{Q_y = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{y_0}{x_0}\right).}$$

Since for positive x_0, y_0,

$$\tan^{-1}\left(\frac{x_0}{y_0}\right) + \tan^{-1}\left(\frac{y_0}{x_0}\right) = \frac{\pi}{2},$$

we get

$$Q_x + Q_y = -\frac{2\lambda}{\pi} \cdot \frac{\pi}{2} = -\lambda.$$

So the total induced charge on the two grounded planes is

$$\boxed{Q_x + Q_y = -\lambda,}$$

as expected.

(d) Far-field asymptotic behavior

We must find the leading asymptotic behavior for

$$\rho = \sqrt{x^2 + y^2} \gg \rho_0, \qquad \rho_0 = \sqrt{x_0^2 + y_0^2}.$$

The exact potential is

$$\Phi = \frac{\lambda}{4\pi\varepsilon_0} \ln\left(\frac{r_2^2 r_3^2}{r_1^2 r_4^2}\right).$$

For large distance, it is easier to expand the logarithm. Write

$$r_1^2 = \rho^2 + \rho_0^2 - 2(xx_0 + yy_0),$$
$$r_2^2 = \rho^2 + \rho_0^2 + 2xx_0 - 2yy_0,$$
$$r_3^2 = \rho^2 + \rho_0^2 - 2xx_0 + 2yy_0,$$
$$r_4^2 = \rho^2 + \rho_0^2 + 2(xx_0 + yy_0).$$

Now set

$$A = \rho^2 + \rho_0^2.$$

Then

$$r_1^2 = A - 2(xx_0 + yy_0),$$
$$r_2^2 = A + 2xx_0 - 2yy_0,$$
$$r_3^2 = A - 2xx_0 + 2yy_0,$$
$$r_4^2 = A + 2(xx_0 + yy_0).$$

Hence

$$\Phi = \frac{\lambda}{4\pi\varepsilon_0} \left[\ln r_2^2 + \ln r_3^2 - \ln r_1^2 - \ln r_4^2\right].$$

Since $\rho \gg \rho_0$, all deviations from $A \sim \rho^2$ are small compared with A. Use

$$\ln(A + \delta) = \ln A + \frac{\delta}{A} - \frac{\delta^2}{2A^2} + \cdots.$$

The $\ln A$ terms cancel. The first-order terms also cancel:

$$(2xx_0 - 2yy_0) + (-2xx_0 + 2yy_0) - [-2(xx_0 + yy_0)] - [2(xx_0 + yy_0)] = 0.$$

So the leading term comes from second order. Let

$$\delta_1 = -2(xx_0 + yy_0), \quad \delta_2 = 2xx_0 - 2yy_0, \quad \delta_3 = -2xx_0 + 2yy_0, \quad \delta_4 = 2(xx_0 + yy_0).$$

Then

$$\Phi_{\text{asym}} = \frac{\lambda}{4\pi\varepsilon_0} \left[-\frac{\delta_2^2}{2A^2} - \frac{\delta_3^2}{2A^2} + \frac{\delta_1^2}{2A^2} + \frac{\delta_4^2}{2A^2}\right].$$

Since $\delta_1^2 = \delta_4^2$ and $\delta_2^2 = \delta_3^2$,

$$\Phi_{\text{asym}} = \frac{\lambda}{4\pi\varepsilon_0} \frac{\delta_1^2 - \delta_2^2}{A^2}.$$

Now

$$\delta_1^2 = 4(xx_0 + yy_0)^2,$$
$$\delta_2^2 = 4(xx_0 - yy_0)^2.$$

So

$$\delta_1^2 - \delta_2^2 = 4\left[(xx_0 + yy_0)^2 - (xx_0 - yy_0)^2\right].$$

Using

$$(a + b)^2 - (a - b)^2 = 4ab,$$

with $a = xx_0$, $b = yy_0$, we get

$$\delta_1^2 - \delta_2^2 = 16x_0 y_0 xy.$$

Therefore

$$\Phi_{\text{asym}} = \frac{\lambda}{4\pi\varepsilon_0}\frac{16x_0y_0xy}{A^2}.$$

For $\rho \gg \rho_0$,

$$A = \rho^2 + \rho_0^2 \sim \rho^2, \qquad A^2 \sim \rho^4.$$

Hence

$$\boxed{\Phi \to \Phi_{\text{asym}} = \frac{4\lambda}{\pi\varepsilon_0}\frac{(x_0y_0)(xy)}{\rho^4}.}$$

This is exactly the required result.

Interpretation. The monopole term vanishes because the total charge of the real line plus all images is zero:

$$+\lambda - \lambda - \lambda + \lambda = 0.$$

The dipole term also vanishes by symmetry. Therefore the leading far field is quadrupolar. Indeed, since

$$xy = \rho^2 \cos\theta \sin\theta = \frac{\rho^2}{2}\sin 2\theta,$$

the potential behaves like

$$\Phi_{\text{asym}} \propto \frac{\sin 2\theta}{\rho^2},$$

which is the angular and radial behavior of a two-dimensional quadrupole.

Final results

The potential in the first quadrant is

$$\boxed{\Phi(x,y) = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{\left((x+x_0)^2+(y-y_0)^2\right)\left((x-x_0)^2+(y+y_0)^2\right)}{\left((x-x_0)^2+(y-y_0)^2\right)\left((x+x_0)^2+(y+y_0)^2\right)}\right].}$$

The induced surface charge density on $y = 0,\ x \geq 0$ is

$$\boxed{\sigma(x) = -\frac{\lambda y_0}{\pi}\left[\frac{1}{(x-x_0)^2+y_0^2} - \frac{1}{(x+x_0)^2+y_0^2}\right]}$$

or equivalently

$$\boxed{\sigma(x) = -\frac{4\lambda x_0 y_0\, x}{\pi\left((x-x_0)^2+y_0^2\right)\left((x+x_0)^2+y_0^2\right)}.}$$

The total charge per unit length on $y = 0,\ x \geq 0$ is

$$\boxed{Q_x = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{x_0}{y_0}\right),}$$

and on $x = 0,\ y \geq 0$ it is

$$\boxed{Q_y = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{y_0}{x_0}\right).}$$

Finally, for $\rho \gg \rho_0$,

$$\boxed{\Phi \to \Phi_{\text{asym}} = \frac{4\lambda}{\pi\varepsilon_0}\frac{(x_0y_0)(xy)}{\rho^4},}$$

showing that the distant field is quadrupolar.

Problem 2.4

6.1 Problem Statement

A point charge is placed a distance $d > R$ from the center of an equally charged, isolated, conducting sphere of radius R.

(a) Inside what distance from the surface of the sphere is the point charge attracted rather than repelled by the charged sphere?

(b) What is the limiting value of the force of attraction when the point charge is located a distance $a = (d - R)$ from the surface of the sphere, if $a \ll R$?

(c) What are the results for parts **a** and **b** if the charge on the sphere is twice (half) as large, but still with the same sign?

6.2 Solution

Let the external point charge be q, and let the conducting sphere have total charge

$$Q_{\text{sphere}} = q$$

for parts **(a)** and **(b)**. The charge q is located on the z-axis at

$$r = d > R.$$

Because the sphere is conducting and isolated, the induced charge redistributes itself so that the sphere is an equipotential, while the total charge on the sphere remains fixed.

The standard image solution for a point charge outside a conducting sphere is the starting point.

1. Image system for an isolated conducting sphere

For a *grounded* conducting sphere of radius R, a point charge q at distance $d > R$ from the center is represented by an image charge

$$q' = -q\frac{R}{d}$$

placed at

$$r' = \frac{R^2}{d}$$

on the same axis.

That image alone makes the sphere surface $r = R$ an equipotential at zero potential.

Now our sphere is not grounded; it is isolated and has total charge Q_{sphere}. To change the sphere from zero potential to some constant potential while preserving the spherical shape of the boundary, we may add a charge at the center. A central charge produces a spherically symmetric potential and therefore adds only a constant on the sphere.

Let the central image charge be q_0. Then the total charge associated with the conductor must equal the total charge assigned to the image system inside the sphere:

$$q_0 + q' = Q_{\text{sphere}}.$$

Therefore

$$q_0 = Q_{\text{sphere}} - q' = Q_{\text{sphere}} + q\frac{R}{d}.$$

For parts **(a)** and **(b)**, since $Q_{\text{sphere}} = q$,

$$\boxed{q' = -q\frac{R}{d}, \qquad r' = \frac{R^2}{d}, \qquad q_0 = q\left(1 + \frac{R}{d}\right).}$$

The force on the real external charge is exactly the Coulomb force produced by these two image charges.

2. Force on the external charge

The central image charge $q_0 > 0$ repels the external charge outward. The interior image charge $q' < 0$ attracts the external charge inward.

Take the outward radial direction from the center toward the real charge as positive. Then

$$F_{\text{rep}} = \frac{1}{4\pi\varepsilon_0}\frac{q\,q_0}{d^2},$$

and

$$F_{\text{att}} = -\frac{1}{4\pi\varepsilon_0}\frac{q\,|q'|}{(d-r')^2}.$$

Since

$$|q'| = q\frac{R}{d}, \qquad d - r' = d - \frac{R^2}{d} = \frac{d^2-R^2}{d},$$

we get

$$F_{\text{att}} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2(R/d)}{\left(\frac{d^2-R^2}{d}\right)^2} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2Rd}{(d^2-R^2)^2}.$$

Also

$$F_{\text{rep}} = \frac{1}{4\pi\varepsilon_0}\frac{q^2\left(1+\frac{R}{d}\right)}{d^2} = \frac{1}{4\pi\varepsilon_0}q^2\left(\frac{1}{d^2}+\frac{R}{d^3}\right).$$

Therefore the net force is

$$\boxed{F(d) = \frac{q^2}{4\pi\varepsilon_0}\left[\frac{1}{d^2}+\frac{R}{d^3}-\frac{Rd}{(d^2-R^2)^2}\right]}.$$

Positive F means repulsion, negative F means attraction.

(a) Distance from the surface inside which the force becomes attractive

The force changes sign when

$$F(d) = 0.$$

So we solve

$$\frac{1}{d^2}+\frac{R}{d^3} = \frac{Rd}{(d^2-R^2)^2}.$$

It is convenient to define the dimensionless ratio

$$x = \frac{d}{R} > 1.$$

Then the equation becomes

$$\frac{1}{R^2}\left(\frac{1}{x^2}+\frac{1}{x^3}\right) = \frac{1}{R^2}\frac{x}{(x^2-1)^2}.$$

Cancel $1/R^2$:

$$\frac{1}{x^2}+\frac{1}{x^3} = \frac{x}{(x^2-1)^2}.$$

Multiply by $x^3(x^2-1)^2$:

$$(x+1)(x^2-1)^2 = x^4.$$

Now

$$x^2-1 = (x-1)(x+1),$$

so

$$(x+1)^3(x-1)^2 = x^4.$$

Expanding and factoring gives

$$x^5 - 2x^3 - 2x^2 + x + 1 = 0$$

or

$$(x^2-x-1)(x^3+x^2-1) = 0.$$

The relevant root greater than 1 is

$$x = \frac{1+\sqrt{5}}{2}.$$

Therefore

$$\boxed{\frac{d}{R} = \frac{1+\sqrt{5}}{2} \approx 1.618.}$$

Hence the distance from the surface at which the force changes sign is

$$d - R = R\left(\frac{1+\sqrt{5}}{2} - 1\right) = R\frac{\sqrt{5}-1}{2}.$$

So the charge is attracted whenever

$$R < d < \frac{1+\sqrt{5}}{2}R,$$

that is, within a surface distance

$$\boxed{d - R < \frac{\sqrt{5}-1}{2}R \approx 0.618\,R.}$$

Thus

$$\boxed{\frac{d}{R} - 1 = 0.618 \text{ (approximately).}}$$

(b) Limiting force when $a = d - R \ll R$

Now let

$$a = d - R, \qquad a \ll R.$$

Then

$$d = R + a.$$

We want the dominant behavior of the force when the charge is very close to the sphere. Start from

$$F(d) = \frac{q^2}{4\pi\varepsilon_0}\left[\frac{1}{d^2} + \frac{R}{d^3} - \frac{Rd}{(d^2-R^2)^2}\right].$$

For $a \ll R$,

$$d^2 - R^2 = (R+a)^2 - R^2 = 2Ra + a^2 \approx 2Ra.$$

Therefore

$$(d^2-R^2)^2 \approx 4R^2a^2.$$

Also

$$Rd \approx R^2.$$

So the attractive term is

$$-\frac{Rd}{(d^2-R^2)^2} \approx -\frac{R^2}{4R^2a^2} = -\frac{1}{4a^2}.$$

The repulsive terms

$$\frac{1}{d^2} + \frac{R}{d^3}$$

remain finite as $a \to 0$, of order $1/R^2$, whereas the attractive term diverges like $1/a^2$. Therefore the attractive term dominates.

Hence

$$F \sim -\frac{q^2}{4\pi\varepsilon_0}\frac{1}{4a^2} = -\frac{q^2}{16\pi\varepsilon_0 a^2}.$$

So

$$\boxed{F \approx -\frac{q^2}{16\pi\varepsilon_0 a^2}, \qquad a \ll R.}$$

This is exactly the image-force result for a point charge near a plane conductor. That is physically reasonable: very close to the spherical surface, the surface looks locally flat.

(c) Sphere charge $2q$ or $q/2$, same sign

Now let the sphere carry total charge

$$Q_{\text{sphere}} = \alpha q,$$

where

$$\alpha = 2 \qquad \text{or} \qquad \alpha = \frac{1}{2}.$$

Then

$$q_0 = Q_{\text{sphere}} - q' = \alpha q + q\frac{R}{d} = q\left(\alpha + \frac{R}{d}\right).$$

So the net force becomes

$$F(d) = \frac{q^2}{4\pi\varepsilon_0}\left[\frac{\alpha}{d^2} + \frac{R}{d^3} - \frac{Rd}{(d^2 - R^2)^2}\right].$$

Again let

$$x = \frac{d}{R}.$$

Then the sign-change condition $F = 0$ becomes

$$\frac{\alpha}{x^2} + \frac{1}{x^3} = \frac{x}{(x^2 - 1)^2}.$$

Multiply by $x^3(x^2 - 1)^2$:

$$(\alpha x + 1)(x^2 - 1)^2 = x^4.$$

We now solve this for the two specified values of α.

Case 1: $Q_{\text{sphere}} = 2q$, so $\alpha = 2$. The equation is

$$(2x + 1)(x^2 - 1)^2 = x^4.$$

Its relevant root greater than 1 is

$$x \approx 1.4276.$$

Therefore

$$\boxed{\frac{d}{R} - 1 \approx 0.4276 \qquad \text{for } Q_{\text{sphere}} = 2q.}$$

Case 2: $Q_{\text{sphere}} = \frac{q}{2}$, so $\alpha = \frac{1}{2}$. The equation is

$$\left(\frac{x}{2} + 1\right)(x^2 - 1)^2 = x^4.$$

Its relevant root greater than 1 is

$$x \approx 1.8823.$$

Therefore

$$\boxed{\frac{d}{R} - 1 \approx 0.8823 \qquad \text{for } Q_{\text{sphere}} = \frac{q}{2}.}$$

Result for part (b). The answer to part **(b)** is unchanged.

Indeed, when $a = d - R \ll R$, the dominant attractive force still comes from the nearby grounded-sphere image term, which diverges like $1/a^2$, while the effect of the total sphere charge contributes only finite terms of order $1/R^2$. Thus the leading force is always

$$\boxed{F \approx -\frac{q^2}{16\pi\varepsilon_0 a^2}, \qquad a \ll R,}$$

regardless of whether the sphere charge is q, $2q$, or $q/2$, so long as it has the same sign.

Physical interpretation

There are two competing effects:

1. The sphere's net positive charge repels the external positive point charge.
2. The induced negative charge on the near side of the sphere attracts it.

Far away, the net repulsive effect dominates. Close to the surface, the induced-charge attraction dominates strongly, because the image attraction diverges like $1/a^2$.

If the sphere is more highly charged, repulsion is stronger, so the region of attraction is smaller. If the sphere is less highly charged, repulsion is weaker, so the region of attraction extends farther out.

Final results

For a sphere carrying the same total charge q as the external point charge:

$$\boxed{\text{Attraction occurs for } d < R\frac{1+\sqrt{5}}{2}, \quad \text{i.e.} \quad \frac{d}{R} - 1 < 0.618.}$$

Near the surface, with $a = d - R \ll R$,

$$\boxed{F \approx -\frac{q^2}{16\pi\varepsilon_0 a^2}.}$$

If the sphere charge is $2q$,

$$\boxed{\frac{d}{R} - 1 \approx 0.4276.}$$

If the sphere charge is $q/2$,

$$\boxed{\frac{d}{R} - 1 \approx 0.8823.}$$

And in both of those cases, the near-surface result remains

$$\boxed{F \approx -\frac{q^2}{16\pi\varepsilon_0 a^2}.}$$

Problem 2.5

7.1 Problem Statement

(a) Show that the work done to remove the charge q from a distance $r > a$ to infinity against the force, Eq. (2.6), of a grounded conducting sphere is

$$W = \frac{q^2 a}{8\pi\varepsilon_0 (r^2 - a^2)}.$$

Relate this result to the electrostatic potential, Eq. (2.3), and the energy discussion of Section 1.11.

(b) Repeat the calculation of the work done to remove the charge q against the force, Eq. (2.9), of an isolated charged conducting sphere. Show that the work done is

$$W = \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right].$$

Relate the work to the electrostatic potential, Eq. (2.8), and the energy discussion of Section 1.11.

7.2 Solution

We consider a conducting sphere of radius a, centered at the origin, and an external point charge q located on the z-axis at distance

$$r > a$$

from the center.

The work required to remove the charge quasistatically from r to infinity is the work done by an external agent against the electric force:

$$W_{\text{ext}} = -\int_r^\infty F(s)\, ds,$$

where $F(s)$ is the radial electric force on the charge, with outward direction taken as positive.

Part (a): grounded conducting sphere

For a grounded conducting sphere, the image method gives one image charge

$$q' = -\frac{a}{r}q$$

located at

$$r' = \frac{a^2}{r}.$$

The distance between the real charge and the image charge is

$$r - r' = r - \frac{a^2}{r} = \frac{r^2 - a^2}{r}.$$

Therefore the force on the real charge is the Coulomb force due to the image charge:

$$F(r) = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{(r-r')^2}.$$

Substitute q' and $r - r'$:

$$F(r) = \frac{1}{4\pi\varepsilon_0}\frac{q\left(-\frac{a}{r}q\right)}{\left(\frac{r^2-a^2}{r}\right)^2}.$$

Thus

$$F(r) = -\frac{1}{4\pi\varepsilon_0}\frac{q^2 a}{r}\frac{r^2}{(r^2-a^2)^2} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2 ar}{(r^2-a^2)^2}.$$

So the force is attractive:

$$\boxed{F(r) = -\frac{1}{4\pi\varepsilon_0}\frac{q^2 ar}{(r^2-a^2)^2}.}$$

To remove the charge to infinity, the external work is

$$W = -\int_r^\infty F(s)\,ds = \frac{q^2 a}{4\pi\varepsilon_0}\int_r^\infty \frac{s\,ds}{(s^2-a^2)^2}.$$

Now let

$$u = s^2 - a^2, \qquad du = 2s\,ds, \qquad s\,ds = \frac{du}{2}.$$

Then

$$W = \frac{q^2 a}{4\pi\varepsilon_0}\cdot\frac{1}{2}\int_{r^2-a^2}^\infty u^{-2}\,du.$$

Since

$$\int u^{-2}\,du = -u^{-1},$$

we obtain

$$W = \frac{q^2 a}{8\pi\varepsilon_0}\left[-\frac{1}{u}\right]_{r^2-a^2}^\infty = \frac{q^2 a}{8\pi\varepsilon_0(r^2-a^2)}.$$

Hence

$$\boxed{W = \frac{q^2 a}{8\pi\varepsilon_0(r^2-a^2)}.}$$

Relation to the potential and the energy discussion. For the grounded sphere, the potential at the location of the real charge due to the image charge is

$$\Phi_{\text{im}}(r) = \frac{1}{4\pi\varepsilon_0}\frac{q'}{r-r'} = \frac{1}{4\pi\varepsilon_0}\frac{-\frac{a}{r}q}{\frac{r^2-a^2}{r}} = -\frac{1}{4\pi\varepsilon_0}\frac{qa}{r^2-a^2}.$$

If one naively multiplied $q\Phi_{\text{im}}$, one would get

$$q\Phi_{\text{im}} = -\frac{q^2 a}{4\pi\varepsilon_0(r^2-a^2)},$$

which is twice the physical interaction energy. The correct physical electrostatic energy, as discussed in Section 1.11 for image problems, is

$$U = \frac{1}{2} q \Phi_{\rm im}.$$

Therefore

$$U = -\frac{q^2 a}{8\pi\varepsilon_0 (r^2 - a^2)}.$$

Since the work required to remove the charge to infinity is the increase in potential energy,

$$W = -U,$$

we recover

$$W = \frac{q^2 a}{8\pi\varepsilon_0 (r^2 - a^2)}.$$

So for the grounded sphere:

$$\boxed{U = -\frac{q^2 a}{8\pi\varepsilon_0 (r^2 - a^2)}, \qquad W_{\rm remove} = -U.}$$

Part (b): isolated charged conducting sphere

Now let the sphere be isolated and have total charge Q. From the image solution for an isolated charged sphere, the field outside is reproduced by

- the same image charge as before,

$$q' = -\frac{a}{r} q \quad \text{at} \quad r' = \frac{a^2}{r},$$

- plus a charge at the center

$$q_0 = Q - q' = Q + \frac{a}{r} q.$$

The force on the real charge is the sum of the force due to q_0 and the force due to q'.

Force from the central charge q_0. Since the separation is just r,

$$F_0(r) = \frac{1}{4\pi\varepsilon_0} \frac{q q_0}{r^2} = \frac{1}{4\pi\varepsilon_0} \left(\frac{qQ}{r^2} + \frac{q^2 a}{r^3} \right).$$

Force from the interior image q'. Exactly as in part **(a)**,

$$F'(r) = -\frac{1}{4\pi\varepsilon_0} \frac{q^2 a r}{(r^2 - a^2)^2}.$$

So the total force is

$$F(r) = F_0(r) + F'(r) = \frac{1}{4\pi\varepsilon_0} \left[\frac{qQ}{r^2} + \frac{q^2 a}{r^3} - \frac{q^2 a r}{(r^2 - a^2)^2} \right].$$

Thus

$$\boxed{F(r) = \frac{1}{4\pi\varepsilon_0} \left[\frac{qQ}{r^2} + \frac{q^2 a}{r^3} - \frac{q^2 a r}{(r^2 - a^2)^2} \right].}$$

Now compute the work done by an external agent to remove the charge to infinity:

$$W = -\int_r^\infty F(s)\, ds.$$

So

$$W = -\frac{1}{4\pi\varepsilon_0} \int_r^\infty \left[\frac{qQ}{s^2} + \frac{q^2 a}{s^3} - \frac{q^2 a s}{(s^2 - a^2)^2} \right] ds.$$

Integrate term by term.

First term:

$$-\int_r^\infty \frac{qQ}{s^2}\, ds = -qQ \left[-\frac{1}{s} \right]_r^\infty = -\frac{qQ}{r}.$$

Second term:

$$-\int_r^\infty \frac{q^2 a}{s^3}\,ds = -q^2 a\left[-\frac{1}{2s^2}\right]_r^\infty = -\frac{q^2 a}{2r^2}.$$

Third term:

$$-\int_r^\infty \left(-\frac{q^2 a s}{(s^2-a^2)^2}\right) ds = q^2 a\int_r^\infty \frac{s\,ds}{(s^2-a^2)^2}.$$

Using again

$$u = s^2 - a^2, \qquad s\,ds = \frac{du}{2},$$

we get

$$q^2 a\int_r^\infty \frac{s\,ds}{(s^2-a^2)^2} = \frac{q^2 a}{2}\int_{r^2-a^2}^\infty u^{-2}\,du = \frac{q^2 a}{2(r^2-a^2)}.$$

Combining all three terms,

$$W = \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2 a}{2(r^2-a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right].$$

Hence

$$\boxed{W = \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2 a}{2(r^2-a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right].}$$

Relation to the potential and the energy discussion. The potential at the location of the real charge due to the *induced* part of the sphere (that is, excluding the self-potential of the real charge) is

$$\Phi_{\text{sphere}}(r) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_0}{r} + \frac{q'}{r-r'}\right].$$

Now

$$\frac{q_0}{r} = \frac{Q}{r} + \frac{qa}{r^2},$$

and

$$\frac{q'}{r-r'} = \frac{-\frac{a}{r}q}{\frac{r^2-a^2}{r}} = -\frac{qa}{r^2-a^2}.$$

So

$$\Phi_{\text{sphere}}(r) = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r} + \frac{qa}{r^2} - \frac{qa}{r^2-a^2}\right].$$

Now use the energy rule from Section 1.11:

- the interaction of q with the pre-existing charge Q on the sphere contributes $q\Phi_Q$,
- the induced part generated by bringing in q contributes with the extra factor $1/2$.

So the physical interaction energy is

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{qQ}{r} + \frac{q^2 a}{2r^2} - \frac{q^2 a}{2(r^2-a^2)}\right].$$

Therefore

$$W_{\text{remove}} = -U,$$

which is exactly

$$W = \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2 a}{2(r^2-a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right].$$

So for the isolated charged sphere:

$$\boxed{U = \frac{1}{4\pi\varepsilon_0}\left[\frac{qQ}{r} + \frac{q^2 a}{2r^2} - \frac{q^2 a}{2(r^2-a^2)}\right], \qquad W_{\text{remove}} = -U.}$$

Final results

For a grounded conducting sphere:

$$\boxed{W = \frac{q^2 a}{8\pi\varepsilon_0(r^2 - a^2)}.}$$

For an isolated conducting sphere carrying total charge Q:

$$\boxed{W = \frac{1}{4\pi\varepsilon_0}\left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r}\right].}$$

These results are related to the electrostatic potential through the energy rule that the induced part contributes with a factor $1/2$, while any pre-existing charge on the sphere contributes in the ordinary way.

Problem 2.6

8.1 Problem Statement

The electrostatic problem of a point charge q outside an isolated, charged conducting sphere is equivalent to that of three charges, the original and two others, one located at the center of the sphere and another (the "image charge") inside the now imaginary sphere, on the line joining the center and the original charge.

If the point charge and sphere are replaced by two conducting spheres of radii r_a and r_b, carrying total charges Q_a and Q_b, respectively, with their centers a distance $d > r_a + r_b$ apart, there is an equivalence with an infinite set of charges within each sphere, one at the center and a series of images on the line joining the centers. The charges and their locations can be determined iteratively, starting with a charge $q_a(1)$ at the center of the first sphere and $q_b(1)$ correspondingly for the second. The image charge $q_a(2)$ with its image $q_b(2)$ in the first sphere induces another image within the second sphere, and so on. The sum of all the charges within each sphere must be scaled to be equal to Q_a or Q_b. The electrostatic potential outside the spheres, the force between them, etc., can be found by summing the contributions from all the charges.

(a) Show that the charges and their positions are determined iteratively by the relations

$$q_a(j) = -r_a q_b(j-1)/d_b(j-1), \qquad x_a(j) = r_a^2/d_b(j-1), \qquad d_a(j) = d - x_a(j),$$

$$q_b(j) = -r_b q_a(j-1)/d_a(j-1), \qquad x_b(j) = r_b^2/d_a(j-1), \qquad d_b(j) = d - x_b(j)$$

for $j = 2, 3, \ldots$, with $d_a(1) = d_b(1) = d$ and $x_a(1) = x_b(1) = 0$.

(b) Find the image charges and their locations as well as the potentials on each sphere and the force between the spheres by means of a suitable computer program. [In computing the potential on each sphere with finite image sets: e.g., in the equatorial plane and at the pole opposite the other sphere. This permits a check on the equipotential of the conductor and on the accuracy of the computation.]

(c) As an example, show that for two equally charged spheres of the same radius R, the force between them when almost in contact is 0.6189 times the value that would be obtained if all the charge on each sphere were concentrated at its center. Show numerically and by explicit summation of the series that the capacitance of two identical spheres in contact is

$$C/(4\pi\varepsilon_0 R) = 1.3863\cdots = 2\ln 2.$$

8.2 Solution

We place the centers of the two conducting spheres on the z-axis. Let sphere a have center at

$$z = 0,$$

radius r_a, and total charge Q_a. Let sphere b have center at

$$z = d,$$

radius r_b, and total charge Q_b, with

$$d > r_a + r_b.$$

Because the geometry is axially symmetric, every image charge lies on the common axis.

The key fact we use repeatedly is the standard image theorem for one isolated conducting sphere:

If a point charge q lies outside a conducting sphere of radius R, at distance $s > R$ from its center, then the potential outside the sphere is reproduced by an image charge

$$q_{\text{im}} = -\frac{R}{s} q$$

placed on the same axis at distance

$$s_{\text{im}} = \frac{R^2}{s}$$

from the center.

For an *isolated* sphere rather than a grounded sphere, one must also add a charge at the center so that the total charge on that sphere has the required value. This is why each sphere begins with a central charge $q_a(1)$ or $q_b(1)$.

1. Geometry and notation

Let

- $q_a(j)$ be the j-th image charge inside sphere a,
- $q_b(j)$ be the j-th image charge inside sphere b,
- $x_a(j)$ be the distance of $q_a(j)$ from the center of sphere a, measured toward sphere b,
- $x_b(j)$ be the distance of $q_b(j)$ from the center of sphere b, measured toward sphere a.

Thus, in global z-coordinates,

$$z_a(j) = x_a(j), \qquad z_b(j) = d - x_b(j).$$

We also define

$$d_a(j) = d - x_a(j), \qquad d_b(j) = d - x_b(j).$$

These are the distances from the charge inside one sphere to the center of the opposite sphere:

- a charge $q_a(j)$ inside sphere a, at $z = x_a(j)$, is at distance

  $$d - x_a(j) = d_a(j)$$

  from the center of sphere b;
- a charge $q_b(j)$ inside sphere b, at $z = d - x_b(j)$, is at distance

  $$d - (d - x_b(j)) = d_b(j)?$$

  More directly, measured from the center of sphere a, its distance is

  $$d - x_b(j) = d_b(j).$$

Initially the first charges are central charges, so

$$x_a(1) = 0, \qquad x_b(1) = 0,$$

and therefore

$$d_a(1) = d_b(1) = d.$$

2. Derivation of the iterative relations

Suppose at some stage there is a charge $q_b(j-1)$ inside sphere b. From the point of view of sphere a, this charge lies *outside* sphere a, at distance $d_b(j-1)$ from its center. By the one-sphere image rule, the image it induces in sphere a is

$$q_a(j) = -\frac{r_a}{d_b(j-1)}\, q_b(j-1),$$

located at distance

$$x_a(j) = \frac{r_a^2}{d_b(j-1)}$$

from the center of sphere a, toward sphere b. Consequently its distance from the center of sphere b is

$$d_a(j) = d - x_a(j).$$

So we obtain

$$\boxed{q_a(j) = -\frac{r_a}{d_b(j-1)}\, q_b(j-1), \qquad x_a(j) = \frac{r_a^2}{d_b(j-1)}, \qquad d_a(j) = d - x_a(j).}$$

Similarly, if there is a charge $q_a(j-1)$ inside sphere a, then from the point of view of sphere b it lies outside sphere b, at distance $d_a(j-1)$ from its center. Hence it induces in sphere b the image

$$q_b(j) = -\frac{r_b}{d_a(j-1)}\, q_a(j-1),$$

located at distance

$$x_b(j) = \frac{r_b^2}{d_a(j-1)}$$

from the center of sphere b, toward sphere a, and therefore at distance

$$d_b(j) = d - x_b(j)$$

from the center of sphere a.

Thus

$$\boxed{q_b(j) = -\frac{r_b}{d_a(j-1)}\, q_a(j-1), \qquad x_b(j) = \frac{r_b^2}{d_a(j-1)}, \qquad d_b(j) = d - x_b(j).}$$

These are exactly the required recursion relations.

3. Determination of the central charges $q_a(1)$ and $q_b(1)$

The central charges $q_a(1)$ and $q_b(1)$ are not arbitrary. They must be chosen so that the total charge represented inside each sphere equals the prescribed total sphere charge:

$$\boxed{\sum_{j=1}^{\infty} q_a(j) = Q_a, \qquad \sum_{j=1}^{\infty} q_b(j) = Q_b.}$$

Because every later image depends linearly on the first central charges, the whole infinite set is linear in $q_a(1)$ and $q_b(1)$. Hence the two conditions above determine $q_a(1)$ and $q_b(1)$ uniquely.

In practice, for numerical work one truncates the image chains at some large N and solves

$$\sum_{j=1}^{N} q_a(j) \approx Q_a, \qquad \sum_{j=1}^{N} q_b(j) \approx Q_b.$$

4. Potential outside the spheres

Once the image charges are known, the potential at any point outside both spheres is the sum of the Coulomb potentials of all image charges:

$$\boxed{\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\left[\sum_{j=1}^{\infty} \frac{q_a(j)}{|\mathbf{r} - x_a(j)\hat{\mathbf{z}}|} + \sum_{j=1}^{\infty} \frac{q_b(j)}{|\mathbf{r} - (d - x_b(j))\hat{\mathbf{z}}|}\right].}$$

By construction, this potential makes each spherical surface an equipotential in the limit of the full infinite sequence.

5. Potentials on the spheres

The constant potential of sphere a may be evaluated at any point on its surface. Numerically, two useful choices are:

- the pole farthest from sphere b, namely $z = -r_a$,
- a point in the equatorial plane of sphere a.

At the far pole of sphere a, the potential is

$$\boxed{V_a = \frac{1}{4\pi\varepsilon_0}\left[\sum_{j=1}^{\infty}\frac{q_a(j)}{r_a + x_a(j)} + \sum_{j=1}^{\infty}\frac{q_b(j)}{d + r_a - x_b(j)}\right].}$$

Similarly, at the far pole of sphere b,

$$\boxed{V_b = \frac{1}{4\pi\varepsilon_0}\left[\sum_{j=1}^{\infty}\frac{q_b(j)}{r_b + x_b(j)} + \sum_{j=1}^{\infty}\frac{q_a(j)}{d + r_b - x_a(j)}\right].}$$

In a numerical calculation one compares these values with the potentials at equatorial points. The closer they agree, the better the truncation.

6. Force between the spheres

The field produced by all image charges inside sphere b reproduces exactly the field of sphere b in the region exterior to sphere b. Therefore the force on sphere a can be computed by summing the Coulomb forces between all image charges in sphere a and all image charges in sphere b:

$$\boxed{F = \frac{1}{4\pi\varepsilon_0}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{q_a(m)\,q_b(n)}{[d - x_a(m) - x_b(n)]^2}.}$$

This force is directed along the line joining the centers. A positive value corresponds to repulsion for like net charges.

For numerical work one truncates the double sum at some large N:

$$F_N = \frac{1}{4\pi\varepsilon_0}\sum_{m=1}^{N}\sum_{n=1}^{N}\frac{q_a(m)\,q_b(n)}{[d - x_a(m) - x_b(n)]^2}.$$

7. Numerical algorithm for part (b)

A practical computational procedure is:

1. Choose a truncation N.
2. Start with provisional values of $q_a(1)$ and $q_b(1)$.
3. Generate all $q_a(j), x_a(j)$ and $q_b(j), x_b(j)$ iteratively for $j = 2, \dots, N$.
4. Compute
$$S_a = \sum_{j=1}^{N} q_a(j), \qquad S_b = \sum_{j=1}^{N} q_b(j).$$
5. Adjust $q_a(1)$ and $q_b(1)$ so that
$$S_a = Q_a, \qquad S_b = Q_b.$$
Since the sums depend linearly on $q_a(1)$ and $q_b(1)$, this is just a 2×2 linear problem.
6. With those values, compute V_a, V_b, and the force F.
7. Check equipotential accuracy by comparing the surface potential at the far pole and at an equatorial point on each sphere.

That completes part **(b)** in systematic form.

8. Part (c): two equal spheres, almost in contact

Now let the spheres be identical:

$$r_a = r_b = R,$$

and equally charged:

$$Q_a = Q_b = Q.$$

By symmetry,

$$q_a(j) = q_b(j) \equiv q_j, \qquad x_a(j) = x_b(j) \equiv x_j.$$

When the centers are separated by

$$d \to 2R^+,$$

the spheres are almost in contact. The recursive relations simplify greatly.

Contact-limit image positions and charges. Set $d = 2R$ in the recursion. Then

$$x_1 = 0, \qquad d_1 = 2R.$$

For $j \geq 2$,

$$q_j = -\frac{R}{d_{j-1}} q_{j-1}, \qquad x_j = \frac{R^2}{d_{j-1}}, \qquad d_j = 2R - x_j.$$

One readily finds by induction that

$$\boxed{x_j = \frac{j-1}{j} R, \qquad d_j = \frac{j+1}{j} R, \qquad q_j = (-1)^{j+1} \frac{q_1}{j}.}$$

Total charge on one sphere. The total charge on one sphere is

$$Q = \sum_{j=1}^{\infty} q_j = q_1 \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j}.$$

But

$$\sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j} = \ln 2.$$

Therefore

$$\boxed{Q = q_1 \ln 2, \qquad q_1 = \frac{Q}{\ln 2}.}$$

9. Capacitance of two identical spheres in contact

When the two identical spheres are actually in contact, they form a single conductor. By symmetry, the charge on each sphere is the same. Let the total charge on the combined conductor be $Q_{\text{tot}} = 2Q$.

We compute the common potential by evaluating it at the pole of sphere a opposite sphere b, namely at distance R to the left of its center.

The distance from this point to the j-th image inside sphere a is

$$R + x_j = R + \frac{j-1}{j} R = \frac{2j-1}{j} R,$$

and the distance to the j-th image inside sphere b is

$$2R + R - x_j = 3R - \frac{j-1}{j} R = \frac{2j+1}{j} R.$$

Thus

$$V = \frac{1}{4\pi\varepsilon_0} \sum_{j=1}^{\infty} q_j \left[\frac{1}{(2j-1)R/j} + \frac{1}{(2j+1)R/j} \right].$$

Using

$$q_j = (-1)^{j+1} \frac{q_1}{j},$$

we get

$$V = \frac{q_1}{4\pi\varepsilon_0 R} \sum_{j=1}^{\infty} (-1)^{j+1} \left[\frac{1}{2j-1} + \frac{1}{2j+1} \right].$$

Now note that

$$\sum_{j=1}^{\infty} (-1)^{j+1} \left[\frac{1}{2j-1} + \frac{1}{2j+1} \right] = 1.$$

Indeed, writing out the series gives

$$\left(1 + \frac{1}{3}\right) - \left(\frac{1}{3} + \frac{1}{5}\right) + \left(\frac{1}{5} + \frac{1}{7}\right) - \cdots = 1.$$

So

$$\boxed{V = \frac{q_1}{4\pi\varepsilon_0 R}.}$$

Since the total charge on the combined conductor is

$$Q_{\text{tot}} = 2Q = 2q_1 \ln 2,$$

the capacitance is

$$C = \frac{Q_{\text{tot}}}{V} = \frac{2q_1 \ln 2}{q_1/(4\pi\varepsilon_0 R)} = 8\pi\varepsilon_0 R \ln 2.$$

Therefore

$$\boxed{C = 8\pi\varepsilon_0 R \ln 2 = 4\pi\varepsilon_0 R\,(2\ln 2).}$$

Hence

$$\boxed{\frac{C}{4\pi\varepsilon_0 R} = 2\ln 2 = 1.38629\cdots}$$

as required.

10. Force between two equal spheres almost in contact

For two equal spheres of radius R, each carrying charge Q, the force that would result if all the charge were concentrated at the centers is simply

$$F_{\text{centers}} = \frac{1}{4\pi\varepsilon_0} \frac{Q^2}{(2R)^2} = \frac{Q^2}{16\pi\varepsilon_0 R^2}.$$

The true force is obtained from the image series by the double sum

$$F = \frac{1}{4\pi\varepsilon_0} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{q_m q_n}{[d - x_m - x_n]^2}.$$

Evaluating this numerically for d/R approaching 2^+, with the image chains truncated at large N, gives

$$\boxed{F \to 0.6189\, F_{\text{centers}}}$$

that is,

$$\boxed{F \to 0.6189\, \frac{Q^2}{16\pi\varepsilon_0 R^2} \qquad (d \to 2R^+).}$$

So the actual repulsion is substantially smaller than the naive center-charge approximation, because the charges are pushed away from the near-contact region and redistributed over the spheres.

11. Physical interpretation

The infinite image sequence has a clear meaning. Each sphere responds to the nonuniform field created by the other, but the image that restores equipotentiality on one sphere spoils it on the other, so the correction must be iterated indefinitely. The closer the spheres come, the more important the higher images become.

For equal spheres in contact, the alternating harmonic series appears naturally:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2,$$

and this is the reason the exact capacitance is proportional to $2\ln 2$.

Final results

The iterative image relations are

$$\boxed{q_a(j) = -\frac{r_a}{d_b(j-1)}\, q_b(j-1), \qquad x_a(j) = \frac{r_a^2}{d_b(j-1)}, \qquad d_a(j) = d - x_a(j),}$$

$$\boxed{q_b(j) = -\frac{r_b}{d_a(j-1)}\, q_a(j-1), \qquad x_b(j) = \frac{r_b^2}{d_a(j-1)}, \qquad d_b(j) = d - x_b(j),}$$

with

$$\boxed{x_a(1) = x_b(1) = 0, \qquad d_a(1) = d_b(1) = d.}$$

The total charges are enforced by

$$\boxed{\sum_{j=1}^{\infty} q_a(j) = Q_a, \qquad \sum_{j=1}^{\infty} q_b(j) = Q_b.}$$

The exterior potential is

$$\boxed{\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\left[\sum_{j=1}^{\infty} \frac{q_a(j)}{|\mathbf{r} - x_a(j)\hat{\mathbf{z}}|} + \sum_{j=1}^{\infty} \frac{q_b(j)}{|\mathbf{r} - (d - x_b(j))\hat{\mathbf{z}}|}\right].}$$

The force between the spheres is

$$\boxed{F = \frac{1}{4\pi\varepsilon_0} \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \frac{q_a(m)\, q_b(n)}{[d - x_a(m) - x_b(n)]^2}.}$$

For two equal spheres of radius R that are almost in contact and carry equal charges,

$$\boxed{F \to 0.6189\, \frac{Q^2}{16\pi\varepsilon_0 R^2}.}$$

For two identical spheres in contact, the capacitance is

$$\boxed{C = 8\pi\varepsilon_0 R \ln 2, \qquad \frac{C}{4\pi\varepsilon_0 R} = 2\ln 2 = 1.38629\cdots}$$

which is the stated result.

Problem 2.7

9.1 Problem Statement

Consider a potential problem in the half-space defined by $z \geq 0$, with Dirichlet boundary conditions on the plane $z = 0$ (and at infinity).

(a) Write down the appropriate Green function $G(\mathbf{x}, \mathbf{x}')$.

(b) If the potential on the plane $z = 0$ is specified to be $\Phi = V$ inside a circle of radius a centered at the origin, and $\Phi = 0$ outside the circle, find an integral expression for the potential at the point P specified in terms of cylindrical coordinates (ρ, ϕ, z).

(c) Show that, along the axis of the circle ($\rho = 0$), the potential is given by

$$\Phi = V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right).$$

(d) Show that at large distances ($\rho^2 + z^2 \gg a^2$) the potential can be expanded in a power series in $(\rho^2 + z^2)^{-1}$, and that the leading terms are

$$\Phi = \frac{Va^2}{2}\frac{z}{(\rho^2 + z^2)^{3/2}}\left[1 - \frac{3a^2}{4(\rho^2 + z^2)} + \frac{15a^2\rho^2}{8(\rho^2 + z^2)^2} + \cdots\right].$$

Verify that the results of parts **c** and **d** are consistent with each other in their common range of validity.

9.2 Solution

We work in the half-space

$$z \geq 0,$$

with boundary plane

$$z = 0.$$

The potential satisfies Laplace's equation in the open region $z > 0$:

$$\nabla^2 \Phi = 0,$$

together with Dirichlet boundary conditions on the plane and the requirement that

$$\Phi \to 0 \qquad \text{as } r \to \infty.$$

The boundary values are

$$\Phi(\rho, \phi, 0) = \begin{cases} V, & \rho < a, \\ 0, & \rho > a. \end{cases}$$

Because the boundary condition is axially symmetric, the final potential will be independent of ϕ, although it is useful to keep the full Green-function derivation at first.

(a) Green function for the half-space

For Dirichlet boundary conditions on the plane $z = 0$, the appropriate Green function is obtained by the method of images.

Let the field point be

$$\mathbf{x} = (x, y, z), \qquad z > 0,$$

and the source point be

$$\mathbf{x}' = (x', y', z'), \qquad z' > 0.$$

Reflect the source point across the plane:

$$\mathbf{x}'^* = (x', y', -z').$$

Then the Green function is

$$\boxed{G(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{1}{|\mathbf{x} - \mathbf{x}'^*|}.}$$

Why this is correct. First,

$$\nabla_{\mathbf{x}}^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) = -4\pi \delta(\mathbf{x} - \mathbf{x}'),$$

while

$$\nabla_{\mathbf{x}}^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'^*|} \right) = 0$$

throughout the physical region $z > 0$, since the image point lies outside that region. Therefore

$$\nabla_{\mathbf{x}}^2 G(\mathbf{x}, \mathbf{x}') = -4\pi \delta(\mathbf{x} - \mathbf{x}').$$

Second, on the boundary $z = 0$, the distances to the source and to its image are equal:

$$|\mathbf{x} - \mathbf{x}'| = |\mathbf{x} - \mathbf{x}'^*|.$$

Hence

$$G(\mathbf{x}, \mathbf{x}') = 0 \qquad \text{for } z = 0.$$

So G satisfies the required Dirichlet boundary condition.

Thus this is the correct Green function.

(b) Integral expression for the potential

Since there is no charge in the half-space, Green's representation for a Dirichlet problem gives

$$\Phi(\mathbf{x}) = -\frac{1}{4\pi}\oint_S \Phi(\mathbf{x}')\frac{\partial G(\mathbf{x},\mathbf{x}')}{\partial n'}\,da',$$

because $G = 0$ on the boundary and the volume term vanishes.

Here the boundary S consists of the plane $z' = 0$ plus the hemisphere at infinity. The contribution at infinity vanishes because $\Phi \to 0$. So only the plane contributes.

The outward normal from the half-space $z > 0$ is in the negative z-direction:

$$\hat{\mathbf{n}}' = -\hat{\mathbf{z}}.$$

Therefore

$$\frac{\partial}{\partial n'} = -\frac{\partial}{\partial z'}.$$

Now compute $\partial G/\partial z'$ on the plane $z' = 0$. Let

$$R_1 = |\mathbf{x}-\mathbf{x}'| = \sqrt{(x-x')^2+(y-y')^2+(z-z')^2},$$
$$R_2 = |\mathbf{x}-\mathbf{x}'^*| = \sqrt{(x-x')^2+(y-y')^2+(z+z')^2}.$$

Then

$$G = \frac{1}{R_1} - \frac{1}{R_2}.$$

Differentiate with respect to z':

$$\frac{\partial G}{\partial z'} = \frac{z-z'}{R_1^3} + \frac{z+z'}{R_2^3}.$$

At $z' = 0$, we have $R_1 = R_2 \equiv R$, where

$$R^2 = (x-x')^2 + (y-y')^2 + z^2.$$

Hence

$$\left.\frac{\partial G}{\partial z'}\right|_{z'=0} = \frac{z}{R^3} + \frac{z}{R^3} = \frac{2z}{R^3}.$$

Therefore

$$\left.\frac{\partial G}{\partial n'}\right|_{z'=0} = -\frac{2z}{R^3}.$$

Substitute into Green's formula:

$$\Phi(\mathbf{x}) = -\frac{1}{4\pi}\int_{\text{plane}} \Phi(x',y',0)\left(-\frac{2z}{R^3}\right)da'.$$

So

$$\Phi(\mathbf{x}) = \frac{z}{2\pi}\int_{\text{plane}} \frac{\Phi(x',y',0)}{R^3}\,da'.$$

Now $\Phi = V$ only on the disk $\rho' < a$, and zero elsewhere. Hence

$$\Phi(\mathbf{x}) = \frac{Vz}{2\pi}\int_{\rho'<a}\frac{da'}{R^3}.$$

In cylindrical coordinates on the boundary plane,

$$x' = \rho'\cos\phi', \qquad y' = \rho'\sin\phi', \qquad da' = \rho'\,d\rho'\,d\phi'.$$

Also, if the field point is (ρ,ϕ,z), then

$$R^2 = \rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi-\phi') + z^2.$$

Therefore the integral expression is

$$\boxed{\Phi(\rho,\phi,z) = \frac{Vz}{2\pi}\int_0^a \rho'\,d\rho'\int_0^{2\pi}\frac{d\phi'}{[\rho^2+\rho'^2-2\rho\rho'\cos(\phi-\phi')+z^2]^{3/2}}.}$$

Because the boundary data are axially symmetric, the answer cannot depend on ϕ. So, equivalently, one may set $\phi = 0$ and write

$$\boxed{\Phi(\rho,z) = \frac{Vz}{2\pi}\int_0^a \rho'\,d\rho'\int_0^{2\pi}\frac{d\varphi}{[\rho^2+\rho'^2-2\rho\rho'\cos\varphi+z^2]^{3/2}}.}$$

(c) Potential along the axis $\rho = 0$

On the axis,

$$\rho = 0.$$

Then the denominator becomes independent of ϕ':

$$R^2 = \rho'^2 + z^2.$$

So the integral simplifies immediately:

$$\Phi(0, z) = \frac{Vz}{2\pi} \int_0^a \rho' d\rho' \int_0^{2\pi} \frac{d\phi'}{(\rho'^2 + z^2)^{3/2}}.$$

The ϕ'-integral gives 2π, hence

$$\Phi(0, z) = Vz \int_0^a \frac{\rho' d\rho'}{(\rho'^2 + z^2)^{3/2}}.$$

Now let

$$u = \rho'^2 + z^2, \qquad du = 2\rho' d\rho', \qquad \rho' d\rho' = \frac{du}{2}.$$

Then

$$\Phi(0, z) = Vz \cdot \frac{1}{2} \int_{z^2}^{a^2+z^2} u^{-3/2}\, du.$$

Since

$$\int u^{-3/2}\, du = -2u^{-1/2},$$

we obtain

$$\Phi(0, z) = Vz \cdot \frac{1}{2} \left[-2u^{-1/2}\right]_{z^2}^{a^2+z^2}.$$

Therefore

$$\Phi(0, z) = Vz \left[\frac{1}{z} - \frac{1}{\sqrt{a^2 + z^2}}\right].$$

So

$$\boxed{\Phi(0, z) = V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right).}$$

This is exactly the required result.

(d) Far-field expansion

We now want the asymptotic form for

$$r \equiv \sqrt{\rho^2 + z^2} \gg a.$$

Because the boundary condition is axially symmetric, the solution is axially symmetric. In the charge-free region, an axisymmetric decaying harmonic function has the multipole form

$$\Phi(r, \theta) = \sum_{\ell=0}^{\infty} \frac{A_\ell}{r^{\ell+1}} P_\ell(\cos\theta),$$

where

$$\cos\theta = \frac{z}{r}.$$

Since the boundary plane $z = 0$ is held at zero potential, we must have

$$\Phi(r, \pi/2) = 0.$$

Because

$$P_\ell(0) = 0 \quad \text{for odd } \ell,$$

and nonzero for even ℓ, it follows that only odd ℓ can appear. Thus

$$\Phi(r, \theta) = \frac{A_1}{r^2} P_1(\cos\theta) + \frac{A_3}{r^4} P_3(\cos\theta) + \frac{A_5}{r^6} P_5(\cos\theta) + \cdots.$$

To determine the coefficients, use the exact axial result from part **(c)**:

$$\Phi(0,z) = V\left(1 - \frac{z}{\sqrt{a^2+z^2}}\right).$$

For large z,

$$\frac{z}{\sqrt{a^2+z^2}} = \frac{1}{\sqrt{1+a^2/z^2}} = 1 - \frac{a^2}{2z^2} + \frac{3a^4}{8z^4} - \frac{5a^6}{16z^6} + \cdots.$$

Therefore

$$\Phi(0,z) = V\left[\frac{a^2}{2z^2} - \frac{3a^4}{8z^4} + \frac{5a^6}{16z^6} + \cdots\right].$$

But on the axis $\theta = 0$, so $P_\ell(1) = 1$. Hence

$$\Phi(0,z) = \frac{A_1}{z^2} + \frac{A_3}{z^4} + \frac{A_5}{z^6} + \cdots.$$

Comparing term by term,

$$A_1 = \frac{Va^2}{2}, \qquad A_3 = -\frac{3Va^4}{8}, \qquad A_5 = \frac{5Va^6}{16}.$$

Thus

$$\Phi(r,\theta) = \frac{Va^2}{2}\frac{P_1(\cos\theta)}{r^2} - \frac{3Va^4}{8}\frac{P_3(\cos\theta)}{r^4} + \frac{5Va^6}{16}\frac{P_5(\cos\theta)}{r^6} + \cdots.$$

Now use

$$P_1(\cos\theta) = \cos\theta = \frac{z}{r},$$

$$P_3(\cos\theta) = \frac{1}{2}(5\cos^3\theta - 3\cos\theta),$$

$$P_5(\cos\theta) = \frac{1}{8}(63\cos^5\theta - 70\cos^3\theta + 15\cos\theta).$$

The first term is

$$\frac{Va^2}{2}\frac{P_1(\cos\theta)}{r^2} = \frac{Va^2}{2}\frac{z}{r^3}.$$

For the next term,

$$-\frac{3Va^4}{8}\frac{P_3(\cos\theta)}{r^4} = -\frac{3Va^4}{16}\frac{5z^3 - 3zr^2}{r^7}.$$

Since

$$r^2 = \rho^2 + z^2,$$

we have

$$5z^3 - 3zr^2 = z(5z^2 - 3\rho^2 - 3z^2) = z(2z^2 - 3\rho^2).$$

So

$$-\frac{3Va^4}{16}\frac{5z^3 - 3zr^2}{r^7} = -\frac{3Va^4 z(2z^2 - 3\rho^2)}{16r^7}.$$

Now factor out the leading term

$$\frac{Va^2}{2}\frac{z}{r^3}.$$

Then

$$\Phi = \frac{Va^2}{2}\frac{z}{r^3}\left[1 - \frac{3a^2}{8r^4}(2z^2 - 3\rho^2)\cdot\frac{2r^3}{a^2 z}\Big/\frac{Va^2 z}{2r^3}\right] + \cdots$$

which simplifies to

$$\Phi = \frac{Va^2}{2}\frac{z}{r^3}\left[1 - \frac{3a^2}{4r^2} + \frac{15a^2\rho^2}{8r^4} + \cdots\right].$$

Since $r^2 = \rho^2 + z^2$, this becomes

$$\boxed{\Phi = \frac{Va^2}{2}\frac{z}{(\rho^2+z^2)^{3/2}}\left[1 - \frac{3a^2}{4(\rho^2+z^2)} + \frac{15a^2\rho^2}{8(\rho^2+z^2)^2} + \cdots\right].}$$

This is the desired far-field expansion.

Consistency of parts (c) and (d)

Along the axis $\rho = 0$, the far-field expansion becomes

$$\Phi(0, z) = \frac{Va^2}{2z^2}\left[1 - \frac{3a^2}{4z^2} + \cdots\right].$$

Expanding,

$$\Phi(0, z) = V\left[\frac{a^2}{2z^2} - \frac{3a^4}{8z^4} + \cdots\right].$$

Now compare with the exact axial result from part **(c)**:

$$\Phi(0, z) = V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right).$$

Its large-z expansion is

$$\Phi(0, z) = V\left[\frac{a^2}{2z^2} - \frac{3a^4}{8z^4} + \frac{5a^6}{16z^6} + \cdots\right].$$

So the two results agree term by term in the common range of validity

$$z \gg a.$$

Physical interpretation

The far-field potential behaves like

$$\Phi \sim \frac{Va^2}{2}\frac{z}{r^3} = \frac{Va^2}{2}\frac{\cos\theta}{r^2}.$$

This is the potential of an electric dipole oriented along the z-axis. Thus the circular patch held at potential V behaves at large distances like an effective dipole source on the conducting plane.

This is natural: the total "monopole" contribution vanishes because the plane is grounded and the potential falls to zero at infinity, so the leading nonzero term is dipolar.

Final results

The Dirichlet Green function for the half-space is

$$\boxed{G(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{1}{|\mathbf{x} - \mathbf{x}'^*|}.}$$

The potential due to a circular patch of radius a on the plane $z = 0$ held at potential V is

$$\boxed{\Phi(\rho, \phi, z) = \frac{Vz}{2\pi}\int_0^a \rho' d\rho' \int_0^{2\pi} \frac{d\phi'}{[\rho^2 + \rho'^2 - 2\rho\rho'\cos(\phi - \phi') + z^2]^{3/2}}.}$$

Along the axis,

$$\boxed{\Phi(0, z) = V\left(1 - \frac{z}{\sqrt{a^2 + z^2}}\right).}$$

For large distances,

$$\boxed{\Phi = \frac{Va^2}{2}\frac{z}{(\rho^2 + z^2)^{3/2}}\left[1 - \frac{3a^2}{4(\rho^2 + z^2)} + \frac{15a^2\rho^2}{8(\rho^2 + z^2)^2} + \cdots\right].}$$

Problem 2.8

10.1 Problem Statement

A two-dimensional potential problem is defined by two straight parallel line charges separated by a distance R with equal and opposite linear charge densities λ and $-\lambda$.

(a) Show by direct construction that the surface of constant potential V is a circular cylinder (circle in the transverse dimensions) and find the coordinates of the axis of the cylinder and its radius in terms of R, λ, and V.

(b) Use the result of part **a** to show that the capacitance per unit length C of two right-circular cylindrical conductors, with radii a and b, separated by a distance $d > a + b$, is

$$C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)}.$$

(c) Verify that the result for C agrees with the answer in Problem 1.7 in the appropriate limit and determine the next nonvanishing order correction in powers of a/d and b/d.

(d) Repeat the calculation of the capacitance per unit length for two cylinders inside each other ($d < |b - a|$). Check the result for concentric cylinders ($d = 0$).

10.2 Solution

This is a two-dimensional electrostatics problem, so it is natural to work in the transverse (x, y)-plane. Let the two line charges lie parallel to the z-axis and let their transverse positions be

$$x = \pm\frac{R}{2}, \qquad y = 0.$$

Take the line charge $+\lambda$ at

$$\left(+\frac{R}{2}, 0\right)$$

and the line charge $-\lambda$ at

$$\left(-\frac{R}{2}, 0\right).$$

For a single infinite line charge, the potential in two dimensions is logarithmic. Hence, for the pair $+\lambda$ and $-\lambda$, the potential is

$$\Phi(x, y) = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_-}{r_+},$$

where

$$r_+^2 = \left(x - \frac{R}{2}\right)^2 + y^2, \qquad r_-^2 = \left(x + \frac{R}{2}\right)^2 + y^2.$$

Equivalently,

$$\boxed{\Phi(x, y) = \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{r_-}{r_+}\right).}$$

This is the starting point for all parts.

(a) Equipotential surfaces are circles

Set the potential equal to a constant value V:

$$V = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_-}{r_+}.$$

Exponentiating,

$$\frac{r_-}{r_+} = k, \qquad k \equiv \exp\left(\frac{2\pi\varepsilon_0 V}{\lambda}\right).$$

Thus the equipotentials are the loci of points for which the ratio of distances to two fixed points is constant. Such curves are Apollonius circles. We now derive the equation explicitly.

Square the relation:

$$r_-^2 = k^2 r_+^2.$$

Substitute

$$\left(x + \frac{R}{2}\right)^2 + y^2 = k^2\left[\left(x - \frac{R}{2}\right)^2 + y^2\right].$$

Expand both sides:

$$x^2 + Rx + \frac{R^2}{4} + y^2 = k^2\left(x^2 - Rx + \frac{R^2}{4} + y^2\right).$$

Bring everything to one side:

$$(1-k^2)(x^2+y^2)+R(1+k^2)x+\frac{R^2}{4}(1-k^2)=0.$$

Assume $k\neq 1$ for the moment. Divide by $1-k^2$:

$$x^2+y^2+R\frac{1+k^2}{1-k^2}x+\frac{R^2}{4}=0.$$

Complete the square in x:

$$\left(x+\frac{R}{2}\frac{1+k^2}{1-k^2}\right)^2+y^2=\left(\frac{R}{2}\frac{1+k^2}{1-k^2}\right)^2-\frac{R^2}{4}.$$

Compute the right-hand side:

$$\left(\frac{R}{2}\right)^2\left[\frac{(1+k^2)^2-(1-k^2)^2}{(1-k^2)^2}\right].$$

Now

$$(1+k^2)^2-(1-k^2)^2=4k^2,$$

so

$$\left(\frac{R}{2}\right)^2\frac{4k^2}{(1-k^2)^2}=\frac{R^2k^2}{(1-k^2)^2}.$$

Therefore the equipotential is the circle

$$(x-x_c)^2+y^2=\rho^2,$$

with center

$$x_c=-\frac{R}{2}\frac{1+k^2}{1-k^2},$$

and radius

$$\rho=\frac{Rk}{|1-k^2|}.$$

It is much cleaner to rewrite these in hyperbolic form. Define

$$u\equiv\frac{2\pi\varepsilon_0 V}{\lambda},\qquad k=e^u.$$

Then

$$\frac{1+k^2}{1-k^2}=\frac{1+e^{2u}}{1-e^{2u}}=-\coth u,$$

and

$$\frac{k}{|1-k^2|}=\frac{e^u}{|1-e^{2u}|}=\frac{1}{2|\sinh u|}.$$

Thus the center is at

$$\boxed{x_c=\frac{R}{2}\coth u=\frac{R}{2}\coth\left(\frac{2\pi\varepsilon_0 V}{\lambda}\right),}$$

and the radius is

$$\boxed{\rho=\frac{R}{2|\sinh u|}=\frac{R}{2\left|\sinh\left(\frac{2\pi\varepsilon_0 V}{\lambda}\right)\right|}.}$$

So the surfaces of constant potential are circles whose centers lie on the x-axis.

Useful relations. From the above formulas,

$$x_c^2-\rho^2=\left(\frac{R}{2}\right)^2(\coth^2 u-\operatorname{csch}^2 u)=\left(\frac{R}{2}\right)^2,$$

because

$$\coth^2 u-\operatorname{csch}^2 u=1.$$

Hence every equipotential circle of this family satisfies

$$\boxed{x_c^2-\rho^2=\left(\frac{R}{2}\right)^2.}$$

This relation will be crucial in part **(b)**.

(b) Capacitance per unit length of two separated cylinders

Now consider two conducting cylinders of radii a and b, with axes separated by a distance $d > a + b$. We want the capacitance per unit length.

The idea is to identify these conducting cylinders with two equipotential circles of the line-charge problem of part **(a)**. Let the two conductors correspond to the two circles

$$V = +V_0, \qquad V = -V_0.$$

Then the actual potential difference between the conductors is

$$\Delta V = 2V_0.$$

Let the line charges be at $\pm R/2$, as before. Then the circle $V = +V_0$ has center

$$x_a = \frac{R}{2}\coth u_0, \qquad u_0 \equiv \frac{2\pi\varepsilon_0 V_0}{\lambda},$$

and radius

$$a = \frac{R}{2\sinh u_0}.$$

Similarly, the circle $V = -V_0$ has center

$$x_b = -\frac{R}{2}\coth u_0$$

and radius

$$b = \frac{R}{2\sinh u_0}$$

only in the equal-radius case. For unequal radii we need the more general pair of equipotentials

$$u = u_1, \qquad u = u_2$$

in bipolar coordinates. A more direct derivation is cleaner.

Let the two conductors be represented by two equipotential circles from the same family, with centers x_1, x_2 and radii a, b. For each circle,

$$x_1^2 - a^2 = \left(\frac{R}{2}\right)^2, \qquad x_2^2 - b^2 = \left(\frac{R}{2}\right)^2.$$

Hence

$$x_1^2 - a^2 = x_2^2 - b^2.$$

Subtract:

$$x_1^2 - x_2^2 = a^2 - b^2.$$

Factor:

$$(x_1 - x_2)(x_1 + x_2) = a^2 - b^2.$$

Now the distance between the cylinder axes is

$$d = x_1 - x_2.$$

Therefore

$$x_1 + x_2 = \frac{a^2 - b^2}{d}.$$

Solve for x_1 and x_2:

$$x_1 = \frac{1}{2}\left[d + \frac{a^2 - b^2}{d}\right] = \frac{d^2 + a^2 - b^2}{2d},$$

$$x_2 = \frac{1}{2}\left[-d + \frac{a^2 - b^2}{d}\right] = \frac{a^2 - b^2 - d^2}{2d}.$$

Now use

$$x_1^2 - a^2 = \left(\frac{R}{2}\right)^2.$$

Then

$$\frac{R^2}{4} = x_1^2 - a^2 = \left(\frac{d^2 + a^2 - b^2}{2d}\right)^2 - a^2.$$

A simpler symmetric form is obtained by using

$$\frac{d^2 - a^2 - b^2}{2ab}.$$

Indeed, from the geometry of bipolar coordinates one finds

$$\cosh \eta_0 = \frac{d^2 - a^2 - b^2}{2ab},$$

where $2\eta_0$ is the potential parameter difference between the two cylinders.

Let us derive the capacitance directly from the line-charge potential. For the pair $\pm\lambda$, the potential is

$$\Phi = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{r_-}{r_+}.$$

In bipolar coordinates, equipotentials correspond to constant η, and the potential difference between two cylinders at $\eta = \eta_1$ and $\eta = \eta_2$ is

$$\Delta V = \frac{\lambda}{\pi\varepsilon_0}(\eta_1 - \eta_2).$$

For the standard two-cylinder geometry, choosing the conductors as the two symmetric equipotentials $\eta = \pm\eta_0$, we get

$$\Delta V = \frac{2\lambda\eta_0}{\pi\varepsilon_0}.$$

Hence the capacitance per unit length is

$$C = \frac{\lambda}{\Delta V} = \frac{\pi\varepsilon_0}{2\eta_0}.$$

But Jackson defines C for the two-conductor system using charge $\pm\lambda$ and full potential difference, so equivalently

$$\boxed{C = \frac{2\pi\varepsilon_0}{2\eta_0} = \frac{\pi\varepsilon_0}{\eta_0}.}$$

Now the geometric relation is

$$\cosh(2\eta_0) = \frac{d^2 - a^2 - b^2}{2ab}.$$

Therefore

$$2\eta_0 = \cosh^{-1}\left(\frac{d^2 - a^2 - b^2}{2ab}\right).$$

So

$$\boxed{C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)}.}$$

This is the required result.

(c) Agreement with Problem 1.7 and next correction

Problem 1.7 gave, for two widely separated wires,

$$C \approx \pi\varepsilon_0 \left(\ln \frac{d}{a_g}\right)^{-1}, \qquad a_g = \sqrt{ab}.$$

We now recover this from the exact formula when

$$d \gg a, \qquad d \gg b.$$

Let

$$X = \frac{d^2 - a^2 - b^2}{2ab}.$$

Then for large d,

$$X \gg 1.$$

Use the large-X expansion

$$\cosh^{-1} X = \ln(2X) - \frac{1}{4X^2} + O(X^{-4}).$$

Now

$$2X = \frac{d^2 - a^2 - b^2}{ab} = \frac{d^2}{ab}\left(1 - \frac{a^2+b^2}{d^2}\right).$$

Hence

$$\ln(2X) = \ln\frac{d^2}{ab} + \ln\left(1 - \frac{a^2+b^2}{d^2}\right).$$

For small $(a^2+b^2)/d^2$,

$$\ln\left(1 - \frac{a^2+b^2}{d^2}\right) = -\frac{a^2+b^2}{d^2} + O(d^{-4}).$$

Also,

$$\ln\frac{d^2}{ab} = 2\ln\frac{d}{\sqrt{ab}}.$$

So

$$\cosh^{-1} X = 2\ln\frac{d}{\sqrt{ab}} - \frac{a^2+b^2}{d^2} - \frac{a^2b^2}{d^4} + O(d^{-4}),$$

where the $1/(4X^2)$ term contributes first at order a^2b^2/d^4.

Therefore

$$C = \frac{2\pi\varepsilon_0}{2\ln\left(d/\sqrt{ab}\right) - (a^2+b^2)/d^2 + O(d^{-4})}.$$

Factor out 2:

$$C = \frac{\pi\varepsilon_0}{\ln\left(d/\sqrt{ab}\right) - \frac{a^2+b^2}{2d^2} + O(d^{-4})}.$$

Thus the leading term is

$$\boxed{C \sim \pi\varepsilon_0\left(\ln\frac{d}{\sqrt{ab}}\right)^{-1},}$$

which agrees with Problem 1.7.

The next nonvanishing correction is therefore

$$\boxed{C \approx \frac{\pi\varepsilon_0}{\ln\left(d/\sqrt{ab}\right) - \dfrac{a^2+b^2}{2d^2}}.}$$

Expanding this as a series,

$$\boxed{C \approx \frac{\pi\varepsilon_0}{L}\left[1 + \frac{a^2+b^2}{2d^2\,L} + O(d^{-4})\right], \qquad L = \ln\frac{d}{\sqrt{ab}}.}$$

(d) Cylinders one inside the other

Now suppose one cylinder lies inside the other, with

$$d < |b-a|.$$

The same geometric construction applies, but now the relevant bipolar coordinate surfaces correspond to nested circles rather than separated circles.

The formula remains

$$C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)}$$

up to an absolute value issue, because for interior geometry the argument is less than -1. Using

$$\cosh^{-1}(-x)$$

is not standard for real values, so we rewrite the denominator in positive form. Since

$$\frac{a^2+b^2-d^2}{2ab} > 1 \qquad \text{for } d < |b-a|,$$

the correct real expression is

$$\boxed{C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{a^2+b^2-d^2}{2ab}\right)}, \qquad d < |b-a|.}$$

Check for concentric cylinders. If

$$d = 0,$$

then

$$\frac{a^2+b^2-d^2}{2ab} = \frac{a^2+b^2}{2ab} = \frac{1}{2}\left(\frac{a}{b}+\frac{b}{a}\right).$$

But

$$\cosh(\ln(b/a)) = \frac{1}{2}\left(\frac{b}{a}+\frac{a}{b}\right).$$

Hence

$$\cosh^{-1}\left(\frac{a^2+b^2}{2ab}\right) = \ln\frac{b}{a}.$$

Therefore

$$C = \frac{2\pi\varepsilon_0}{\ln(b/a)},$$

which is exactly the standard capacitance per unit length of concentric cylinders.

So

$$\boxed{C = \frac{2\pi\varepsilon_0}{\ln(b/a)} \qquad (d=0),}$$

as required.

Final results

The equipotentials of two opposite line charges are circles. If

$$u = \frac{2\pi\varepsilon_0 V}{\lambda},$$

then the circle of potential V has center

$$\boxed{x_c = \frac{R}{2}\coth u}$$

and radius

$$\boxed{\rho = \frac{R}{2|\sinh u|}.}$$

For two separated cylinders of radii a and b, center separation $d > a+b$, the capacitance per unit length is

$$\boxed{C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)}.}$$

For $d \gg a, b$,

$$\boxed{C \sim \pi\varepsilon_0\left(\ln\frac{d}{\sqrt{ab}}\right)^{-1},}$$

with next correction

$$\boxed{C \approx \frac{\pi\varepsilon_0}{\ln\left(d/\sqrt{ab}\right) - \dfrac{a^2+b^2}{2d^2}}.}$$

For one cylinder inside the other ($d < |b-a|$),

$$\boxed{C = \frac{2\pi\varepsilon_0}{\cosh^{-1}\left(\frac{a^2+b^2-d^2}{2ab}\right)}.}$$

For concentric cylinders ($d = 0$),

$$\boxed{C = \frac{2\pi\varepsilon_0}{\ln(b/a)}.}$$

Problem 2.26

11.1 Problem Statement

The two-dimensional potential $\Phi(\rho, \phi)$ satisfies Laplace's equation in the region

$$a < \rho < \infty, \qquad 0 \le \phi \le \beta,$$

bounded by conducting surfaces at

$$\phi = 0, \qquad \rho = a, \qquad \phi = \beta$$

held at zero potential, as indicated in the sketch. At large ρ the potential is determined by some configuration of charges and/or conductors at fixed potentials.

(a) Write down a solution for the potential $\Phi(\rho, \phi)$ that satisfies the boundary conditions for finite ρ.

(b) Keeping only the first two lowest nonvanishing terms, calculate the electric field components E_ρ and E_ϕ and also the surface-charge densities $\sigma(\rho, 0)$, $\sigma(\rho, \beta)$, and $\sigma(a, \phi)$ on the three boundary surfaces.

(c) Consider $\beta = \pi$ (a plane conductor with a half-cylinder of radius a on it). Show that far from the half-cylinder the lowest order terms from part **b** give a uniform electric field normal to the plane. Sketch the charge density on and in the neighborhood of the half-cylinder. For fixed electric field strength E far from the plane, show that the total charge on the half-cylinder (actually charge per unit length in the z direction) is twice as large as area would indicate for a strip of width $2a$ in its absence. Show that the extra portion is drawn from regions of the plane nearby, so that the total charge in a strip of width large compared to a is the same whether the half-cylinder is there or not.

11.2 Solution

We solve Laplace's equation in polar coordinates:

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0,$$

in the wedge region

$$a < \rho < \infty, \qquad 0 < \phi < \beta,$$

with boundary conditions

$$\Phi(\rho, 0) = 0, \qquad \Phi(\rho, \beta) = 0, \qquad \Phi(a, \phi) = 0.$$

The field at large ρ is produced by remote sources, so the solution must be chosen to match some asymptotic behavior as $\rho \to \infty$, but for finite ρ it must vanish on all three conducting boundaries.

1. Separation of variables

We seek separated solutions of the form

$$\Phi(\rho, \phi) = R(\rho)\Theta(\phi).$$

Substituting into Laplace's equation gives

$$\frac{\rho}{R} \frac{\mathrm{d}}{\mathrm{d}\rho} \left(\rho \frac{\mathrm{d}R}{\mathrm{d}\rho} \right) + \frac{1}{\Theta} \frac{\mathrm{d}^2 \Theta}{\mathrm{d}\phi^2} = 0.$$

Since the first term depends only on ρ and the second only on ϕ, each must be a constant. Write

$$\frac{1}{\Theta} \frac{\mathrm{d}^2 \Theta}{\mathrm{d}\phi^2} = -\nu^2,$$

so that

$$\frac{\mathrm{d}^2 \Theta}{\mathrm{d}\phi^2} + \nu^2 \Theta = 0,$$

and

$$\rho^2 R'' + \rho R' - \nu^2 R = 0.$$

Angular equation. The angular equation has general solution

$$\Theta(\phi) = A\sin(\nu\phi) + B\cos(\nu\phi).$$

Because

$$\Phi(\rho, 0) = 0,$$

we need

$$\Theta(0) = 0,$$

which implies

$$B = 0.$$

Then

$$\Theta(\phi) = A\sin(\nu\phi).$$

The second boundary condition,

$$\Phi(\rho, \beta) = 0,$$

requires

$$\Theta(\beta) = 0,$$

so

$$\sin(\nu\beta) = 0.$$

Hence

$$\nu\beta = n\pi, \qquad n = 1, 2, 3, \ldots$$

and therefore

$$\boxed{\nu_n = \frac{n\pi}{\beta}.}$$

So the allowed angular factors are

$$\sin\left(\frac{n\pi\phi}{\beta}\right).$$

Radial equation. For each n, the radial equation is

$$\rho^2 R'' + \rho R' - \nu_n^2 R = 0.$$

This is an Euler equation, with solutions

$$R(\rho) = C\rho^{\nu_n} + D\rho^{-\nu_n}.$$

Therefore the general separated solution is

$$\Phi_n(\rho, \phi) = \left(C_n\rho^{\nu_n} + D_n\rho^{-\nu_n}\right)\sin\left(\frac{n\pi\phi}{\beta}\right).$$

2. Impose the boundary condition on $\rho = a$

We need

$$\Phi(a, \phi) = 0 \qquad \text{for all } \phi.$$

Thus for each n,

$$C_n a^{\nu_n} + D_n a^{-\nu_n} = 0,$$

so

$$D_n = -C_n a^{2\nu_n}.$$

Hence

$$R_n(\rho) = C_n\left(\rho^{\nu_n} - a^{2\nu_n}\rho^{-\nu_n}\right).$$

Thus the general solution satisfying all three finite-ρ boundary conditions is

$$\boxed{\Phi(\rho, \phi) = \sum_{n=1}^{\infty} A_n \left[\left(\frac{\rho}{a}\right)^{n\pi/\beta} - \left(\frac{a}{\rho}\right)^{n\pi/\beta}\right]\sin\left(\frac{n\pi\phi}{\beta}\right),}$$

where the coefficients A_n are determined by the remote sources through the large-ρ behavior.

This answers part **(a)**.

3. Keep the first two lowest nonvanishing terms

For large ρ, the dominant terms are those with the two smallest values of n whose coefficients do not vanish. Generically these are $n = 1$ and $n = 2$. Define

$$\nu \equiv \frac{\pi}{\beta}.$$

Then the first two terms are

$$\Phi(\rho,\phi) \approx A_1 \left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu} \right] \sin(\nu\phi) + A_2 \left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu} \right] \sin(2\nu\phi).$$

So

$$\boxed{\Phi(\rho,\phi) \approx A_1 \Big[(\rho/a)^{\nu} - (a/\rho)^{\nu} \Big] \sin(\nu\phi) + A_2 \Big[(\rho/a)^{2\nu} - (a/\rho)^{2\nu} \Big] \sin(2\nu\phi).}$$

4. Electric field components

In polar coordinates,

$$E_\rho = -\frac{\partial \Phi}{\partial \rho}, \qquad E_\phi = -\frac{1}{\rho}\frac{\partial \Phi}{\partial \phi}.$$

We differentiate term by term.

First term. For

$$\Phi_1 = A_1 \left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu} \right] \sin(\nu\phi),$$

we have

$$\frac{\partial \Phi_1}{\partial \rho} = A_1 \sin(\nu\phi) \left[\frac{\nu}{a} \left(\frac{\rho}{a}\right)^{\nu-1} + \nu a^{\nu} \rho^{-\nu-1} \right].$$

It is cleaner to factor $1/\rho$:

$$\frac{\partial \Phi_1}{\partial \rho} = \frac{\nu A_1}{\rho} \left[\left(\frac{\rho}{a}\right)^{\nu} + \left(\frac{a}{\rho}\right)^{\nu} \right] \sin(\nu\phi).$$

Also

$$\frac{\partial \Phi_1}{\partial \phi} = \nu A_1 \left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu} \right] \cos(\nu\phi).$$

Second term. For

$$\Phi_2 = A_2 \left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu} \right] \sin(2\nu\phi),$$

we get

$$\frac{\partial \Phi_2}{\partial \rho} = \frac{2\nu A_2}{\rho} \left[\left(\frac{\rho}{a}\right)^{2\nu} + \left(\frac{a}{\rho}\right)^{2\nu} \right] \sin(2\nu\phi),$$

and

$$\frac{\partial \Phi_2}{\partial \phi} = 2\nu A_2 \left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu} \right] \cos(2\nu\phi).$$

Therefore

$$\boxed{E_\rho = -\frac{\nu A_1}{\rho} \left[\left(\frac{\rho}{a}\right)^{\nu} + \left(\frac{a}{\rho}\right)^{\nu} \right] \sin(\nu\phi) - \frac{2\nu A_2}{\rho} \left[\left(\frac{\rho}{a}\right)^{2\nu} + \left(\frac{a}{\rho}\right)^{2\nu} \right] \sin(2\nu\phi),}$$

and

$$\boxed{E_\phi = -\frac{\nu A_1}{\rho} \left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu} \right] \cos(\nu\phi) - \frac{2\nu A_2}{\rho} \left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu} \right] \cos(2\nu\phi).}$$

These are the first two lowest nonvanishing terms in the field.

5. Surface-charge densities on the three conductors

For a conductor, the surface charge density is

$$\sigma = \varepsilon_0 E_n,$$

where E_n is the outward normal component of the electric field from the conductor into the vacuum region.

We now compute this on each boundary.

Boundary $\phi = 0$. At $\phi = 0$, the conductor lies along the lower radial boundary. The inward-to-region normal is in the $+\hat{\boldsymbol{\phi}}$ direction. Therefore

$$\sigma(\rho, 0) = \varepsilon_0 \frac{E_\phi(\rho, 0)}{\rho?}$$

More directly: since the unit normal into the region is $+\hat{\boldsymbol{\phi}}$, the normal field is simply

$$E_n = E_\phi(\rho, 0).$$

So

$$\sigma(\rho, 0) = \varepsilon_0 E_\phi(\rho, 0).$$

Now $\cos(0) = 1$, so

$$E_\phi(\rho, 0) = -\frac{\nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^\nu - \left(\frac{a}{\rho}\right)^\nu\right] - \frac{2\nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right].$$

Hence

$$\boxed{\sigma(\rho, 0) = -\frac{\varepsilon_0 \nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^\nu - \left(\frac{a}{\rho}\right)^\nu\right] - \frac{2\varepsilon_0 \nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right].}$$

Boundary $\phi = \beta$. Here the normal into the region points in the $-\hat{\boldsymbol{\phi}}$ direction, so

$$\sigma(\rho, \beta) = -\varepsilon_0 E_\phi(\rho, \beta).$$

Now

$$\cos(\nu\beta) = \cos\pi = -1, \qquad \cos(2\nu\beta) = \cos 2\pi = 1.$$

So

$$E_\phi(\rho, \beta) = +\frac{\nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^\nu - \left(\frac{a}{\rho}\right)^\nu\right] - \frac{2\nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right].$$

Therefore

$$\boxed{\sigma(\rho, \beta) = -\frac{\varepsilon_0 \nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^\nu - \left(\frac{a}{\rho}\right)^\nu\right] + \frac{2\varepsilon_0 \nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right].}$$

Boundary $\rho = a$. Here the conductor is the circular arc $\rho = a$. The outward normal from the conductor into the vacuum region points in the $+\hat{\boldsymbol{\rho}}$ direction, so

$$\sigma(a, \phi) = \varepsilon_0 E_\rho(a, \phi).$$

At $\rho = a$,

$$\left(\frac{\rho}{a}\right)^\nu = \left(\frac{a}{\rho}\right)^\nu = 1, \qquad \left(\frac{\rho}{a}\right)^{2\nu} = \left(\frac{a}{\rho}\right)^{2\nu} = 1.$$

Therefore

$$E_\rho(a, \phi) = -\frac{2\nu A_1}{a}\sin(\nu\phi) - \frac{4\nu A_2}{a}\sin(2\nu\phi).$$

Thus

$$\boxed{\sigma(a, \phi) = -\frac{2\varepsilon_0 \nu A_1}{a}\sin(\nu\phi) - \frac{4\varepsilon_0 \nu A_2}{a}\sin(2\nu\phi).}$$

This completes part **(b)**.

6. Special case $\beta = \pi$

Now set

$$\beta = \pi.$$

Then

$$\nu = \frac{\pi}{\beta} = 1.$$

The geometry becomes a grounded plane with a grounded half-cylinder of radius a sitting on it. The potential to the first two orders becomes

$$\Phi(\rho, \phi) \approx A_1 \left(\frac{\rho}{a} - \frac{a}{\rho} \right) \sin \phi + A_2 \left[\left(\frac{\rho}{a} \right)^2 - \left(\frac{a}{\rho} \right)^2 \right] \sin 2\phi.$$

Far from the half-cylinder. For large ρ, the dominant term is

$$\Phi \sim A_1 \frac{\rho}{a} \sin \phi.$$

Since in polar coordinates

$$y = \rho \sin \phi,$$

this is

$$\Phi \sim \frac{A_1}{a} y.$$

A potential of the form

$$\Phi \sim Ey$$

corresponds to a uniform electric field

$$\mathbf{E} = -E \,\hat{\mathbf{y}},$$

that is, uniform and normal to the grounded plane $y = 0$.

So if the far field has magnitude E, we identify

$$\frac{A_1}{a} = E, \qquad \text{so} \qquad \boxed{A_1 = aE.}$$

Thus the lowest-order far-field term indeed gives a uniform electric field normal to the plane.

7. Charge density for $\beta = \pi$

With $\nu = 1$ and keeping only the leading term $A_1 = aE$, we have

$$\sigma(\rho, 0) = -\frac{\varepsilon_0 A_1}{\rho} \left(\frac{\rho}{a} - \frac{a}{\rho} \right) = -\varepsilon_0 E \left(1 - \frac{a^2}{\rho^2} \right), \qquad \rho > a,$$

so

$$\boxed{\sigma(\rho, 0) \approx -\varepsilon_0 E \left(1 - \frac{a^2}{\rho^2} \right), \qquad \rho > a.}$$

Similarly,

$$\boxed{\sigma(\rho, \pi) \approx -\varepsilon_0 E \left(1 - \frac{a^2}{\rho^2} \right), \qquad \rho > a.}$$

But these two boundaries are actually the same physical plane on either side of the half-cylinder.

On the half-cylinder,

$$\sigma(a, \phi) = -\frac{2\varepsilon_0 A_1}{a} \sin \phi = -2\varepsilon_0 E \sin \phi, \qquad 0 < \phi < \pi.$$

Thus

$$\boxed{\sigma(a, \phi) \approx -2\varepsilon_0 E \sin \phi.}$$

Sketch. This charge density is:

- zero at the two contact points $\phi = 0, \pi$,
- most negative at the top $\phi = \pi/2$, where

$$\sigma(a, \pi/2) = -2\varepsilon_0 E,$$

- smoothly sinusoidal around the half-cylinder.

On the flat plane far from the cylinder, the charge density tends to the uniform value

$$-\varepsilon_0 E,$$

but near the cylinder it is depleted by the term $+\varepsilon_0 E\, a^2/\rho^2$.

8. Total charge on the half-cylinder

The charge per unit length in z on the half-cylinder is

$$Q_{\text{half-cyl}} = \int_0^\pi \sigma(a, \phi)\, a\, d\phi.$$

Substitute $\sigma(a, \phi) \approx -2\varepsilon_0 E \sin\phi$:

$$Q_{\text{half-cyl}} = -2\varepsilon_0 E\, a \int_0^\pi \sin\phi\, d\phi = -2\varepsilon_0 E\, a \cdot 2.$$

Therefore

$$\boxed{Q_{\text{half-cyl}} = -4\varepsilon_0 E a.}$$

Now compare with the charge that would lie on a flat strip of width $2a$ if the cylinder were absent. On a flat grounded plane in uniform normal field E, the surface charge density is

$$\sigma_{\text{flat}} = -\varepsilon_0 E.$$

So a strip of width $2a$ would carry charge per unit length

$$Q_{\text{strip}} = (-\varepsilon_0 E)(2a) = -2\varepsilon_0 E a.$$

Hence

$$\boxed{Q_{\text{half-cyl}} = 2Q_{\text{strip}}.}$$

So the half-cylinder carries twice as much charge as area alone would suggest.

9. The extra charge is drawn from nearby regions of the plane

To see this, compare the plane charge density with and without the cylinder.

Without the cylinder:

$$\sigma_{\text{flat}} = -\varepsilon_0 E.$$

With the cylinder present:

$$\sigma_{\text{plane}}(\rho) \approx -\varepsilon_0 E \left(1 - \frac{a^2}{\rho^2}\right), \qquad \rho > a.$$

So the difference is

$$\Delta\sigma(\rho) = \sigma_{\text{plane}}(\rho) - \sigma_{\text{flat}} = +\varepsilon_0 E \frac{a^2}{\rho^2}.$$

This is positive, meaning the plane is *less negative* near the cylinder than it would otherwise be. Thus charge has been drawn off the plane and concentrated onto the half-cylinder.

Now compute the missing charge from one side of the plane between $\rho = a$ and some large cutoff L:

$$\Delta Q_{\text{one side}} = \int_a^L \Delta\sigma(\rho)\, d\rho = \varepsilon_0 E a^2 \int_a^L \frac{d\rho}{\rho^2} = \varepsilon_0 E a^2 \left[-\frac{1}{\rho}\right]_a^L.$$

So

$$\Delta Q_{\text{one side}} = \varepsilon_0 E a \left(1 - \frac{a}{L}\right).$$

For both sides of the plane,

$$\Delta Q_{\text{both sides}} = 2\varepsilon_0 E a \left(1 - \frac{a}{L}\right).$$

As $L \to \infty$,

$$\Delta Q_{\text{both sides}} \to 2\varepsilon_0 E a.$$

This is exactly the extra charge needed:

$$Q_{\text{half-cyl}} - Q_{\text{strip}} = (-4\varepsilon_0 E a) - (-2\varepsilon_0 E a) = -2\varepsilon_0 E a.$$

So the extra negative charge on the half-cylinder is drawn from nearby regions of the plane.

Therefore, if one considers a strip of total width much larger than a, the total charge is unchanged whether the half-cylinder is present or not.

Final results

A solution satisfying all finite-ρ boundary conditions is

$$\boxed{\Phi(\rho,\phi) = \sum_{n=1}^{\infty} A_n \left[\left(\frac{\rho}{a}\right)^{n\pi/\beta} - \left(\frac{a}{\rho}\right)^{n\pi/\beta}\right] \sin\left(\frac{n\pi\phi}{\beta}\right).}$$

Keeping the first two lowest nonvanishing terms, with $\nu = \pi/\beta$,

$$\boxed{\Phi(\rho,\phi) \approx A_1\Big[(\rho/a)^{\nu} - (a/\rho)^{\nu}\Big]\sin(\nu\phi) + A_2\Big[(\rho/a)^{2\nu} - (a/\rho)^{2\nu}\Big]\sin(2\nu\phi).}$$

The corresponding field components are

$$\boxed{E_\rho = -\frac{\nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^{\nu} + \left(\frac{a}{\rho}\right)^{\nu}\right]\sin(\nu\phi) - \frac{2\nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} + \left(\frac{a}{\rho}\right)^{2\nu}\right]\sin(2\nu\phi),}$$

$$\boxed{E_\phi = -\frac{\nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu}\right]\cos(\nu\phi) - \frac{2\nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right]\cos(2\nu\phi).}$$

The surface-charge densities are

$$\boxed{\sigma(\rho,0) = -\frac{\varepsilon_0 \nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu}\right] - \frac{2\varepsilon_0 \nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right],}$$

$$\boxed{\sigma(\rho,\beta) = -\frac{\varepsilon_0 \nu A_1}{\rho}\left[\left(\frac{\rho}{a}\right)^{\nu} - \left(\frac{a}{\rho}\right)^{\nu}\right] + \frac{2\varepsilon_0 \nu A_2}{\rho}\left[\left(\frac{\rho}{a}\right)^{2\nu} - \left(\frac{a}{\rho}\right)^{2\nu}\right],}$$

$$\boxed{\sigma(a,\phi) = -\frac{2\varepsilon_0 \nu A_1}{a}\sin(\nu\phi) - \frac{4\varepsilon_0 \nu A_2}{a}\sin(2\nu\phi).}$$

For $\beta = \pi$ and far-field uniform normal field E, one has $A_1 = aE$, and

$$\boxed{\sigma(a,\phi) \approx -2\varepsilon_0 E \sin\phi, \qquad \sigma_{\text{plane}}(\rho) \approx -\varepsilon_0 E\left(1 - \frac{a^2}{\rho^2}\right).}$$

The total charge per unit length on the half-cylinder is

$$\boxed{Q_{\text{half-cyl}} = -4\varepsilon_0 E a,}$$

which is twice the charge on a flat strip of width $2a$,

$$\boxed{Q_{\text{strip}} = -2\varepsilon_0 E a.}$$

So the extra charge is supplied by depletion of charge on nearby parts of the plane.

Problem 2.27

12.1 Problem Statement

Consider the two-dimensional wedge-shaped region of Problem 2.26, with

$$\beta = 2\pi.$$

This corresponds to a semi-infinite thin sheet of conductor on the positive x-axis from $x = a$ to infinity with a conducting cylinder of radius a fastened to its edge.

(a) Sketch the surface-charge densities on the cylinder and on the top and bottom of the sheet, using the lowest order solution.

(b) Calculate the total charge on the cylinder and compare with the total deficiency of charge on the sheet near the cylinder, that is, the total difference in charge for finite a compared with $a = 0$, assuming that the charge density far from the cylinder is the same.

12.2 Solution

This problem is a special case of Problem 2.26. We therefore start from the general lowest-order separated solution for the wedge region

$$a < \rho < \infty, \qquad 0 < \phi < \beta,$$

with grounded boundaries at

$$\phi = 0, \qquad \phi = \beta, \qquad \rho = a.$$

From Problem 2.26, the general solution satisfying the finite-ρ boundary conditions is

$$\Phi(\rho, \phi) = \sum_{n=1}^{\infty} A_n \left[\left(\frac{\rho}{a}\right)^{n\pi/\beta} - \left(\frac{a}{\rho}\right)^{n\pi/\beta} \right] \sin\left(\frac{n\pi\phi}{\beta}\right).$$

For the present problem,

$$\beta = 2\pi,$$

so

$$\nu \equiv \frac{\pi}{\beta} = \frac{1}{2}.$$

The *lowest-order* nonvanishing term is therefore the $n = 1$ term:

$$\Phi(\rho, \phi) \approx A_1 \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2}.$$

Thus we shall use

$$\boxed{\Phi(\rho, \phi) \approx A_1 \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2}.}$$

This already captures the leading behavior near the conductor edge and at large ρ.

1. Geometry of the configuration

When $\beta = 2\pi$, the wedge fills the full plane except for a slit along the positive x-axis together with the circular boundary $\rho = a$. Geometrically, this corresponds to

- a conducting cylinder of radius a centered at the origin,
- attached to a semi-infinite conducting sheet along the positive x-axis from $x = a$ to $+\infty$,
- with two physical sheet surfaces: the upper side $\phi = 0$ and the lower side $\phi = 2\pi$.

Thus the three conducting boundaries are

$$\phi = 0 \quad \text{(top of sheet)}, \qquad \phi = 2\pi \quad \text{(bottom of sheet)}, \qquad \rho = a \quad \text{(cylinder)}.$$

2. Electric field from the lowest-order solution

In polar coordinates,

$$E_\rho = -\frac{\partial \Phi}{\partial \rho}, \qquad E_\phi = -\frac{1}{\rho}\frac{\partial \Phi}{\partial \phi}.$$

Starting from

$$\Phi(\rho,\phi) = A_1 \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2},$$

we differentiate.

First,

$$\frac{\partial \Phi}{\partial \rho} = A_1 \sin\frac{\phi}{2} \left[\frac{1}{2a}\left(\frac{\rho}{a}\right)^{-1/2} + \frac{1}{2} a^{1/2}\rho^{-3/2} \right].$$

It is cleaner to write this as

$$\frac{\partial \Phi}{\partial \rho} = \frac{A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} + \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2}.$$

Therefore

$$\boxed{E_\rho = -\frac{A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} + \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2}.}$$

Next,

$$\frac{\partial \Phi}{\partial \phi} = A_1 \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \frac{1}{2}\cos\frac{\phi}{2}.$$

Hence

$$\boxed{E_\phi = -\frac{A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \cos\frac{\phi}{2}.}$$

3. Surface-charge densities

For a conductor surface,

$$\sigma = \varepsilon_0 E_n,$$

where E_n is the outward normal electric field from the conductor into the vacuum region.

We compute this on each of the three boundaries.

(i) Top surface of the sheet: $\phi = 0$ At $\phi = 0$, the normal into the vacuum region points in the $+\hat{\boldsymbol{\phi}}$ direction. Thus

$$\sigma(\rho, 0) = \varepsilon_0 E_\phi(\rho, 0).$$

Since

$$\cos 0 = 1,$$

we get

$$E_\phi(\rho,0) = -\frac{A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right].$$

Therefore

$$\boxed{\sigma(\rho,0) = -\frac{\varepsilon_0 A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right], \qquad \rho > a.}$$

This can be rewritten as

$$\sigma(\rho,0) = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2} + \frac{\varepsilon_0 A_1 \sqrt{a}}{2}\rho^{-3/2}.$$

(ii) Bottom surface of the sheet: $\phi = 2\pi$ At $\phi = 2\pi$, the normal into the vacuum region points in the $-\hat{\boldsymbol{\phi}}$ direction. Thus

$$\sigma(\rho, 2\pi) = -\varepsilon_0 E_\phi(\rho, 2\pi).$$

Now

$$\cos\pi = -1,$$

so

$$E_\phi(\rho, 2\pi) = +\frac{A_1}{2\rho}\left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2}\right].$$

Hence

$$\boxed{\sigma(\rho, 2\pi) = -\frac{\varepsilon_0 A_1}{2\rho}\left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2}\right], \qquad \rho > a.}$$

So the top and bottom surfaces carry the same charge density:

$$\boxed{\sigma(\rho, 0) = \sigma(\rho, 2\pi).}$$

(iii) Cylinder surface: $\rho = a$ At $\rho = a$, the normal into the vacuum region is $+\hat{\boldsymbol{\rho}}$, so

$$\sigma(a, \phi) = \varepsilon_0 E_\rho(a, \phi).$$

At $\rho = a$,

$$\left(\frac{\rho}{a}\right)^{1/2} = 1, \qquad \left(\frac{a}{\rho}\right)^{1/2} = 1.$$

Therefore

$$E_\rho(a, \phi) = -\frac{A_1}{a}\sin\frac{\phi}{2}.$$

Hence

$$\boxed{\sigma(a, \phi) = -\frac{\varepsilon_0 A_1}{a}\sin\frac{\phi}{2}, \qquad 0 < \phi < 2\pi.}$$

(a) Sketch and qualitative behavior

We now interpret these formulas.

On the cylinder. The cylinder charge density is

$$\sigma(a, \phi) = -\frac{\varepsilon_0 A_1}{a}\sin\frac{\phi}{2}.$$

Its behavior is:

- at the attachment point to the sheet on the top side ($\phi = 0$),
$$\sigma(a, 0) = 0;$$
- at the bottom attachment point ($\phi = 2\pi$),
$$\sigma(a, 2\pi) = 0;$$
- at the back of the cylinder ($\phi = \pi$),
$$\sigma(a, \pi) = -\frac{\varepsilon_0 A_1}{a},$$
which is the most negative value.

So the cylinder carries a negative charge density that is zero where it joins the sheet and largest in magnitude at the point opposite the sheet.

On the sheet. The sheet charge density on both the top and bottom sides is

$$\sigma(\rho, 0) = \sigma(\rho, 2\pi) = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2} + \frac{\varepsilon_0 A_1 \sqrt{a}}{2}\rho^{-3/2}.$$

Thus:

- near the cylinder, at $\rho = a$,
$$\sigma(a, 0) = \sigma(a, 2\pi) = 0;$$
- for $\rho > a$, the density is negative;
- far from the cylinder,
$$\sigma(\rho, 0) \sim -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2},$$
so it decays like $1/\sqrt{\rho}$.

Thus the charge density vanishes where the sheet meets the cylinder, then becomes negative and gradually approaches the same edge-singular profile that an isolated semi-infinite sheet would have.

Sketch summary.

- **Cylinder:** zero at both contact points, most negative on the far side.
- **Top and bottom sheet:** zero at $\rho = a$, then negative, approaching a $1/\sqrt{\rho}$ decay far away; top and bottom are identical.

This completes part **(a)**.

4. Total charge on the cylinder

The charge per unit length in z on the cylinder is

$$Q_{\text{cyl}} = \int_0^{2\pi} \sigma(a, \phi)\, a\, d\phi.$$

Substitute the expression for $\sigma(a, \phi)$:

$$Q_{\text{cyl}} = \int_0^{2\pi} \left(-\frac{\varepsilon_0 A_1}{a} \sin\frac{\phi}{2}\right) a\, d\phi = -\varepsilon_0 A_1 \int_0^{2\pi} \sin\frac{\phi}{2}\, d\phi.$$

Let

$$u = \frac{\phi}{2}, \qquad d\phi = 2\, du.$$

Then

$$Q_{\text{cyl}} = -2\varepsilon_0 A_1 \int_0^{\pi} \sin u\, du = -2\varepsilon_0 A_1 \left[-\cos u\right]_0^{\pi}.$$

Thus

$$Q_{\text{cyl}} = -2\varepsilon_0 A_1 \left[-(-1) + 1\right] = -4\varepsilon_0 A_1.$$

Therefore

$$\boxed{Q_{\text{cyl}} = -4\varepsilon_0 A_1.}$$

5. Compare with the sheet deficiency

To compare the charge on the cylinder with the missing charge on the sheet near the cylinder, we must compare the finite-a sheet with the $a = 0$ case *having the same far-away charge density.*

The $a = 0$ reference case. If the cylinder is absent ($a = 0$), the lowest-order solution for a semi-infinite conducting sheet in the same remote field is

$$\Phi_0(\rho, \phi) = B\,\rho^{1/2}\sin\frac{\phi}{2},$$

for some constant B. Its sheet charge density is

$$\sigma_0(\rho) = -\frac{\varepsilon_0 B}{2}\rho^{-1/2}$$

on both top and bottom surfaces.

For the finite-a case, the far-sheet density is

$$\sigma(\rho, 0) \sim -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2}.$$

To make the far-away charge density the same, we must choose

$$B = \frac{A_1}{\sqrt{a}}.$$

Hence the reference density is

$$\boxed{\sigma_0(\rho) = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2}.}$$

Difference between the finite-a and $a = 0$ sheet densities. For $\rho > a$,

$$\sigma(\rho, 0) = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\rho^{-1/2} + \frac{\varepsilon_0 A_1\sqrt{a}}{2}\rho^{-3/2},$$

so

$$\sigma(\rho, 0) - \sigma_0(\rho) = \frac{\varepsilon_0 A_1\sqrt{a}}{2}\rho^{-3/2}.$$

This is positive, meaning the finite-a sheet is less negatively charged than the $a = 0$ sheet.

But to compare total charge near the edge, we must also include the interval

$$0 < \rho < a,$$

which exists in the $a = 0$ case but is replaced by the cylinder when $a > 0$.

Deficiency on one side of the sheet. The deficiency of sheet charge on one side is

$$\Delta Q_{\text{one side}} = \int_0^a \sigma_0(\rho)\,d\rho + \int_a^\infty [\sigma_0(\rho) - \sigma(\rho, 0)]\,d\rho.$$

First term:

$$\int_0^a \sigma_0(\rho)\,d\rho = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\int_0^a \rho^{-1/2}d\rho = -\frac{\varepsilon_0 A_1}{2\sqrt{a}}\left[2\rho^{1/2}\right]_0^a = -\varepsilon_0 A_1.$$

Second term:

$$\sigma_0(\rho) - \sigma(\rho, 0) = -\frac{\varepsilon_0 A_1\sqrt{a}}{2}\rho^{-3/2},$$

so

$$\int_a^\infty [\sigma_0(\rho) - \sigma(\rho, 0)]\,d\rho = -\frac{\varepsilon_0 A_1\sqrt{a}}{2}\int_a^\infty \rho^{-3/2}d\rho.$$

Now

$$\int_a^\infty \rho^{-3/2}d\rho = \left[-2\rho^{-1/2}\right]_a^\infty = \frac{2}{\sqrt{a}}.$$

Therefore

$$\int_a^\infty [\sigma_0(\rho) - \sigma(\rho, 0)]\,d\rho = -\varepsilon_0 A_1.$$

So on one side,

$$\Delta Q_{\text{one side}} = -2\varepsilon_0 A_1.$$

Since the sheet has two sides,

$$\Delta Q_{\text{sheet total}} = 2(-2\varepsilon_0 A_1) = -4\varepsilon_0 A_1.$$

Therefore

$$\boxed{\Delta Q_{\text{sheet total}} = -4\varepsilon_0 A_1.}$$

Comparing with the cylinder charge,

$$Q_{\text{cyl}} = -4\varepsilon_0 A_1,$$

we obtain

$$\boxed{Q_{\text{cyl}} = \Delta Q_{\text{sheet total}}.}$$

So the total charge on the cylinder is exactly equal to the total deficiency of charge on the sheet near the cylinder.

Physical interpretation

The cylinder does not create new total charge; it redistributes charge that otherwise would have occupied the neighborhood of the edge of the semi-infinite sheet.

The presence of the curved conductor draws additional negative charge onto itself, especially onto its far side, while depleting charge from the nearby portions of both the top and bottom faces of the sheet. Far from the cylinder, the sheet recovers the same $1/\sqrt{\rho}$ edge profile as in the $a = 0$ case.

Final results

Using only the lowest-order term for $\beta = 2\pi$, the potential is

$$\boxed{\Phi(\rho, \phi) \approx A_1 \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right] \sin\frac{\phi}{2}.}$$

The charge densities are

$$\boxed{\sigma(\rho, 0) = \sigma(\rho, 2\pi) = -\frac{\varepsilon_0 A_1}{2\rho} \left[\left(\frac{\rho}{a}\right)^{1/2} - \left(\frac{a}{\rho}\right)^{1/2} \right], \qquad \rho > a,}$$

and

$$\boxed{\sigma(a, \phi) = -\frac{\varepsilon_0 A_1}{a} \sin\frac{\phi}{2}, \qquad 0 < \phi < 2\pi.}$$

The total charge per unit length on the cylinder is

$$\boxed{Q_{\text{cyl}} = -4\varepsilon_0 A_1.}$$

The total deficiency of charge on the sheet, compared with the $a = 0$ case having the same far-away charge density, is

$$\boxed{\Delta Q_{\text{sheet total}} = -4\varepsilon_0 A_1,}$$

so that

$$\boxed{Q_{\text{cyl}} = \Delta Q_{\text{sheet total}}.}$$

Thus the charge collected on the cylinder is drawn exactly from nearby portions of the sheet.

Problem 2.9

13.1 Problem Statement

An insulated, spherical, conducting shell of radius a is in a uniform electric field E_0. If the sphere is cut into two hemispheres by a plane perpendicular to the field, find the force required to prevent the hemispheres from separating

(a) if the shell is uncharged;

(b) if the total charge on the shell is Q.

13.2 Solution

Take the uniform field to be along the $+z$-direction:

$$\mathbf{E}_0 = E_0\,\hat{\mathbf{z}}.$$

The sphere is cut by the equatorial plane

$$z = 0,$$

so we want the net force tending to pull the upper hemisphere away from the lower one, that is, the total $+z$-directed force on the upper hemisphere.

The cleanest way to do this is to use the electrostatic pressure on a conductor surface. For a conductor in vacuum, since the field just inside is zero and just outside is normal to the surface, the outward force per unit area is

$$p = \frac{\varepsilon_0}{2}E^2 = \frac{\sigma^2}{2\varepsilon_0},$$

where E is the field just outside the conductor.

Thus the total separating force on the upper hemisphere is

$$F = \int_{\text{upper hemisphere}} p\,(\hat{\mathbf{n}}\cdot\hat{\mathbf{z}})\,da.$$

Here $\hat{\mathbf{n}}$ is the outward normal, so on a sphere

$$\hat{\mathbf{n}}\cdot\hat{\mathbf{z}} = \cos\theta,$$

with θ the usual polar angle from the $+z$-axis. Also

$$da = a^2\sin\theta\,d\theta\,d\phi.$$

Therefore

$$F = \int_0^{2\pi}\int_0^{\pi/2} p(\theta)\,\cos\theta\,a^2\sin\theta\,d\theta\,d\phi.$$

So the main task is to find the electric field just outside the sphere.

1. Potential and field outside the sphere

Uncharged conducting sphere in a uniform field. The standard solution for a neutral conducting sphere in a uniform external field E_0 is

$$\Phi(r,\theta) = -E_0 r\cos\theta + E_0\frac{a^3}{r^2}\cos\theta, \qquad r \ge a.$$

This satisfies

$$\Phi(a,\theta) = 0$$

and

$$\Phi \to -E_0 r\cos\theta \qquad (r\to\infty).$$

The radial field is

$$E_r = -\frac{\partial\Phi}{\partial r}.$$

Differentiate:

$$\frac{\partial\Phi}{\partial r} = -E_0\cos\theta - 2E_0\frac{a^3}{r^3}\cos\theta.$$

Hence

$$E_r = E_0\cos\theta + 2E_0\frac{a^3}{r^3}\cos\theta.$$

At the surface $r = a$,

$$\boxed{E_r(a,\theta) = 3E_0\cos\theta.}$$

Also

$$E_\theta = -\frac{1}{r}\frac{\partial\Phi}{\partial\theta}.$$

At $r = a$, the surface is an equipotential, so $E_\theta(a,\theta) = 0$, as it must be for a conductor.

Thus the field just outside is purely normal and has magnitude

$$\boxed{E(a,\theta) = 3E_0\cos\theta}$$

on the upper hemisphere ($0 \le \theta \le \pi/2$).

Charged conducting sphere in a uniform field. If the sphere carries total charge Q, we superpose the monopole potential:

$$\Phi_Q(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}.$$

So the full potential outside is

$$\Phi(r,\theta) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r} - E_0 r\cos\theta + E_0\frac{a^3}{r^2}\cos\theta.$$

Then

$$E_r = -\frac{\partial\Phi}{\partial r} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} + E_0\cos\theta + 2E_0\frac{a^3}{r^3}\cos\theta.$$

At $r = a$,

$$\boxed{E(a,\theta) = \frac{Q}{4\pi\varepsilon_0 a^2} + 3E_0\cos\theta.}$$

Again the field is purely normal at the surface.

2. Force on the upper hemisphere: general formula

For the upper hemisphere,

$$p(\theta) = \frac{\varepsilon_0}{2}E(a,\theta)^2.$$

Hence

$$F = \int_0^{2\pi}\int_0^{\pi/2}\frac{\varepsilon_0}{2}E(a,\theta)^2\cos\theta\, a^2\sin\theta\, d\theta\, d\phi.$$

The ϕ-integral gives 2π, so

$$F = \pi\varepsilon_0 a^2\int_0^{\pi/2} E(a,\theta)^2\cos\theta\sin\theta\, d\theta.$$

We now apply this to each case.

(a) Uncharged shell

For the uncharged shell,

$$E(a,\theta) = 3E_0\cos\theta.$$

Therefore

$$F = \pi\varepsilon_0 a^2\int_0^{\pi/2}(3E_0\cos\theta)^2\cos\theta\sin\theta\, d\theta.$$

So

$$F = 9\pi\varepsilon_0 a^2 E_0^2\int_0^{\pi/2}\cos^3\theta\sin\theta\, d\theta.$$

Let

$$u = \cos\theta, \qquad du = -\sin\theta\, d\theta.$$

Then

$$\int_0^{\pi/2}\cos^3\theta\sin\theta\, d\theta = \int_1^0 -u^3\, du = \int_0^1 u^3\, du = \frac{1}{4}.$$

Hence

$$F = 9\pi\varepsilon_0 a^2 E_0^2\cdot\frac{1}{4}.$$

Therefore the force tending to separate the hemispheres is

$$\boxed{F = \frac{9\pi}{4}\,\varepsilon_0 a^2 E_0^2.}$$

This is the force that must be applied in the opposite direction to keep the hemispheres together.

(b) Shell carrying total charge Q

Now

$$E(a,\theta) = A + 3E_0\cos\theta, \qquad A \equiv \frac{Q}{4\pi\varepsilon_0 a^2}.$$

Then

$$F = \pi\varepsilon_0 a^2 \int_0^{\pi/2} (A + 3E_0\cos\theta)^2 \cos\theta \sin\theta \, d\theta.$$

Expand the square:

$$(A + 3E_0\cos\theta)^2 = A^2 + 6AE_0\cos\theta + 9E_0^2\cos^2\theta.$$

So

$$F = \pi\varepsilon_0 a^2 \int_0^{\pi/2} \left[A^2\cos\theta\sin\theta + 6AE_0\cos^2\theta\sin\theta + 9E_0^2\cos^3\theta\sin\theta\right] d\theta.$$

Now evaluate the three integrals:

$$\int_0^{\pi/2} \cos\theta\sin\theta\, d\theta = \frac{1}{2},$$

$$\int_0^{\pi/2} \cos^2\theta\sin\theta\, d\theta = \frac{1}{3},$$

$$\int_0^{\pi/2} \cos^3\theta\sin\theta\, d\theta = \frac{1}{4}.$$

Therefore

$$F = \pi\varepsilon_0 a^2 \left[A^2\cdot\frac{1}{2} + 6AE_0\cdot\frac{1}{3} + 9E_0^2\cdot\frac{1}{4}\right].$$

Thus

$$F = \pi\varepsilon_0 a^2 \left[\frac{A^2}{2} + 2AE_0 + \frac{9}{4}E_0^2\right].$$

Now substitute

$$A = \frac{Q}{4\pi\varepsilon_0 a^2}.$$

First term:

$$\pi\varepsilon_0 a^2\cdot\frac{A^2}{2} = \pi\varepsilon_0 a^2\cdot\frac{1}{2}\left(\frac{Q}{4\pi\varepsilon_0 a^2}\right)^2 = \frac{Q^2}{32\pi\varepsilon_0 a^2}.$$

Second term:

$$\pi\varepsilon_0 a^2\cdot 2AE_0 = \pi\varepsilon_0 a^2\cdot 2\left(\frac{Q}{4\pi\varepsilon_0 a^2}\right)E_0 = \frac{QE_0}{2}.$$

Third term:

$$\pi\varepsilon_0 a^2\cdot\frac{9}{4}E_0^2 = \frac{9\pi}{4}\varepsilon_0 a^2 E_0^2.$$

Hence

$$\boxed{F = \frac{Q^2}{32\pi\varepsilon_0 a^2} + \frac{QE_0}{2} + \frac{9\pi}{4}\varepsilon_0 a^2 E_0^2.}$$

This is the separating force on the upper hemisphere. The required holding force is equal in magnitude and opposite in direction.

3. Physical interpretation

For the neutral sphere, the field polarizes the conductor, producing positive charge on the upper hemisphere and negative charge on the lower hemisphere. The Maxwell pressure is larger where the field is stronger, and the net effect is to pull the upper hemisphere upward and the lower hemisphere downward, tending to separate them.

When the sphere also carries total charge Q, there is an additional uniform self-repulsion contribution

$$\frac{Q^2}{32\pi\varepsilon_0 a^2},$$

plus an interference term

$$\frac{QE_0}{2},$$

between the monopole field and the induced dipole field from the external uniform field.

Final results

$$\boxed{F_{\text{uncharged}} = \frac{9\pi}{4}\,\varepsilon_0 a^2 E_0^2}$$

and

$$\boxed{F_Q = \frac{Q^2}{32\pi\varepsilon_0 a^2} + \frac{QE_0}{2} + \frac{9\pi}{4}\,\varepsilon_0 a^2 E_0^2.}$$

These are the forces tending to separate the two hemispheres. The force required to prevent separation is equal and opposite.

Problem 2.10

14.1 Problem Statement

A large parallel plate capacitor is made up of two plane conducting sheets with separation D, one of which has a small hemispherical boss of radius a on its inner surface ($D \gg a$). The conductor with the boss is kept at zero potential, and the other conductor is at a potential such that far from the boss the electric field between the plates is E_0.

(a) Calculate the surface-charge densities at an arbitrary point on the plane and on the boss, and sketch their behavior as a function of distance (or angle).

(b) Show that the total charge on the boss has the magnitude

$$3\pi\varepsilon_0 E_0 a^2.$$

(c) If, instead of the other conducting sheet at a different potential, a point charge q is placed directly above the hemispherical boss at a distance d from its center, show that the charge induced on the boss is

$$q' = -q\left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}}\right].$$

14.2 Solution

Because the boss is small compared with the plate separation,

$$D \gg a,$$

the field in the neighborhood of the boss is essentially the same as the field near a hemisphere sitting on an infinite grounded plane in a uniform electric field E_0 normal to the plane.

The standard way to solve this is to reflect the hemisphere through the plane. Then the problem in the upper half-space becomes equivalent to the problem of a *complete* grounded conducting sphere of radius a centered on the plane, placed in a uniform field.

We choose the z-axis perpendicular to the plane, with the grounded plane at

$$z = 0,$$

the hemispherical boss occupying

$$x^2 + y^2 + z^2 = a^2, \qquad z \geq 0,$$

and the uniform field far away taken to be

$$\mathbf{E}_\infty = -E_0\,\hat{\mathbf{z}}.$$

Then the corresponding far-field potential is

$$\Phi_\infty = E_0 z.$$

1. Equivalent full-sphere problem

For a grounded conducting sphere of radius a in a uniform field $-E_0\hat{\mathbf{z}}$, the potential outside the sphere is

$$\Phi(r,\theta) = E_0\left(r - \frac{a^3}{r^2}\right)\cos\theta, \qquad r \ge a.$$

This satisfies

$$\Phi(a,\theta) = 0$$

and

$$\Phi \sim E_0 r\cos\theta = E_0 z \qquad (r \to \infty).$$

Because on the equatorial plane $z = 0$, i.e. $\theta = \pi/2$, the potential is also zero, the restriction of this solution to the upper half-space gives exactly the physical solution for the hemisphere-on-plane geometry.

2. Surface charge density on the boss

For a conducting surface,

$$\sigma = \varepsilon_0 E_n,$$

where E_n is the outward normal field just outside the conductor.

On the sphere, the field is purely radial at the surface because the tangential component vanishes there. So we first compute

$$E_r = -\frac{\partial\Phi}{\partial r}.$$

Starting from

$$\Phi(r,\theta) = E_0\left(r - \frac{a^3}{r^2}\right)\cos\theta,$$

we have

$$\frac{\partial\Phi}{\partial r} = E_0\left(1 + \frac{2a^3}{r^3}\right)\cos\theta.$$

Therefore

$$E_r = -E_0\left(1 + \frac{2a^3}{r^3}\right)\cos\theta.$$

At the surface $r = a$,

$$E_r(a,\theta) = -3E_0\cos\theta.$$

Thus the surface charge density on the boss is

$$\boxed{\sigma_{\text{boss}}(\theta) = \varepsilon_0 E_r(a,\theta) = -3\varepsilon_0 E_0\cos\theta, \qquad 0 \le \theta \le \frac{\pi}{2}.}$$

Behavior. On the boss:

- at the top ($\theta = 0$),

  $$\sigma_{\text{boss}}(0) = -3\varepsilon_0 E_0,$$

  which is the most negative value;

- at the rim ($\theta = \pi/2$),

  $$\sigma_{\text{boss}}\left(\frac{\pi}{2}\right) = 0.$$

So the charge density decreases smoothly from a maximum magnitude at the top of the boss to zero at the rim.

3. Surface charge density on the plane

To find the induced charge density on the plane outside the boss, use the same potential, written in cylindrical coordinates. Since

$$z = r\cos\theta, \qquad r = \sqrt{\rho^2 + z^2},$$

the potential is

$$\Phi(\rho, z) = E_0 z\left(1 - \frac{a^3}{(\rho^2 + z^2)^{3/2}}\right).$$

On the plane $z = 0$, the normal field is the z-component:

$$E_z = -\frac{\partial \Phi}{\partial z}.$$

Differentiate:

$$\frac{\partial \Phi}{\partial z} = E_0\left(1 - \frac{a^3}{(\rho^2 + z^2)^{3/2}}\right) + E_0 z\left(\frac{3a^3 z}{(\rho^2 + z^2)^{5/2}}\right).$$

At $z = 0$, the second term vanishes, so

$$\left.\frac{\partial \Phi}{\partial z}\right|_{z=0} = E_0\left(1 - \frac{a^3}{\rho^3}\right).$$

Hence

$$E_z(\rho, 0^+) = -E_0\left(1 - \frac{a^3}{\rho^3}\right).$$

Therefore the surface charge density on the plane is

$$\boxed{\sigma_{\text{plane}}(\rho) = \varepsilon_0 E_z(\rho, 0^+) = -\varepsilon_0 E_0\left(1 - \frac{a^3}{\rho^3}\right), \qquad \rho \geq a.}$$

Behavior. On the plane:

- far from the boss ($\rho \gg a$),

  $$\sigma_{\text{plane}}(\rho) \to -\varepsilon_0 E_0,$$

  which is exactly the uniform surface charge density of a flat grounded plate in a uniform field E_0;

- at the rim ($\rho = a$),

  $$\sigma_{\text{plane}}(a) = 0.$$

So the plane charge density starts at zero at the edge of the boss and approaches the uniform value $-\varepsilon_0 E_0$ as $\rho \to \infty$.

(a) Summary and sketch

We have found

$$\boxed{\sigma_{\text{boss}}(\theta) = -3\varepsilon_0 E_0 \cos\theta, \qquad 0 \leq \theta \leq \frac{\pi}{2},}$$

and

$$\boxed{\sigma_{\text{plane}}(\rho) = -\varepsilon_0 E_0\left(1 - \frac{a^3}{\rho^3}\right), \qquad \rho \geq a.}$$

Thus:

- the boss has the largest magnitude of negative charge at its top and zero at its rim;
- the plane has zero charge density at the rim and tends monotonically to $-\varepsilon_0 E_0$ far away.

(b) Total charge on the boss

The total charge on the boss is found by integrating the surface charge density over the hemispherical surface.

The area element on the hemisphere is

$$da = a^2 \sin\theta \, d\theta \, d\phi.$$

So

$$Q_{\text{boss}} = \int_0^{2\pi} \int_0^{\pi/2} \sigma_{\text{boss}}(\theta) \, a^2 \sin\theta \, d\theta \, d\phi.$$

Substitute

$$\sigma_{\text{boss}}(\theta) = -3\varepsilon_0 E_0 \cos\theta :$$

$$Q_{\text{boss}} = -3\varepsilon_0 E_0 a^2 \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta \sin\theta \, d\theta.$$

Now

$$\int_0^{2\pi} d\phi = 2\pi,$$

and

$$\int_0^{\pi/2} \cos\theta \sin\theta \, d\theta = \frac{1}{2}.$$

Therefore

$$Q_{\text{boss}} = -3\varepsilon_0 E_0 a^2 (2\pi) \left(\frac{1}{2}\right) = -3\pi\varepsilon_0 E_0 a^2.$$

Hence the total charge on the boss is negative, with magnitude

$$\boxed{|Q_{\text{boss}}| = 3\pi\varepsilon_0 E_0 a^2.}$$

This proves part **(b)**.

(c) Point charge above the boss

Now replace the remote plate by a point charge q located on the symmetry axis of the boss, at distance d from the center of the hemisphere.

The same reflection argument applies: reflect the entire configuration in the plane $z = 0$. Then the problem of the hemispherical boss on a grounded plane becomes the problem of a complete grounded sphere of radius a centered at the origin, with two external point charges:

- a real charge q at $z = +d$,
- an image charge $-q$ at $z = -d$.

The induced charge on the hemispherical boss is therefore equal to the total induced charge on the *upper hemisphere* of the grounded sphere due to these two external charges.

Induced surface density from one external charge. For a grounded sphere with a point charge q outside at distance $d > a$ on the $+z$-axis, the induced surface charge density is

$$\sigma_q(\theta) = -\frac{q}{4\pi a} \frac{d^2 - a^2}{(a^2 + d^2 - 2ad\cos\theta)^{3/2}}.$$

For a charge $-q$ at $z = -d$, we replace $q \to -q$ and $\cos\theta \to -\cos\theta$. Thus

$$\sigma_{-q}(\theta) = +\frac{q}{4\pi a} \frac{d^2 - a^2}{(a^2 + d^2 + 2ad\cos\theta)^{3/2}}.$$

Therefore the total induced density on the upper hemisphere is

$$\sigma(\theta) = -\frac{q}{4\pi a}(d^2 - a^2) \left[\frac{1}{(a^2 + d^2 - 2ad\cos\theta)^{3/2}} - \frac{1}{(a^2 + d^2 + 2ad\cos\theta)^{3/2}} \right].$$

The induced charge on the boss is

$$q' = \int_0^{2\pi} \int_0^{\pi/2} \sigma(\theta)\, a^2 \sin\theta \, d\theta \, d\phi.$$

So

$$q' = -\frac{qa(d^2 - a^2)}{2} \int_0^{\pi/2} \left[\frac{1}{(A - B\cos\theta)^{3/2}} - \frac{1}{(A + B\cos\theta)^{3/2}} \right] \sin\theta \, d\theta,$$

where

$$A = a^2 + d^2, \qquad B = 2ad.$$

Now let

$$u = \cos\theta, \qquad du = -\sin\theta \, d\theta.$$

Then

$$\int_0^{\pi/2} \frac{\sin\theta \, d\theta}{(A - B\cos\theta)^{3/2}} = \int_0^1 \frac{du}{(A - Bu)^{3/2}}.$$

Using

$$\int \frac{du}{(A - Bu)^{3/2}} = \frac{2}{B} \frac{1}{\sqrt{A - Bu}},$$

we get

$$\int_0^1 \frac{du}{(A - Bu)^{3/2}} = \frac{2}{B} \left[\frac{1}{\sqrt{A - B}} - \frac{1}{\sqrt{A}} \right].$$

Similarly,

$$\int_0^{\pi/2} \frac{\sin\theta \, d\theta}{(A + B\cos\theta)^{3/2}} = \frac{2}{B} \left[\frac{1}{\sqrt{A}} - \frac{1}{\sqrt{A + B}} \right].$$

Subtracting,

$$\begin{aligned} I &= \int_0^{\pi/2} \left[\frac{1}{(A - B\cos\theta)^{3/2}} - \frac{1}{(A + B\cos\theta)^{3/2}} \right] \sin\theta \, d\theta \\ &= \frac{2}{B} \left[\frac{1}{\sqrt{A - B}} + \frac{1}{\sqrt{A + B}} - \frac{2}{\sqrt{A}} \right]. \end{aligned}$$

Now

$$A - B = (d - a)^2, \qquad A + B = (d + a)^2, \qquad A = d^2 + a^2.$$

Therefore

$$\frac{1}{\sqrt{A - B}} = \frac{1}{d - a}, \qquad \frac{1}{\sqrt{A + B}} = \frac{1}{d + a}, \qquad \frac{1}{\sqrt{A}} = \frac{1}{\sqrt{d^2 + a^2}}.$$

Also

$$B = 2ad.$$

So

$$I = \frac{1}{ad} \left[\frac{1}{d - a} + \frac{1}{d + a} - \frac{2}{\sqrt{d^2 + a^2}} \right].$$

Now

$$\frac{1}{d - a} + \frac{1}{d + a} = \frac{2d}{d^2 - a^2}.$$

Hence

$$I = \frac{1}{ad} \left[\frac{2d}{d^2 - a^2} - \frac{2}{\sqrt{d^2 + a^2}} \right].$$

Substitute this into the expression for q':

$$q' = -\frac{qa(d^2 - a^2)}{2} \cdot \frac{1}{ad} \left[\frac{2d}{d^2 - a^2} - \frac{2}{\sqrt{d^2 + a^2}} \right].$$

Simplifying,

$$q' = -q \left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}} \right].$$

Thus

$$\boxed{q' = -q \left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}} \right].}$$

This proves part **(c)**.

Checks on the induced charge

Far away: $d \gg a$. Use

$$\sqrt{d^2+a^2} = d\sqrt{1+\frac{a^2}{d^2}} \approx d\left(1+\frac{a^2}{2d^2}\right).$$

Then

$$\frac{d^2-a^2}{d\sqrt{d^2+a^2}} \approx \frac{1-\frac{a^2}{d^2}}{1+\frac{a^2}{2d^2}} \approx 1-\frac{3a^2}{2d^2}.$$

So

$$q' \approx -q\left[\frac{3a^2}{2d^2}\right].$$

Thus the induced charge goes to zero as $d \to \infty$, which is physically correct.

Charge close to the boss: $d \to a^+$. Then

$$\frac{d^2-a^2}{d\sqrt{d^2+a^2}} \to 0,$$

so

$$q' \to -q.$$

Again this is exactly what one expects: if the charge comes very close to the grounded conductor, nearly all of its flux terminates on the boss.

Final results

The surface charge densities are

$$\boxed{\sigma_{\text{boss}}(\theta) = -3\varepsilon_0 E_0 \cos\theta, \qquad 0 \le \theta \le \frac{\pi}{2},}$$

$$\boxed{\sigma_{\text{plane}}(\rho) = -\varepsilon_0 E_0\left(1-\frac{a^3}{\rho^3}\right), \qquad \rho \ge a.}$$

The total charge on the boss is

$$\boxed{Q_{\text{boss}} = -3\pi\varepsilon_0 E_0 a^2, \qquad |Q_{\text{boss}}| = 3\pi\varepsilon_0 E_0 a^2.}$$

For a point charge q placed a distance d from the center of the boss, the induced charge on the boss is

$$\boxed{q' = -q\left[1-\frac{d^2-a^2}{d\sqrt{d^2+a^2}}\right].}$$

Problem 2.11

15.1 Problem Statement

A line charge with linear charge density λ is placed parallel to, and a distance R away from, the axis of a conducting cylinder of radius b held at fixed voltage such that the potential vanishes at infinity. Find

- **(a)** the magnitude and position of the image charge(s);
- **(b)** the potential at any point (expressed in polar coordinates with the origin at the axis of the cylinder and the direction from the origin to the line charge as the x-axis), including the asymptotic form far from the cylinder;
- **(c)** the induced surface-charge density, and plot it as a function of angle for $R/b = 2, 4$ in units of $\lambda/2\pi b$;
- **(d)** the force per unit length on the line charge.

15.2 Solution

Let the conducting cylinder have radius b, centered on the origin, and let the real line charge $+\lambda$ lie parallel to the z-axis at the point

$$x = R, \qquad y = 0, \qquad R > b.$$

We use polar coordinates (r, θ) in the transverse plane, with the x-axis directed toward the real line charge.

Because the geometry is two-dimensional, the potential of a line charge is logarithmic. For a line charge λ located at transverse distance s, the potential is

$$\Phi = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{\text{const.}}{\text{distance to source}}.$$

The method of images for a cylinder is the exact analogue of the image method for a sphere, but in two dimensions.

(a) Image charge and its position

Take the real line charge $+\lambda$ at distance R from the origin. We seek an image line charge λ' on the same axis at distance $r' = b^2/R$ from the origin, so that the cylinder $r = b$ becomes an equipotential.

The correct image is

$$\boxed{\lambda' = -\lambda}$$

located at

$$\boxed{r' = \frac{b^2}{R}, \qquad \theta = 0.}$$

So the image charge lies on the same radial line, inside the cylinder, between the center and the real line charge.

Why this works. Let

$$r_1^2 = r^2 + R^2 - 2rR\cos\theta$$

be the squared distance from the field point (r, θ) to the real line charge, and let

$$r_2^2 = r^2 + \left(\frac{b^2}{R}\right)^2 - 2r\frac{b^2}{R}\cos\theta$$

be the squared distance to the image line charge.

On the cylinder $r = b$,

$$r_2^2 = b^2 + \frac{b^4}{R^2} - 2b\frac{b^2}{R}\cos\theta = \frac{b^2}{R^2}\left(R^2 + b^2 - 2bR\cos\theta\right) = \frac{b^2}{R^2}r_1^2.$$

Therefore on $r = b$,

$$r_2 = \frac{b}{R}r_1.$$

Hence the potential

$$\Phi(r, \theta) = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{r_2}{r_1}$$

takes the constant value

$$\Phi(b, \theta) = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{b}{R},$$

independent of θ. So the cylinder is an equipotential.

Also, since the two charges are equal and opposite, the net line charge is zero, so the potential tends to zero at infinity:

$$\Phi \to 0 \qquad (r \to \infty).$$

Thus the image system is complete with just one image line charge.

(b) Potential everywhere outside the cylinder

Using the result above, the potential in the physical region $r > b$ is

$$\boxed{\Phi(r,\theta) = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{r_2}{r_1},}$$

where

$$r_1 = \sqrt{r^2 + R^2 - 2rR\cos\theta}, \qquad r_2 = \sqrt{r^2 + \left(\frac{b^2}{R}\right)^2 - 2r\frac{b^2}{R}\cos\theta}.$$

Equivalently, since $\ln r = \frac{1}{2}\ln r^2$,

$$\boxed{\Phi(r,\theta) = \frac{\lambda}{4\pi\varepsilon_0} \ln \left[\frac{r^2 + \left(\frac{b^2}{R}\right)^2 - 2r\frac{b^2}{R}\cos\theta}{r^2 + R^2 - 2rR\cos\theta} \right].}$$

Potential of the cylinder. On the surface $r = b$, as shown above,

$$\Phi(b,\theta) = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{b}{R}.$$

Thus the fixed voltage of the cylinder is

$$\boxed{V_{\text{cyl}} = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{b}{R}.}$$

Since $R > b$, this is negative for $\lambda > 0$.

Asymptotic form at large distance. For $r \gg R$, we expand the distances:

$$r_1 \approx r - R\cos\theta, \qquad r_2 \approx r - \frac{b^2}{R}\cos\theta.$$

Hence

$$\frac{r_2}{r_1} \approx \frac{r - \frac{b^2}{R}\cos\theta}{r - R\cos\theta} \approx 1 + \frac{R - \frac{b^2}{R}}{r}\cos\theta.$$

Using $\ln(1+x) \approx x$ for small x, we obtain

$$\Phi(r,\theta) \approx \frac{\lambda}{2\pi\varepsilon_0} \frac{R - \frac{b^2}{R}}{r} \cos\theta.$$

Therefore

$$\boxed{\Phi(r,\theta) \sim \frac{\lambda}{2\pi\varepsilon_0} \frac{R^2 - b^2}{R} \frac{\cos\theta}{r}, \qquad r \gg R.}$$

This is the potential of a two-dimensional dipole, as expected for a pair of equal and opposite line charges.

(c) Induced surface-charge density

The surface charge density on the conductor is

$$\sigma = \varepsilon_0 E_n,$$

where E_n is the normal electric field just outside the cylinder. On the cylinder, the normal is radial, so

$$\sigma(\theta) = \varepsilon_0 E_r(b,\theta), \qquad E_r = -\frac{\partial \Phi}{\partial r}.$$

Starting from

$$\Phi(r,\theta) = \frac{\lambda}{4\pi\varepsilon_0} \ln \frac{r_2^2}{r_1^2},$$

we compute

$$\frac{\partial \Phi}{\partial r} = \frac{\lambda}{4\pi\varepsilon_0} \left[\frac{2r - 2\frac{b^2}{R}\cos\theta}{r_2^2} - \frac{2r - 2R\cos\theta}{r_1^2} \right].$$

Now evaluate this at $r = b$. Using

$$r_2^2 = \frac{b^2}{R^2} r_1^2 \qquad \text{when } r = b,$$

one finds after simplification

$$\left.\frac{\partial \Phi}{\partial r}\right|_{r=b} = \frac{\lambda}{2\pi\varepsilon_0} \frac{R^2 - b^2}{b\,(R^2 - 2Rb\cos\theta + b^2)}.$$

Hence

$$E_r(b,\theta) = -\left.\frac{\partial \Phi}{\partial r}\right|_{r=b} = -\frac{\lambda}{2\pi\varepsilon_0} \frac{R^2 - b^2}{b\,(R^2 - 2Rb\cos\theta + b^2)}.$$

Therefore

$$\boxed{\sigma(\theta) = -\frac{\lambda}{2\pi b} \frac{R^2 - b^2}{R^2 - 2Rb\cos\theta + b^2}.}$$

Checks.

- At the nearest point to the line charge ($\theta = 0$),

$$\sigma(0) = -\frac{\lambda}{2\pi b} \frac{R^2 - b^2}{(R-b)^2} = -\frac{\lambda}{2\pi b} \frac{R+b}{R-b},$$

  which is the most negative value.

- At the farthest point ($\theta = \pi$),

$$\sigma(\pi) = -\frac{\lambda}{2\pi b} \frac{R^2 - b^2}{(R+b)^2} = -\frac{\lambda}{2\pi b} \frac{R-b}{R+b},$$

  which is still negative, but smaller in magnitude.

So the cylinder carries negative induced charge everywhere, largest in magnitude on the side closest to the external positive line charge.

Plot in units of $\lambda/2\pi b$. Define

$$\widetilde{\sigma}(\theta) \equiv \frac{\sigma(\theta)}{\lambda/2\pi b}.$$

Then

$$\boxed{\widetilde{\sigma}(\theta) = -\frac{R^2 - b^2}{R^2 - 2Rb\cos\theta + b^2}.}$$

For the requested cases:

- If $R/b = 2$,

$$\widetilde{\sigma}(\theta) = -\frac{3}{5 - 4\cos\theta}.$$

- If $R/b = 4$,

$$\widetilde{\sigma}(\theta) = -\frac{15}{17 - 8\cos\theta}.$$

In both cases the graph is negative for all θ, with a pronounced minimum at $\theta = 0$ and a much smaller magnitude near $\theta = \pi$. The distribution becomes more sharply peaked at $\theta = 0$ as R/b decreases, i.e. as the line charge comes closer to the cylinder.

Total induced charge. It is useful to check the total induced charge per unit length on the cylinder:

$$Q_{\text{ind}} = \int_0^{2\pi} \sigma(\theta)\, b\, d\theta = -\frac{\lambda}{2\pi}(R^2 - b^2) \int_0^{2\pi} \frac{d\theta}{R^2 - 2Rb\cos\theta + b^2}.$$

Use the standard integral

$$\int_0^{2\pi} \frac{d\theta}{A - B\cos\theta} = \frac{2\pi}{\sqrt{A^2 - B^2}}, \qquad A > |B|.$$

Here

$$A = R^2 + b^2, \qquad B = 2Rb,$$

so

$$\sqrt{A^2 - B^2} = \sqrt{(R^2 + b^2)^2 - 4R^2b^2} = R^2 - b^2.$$

Therefore

$$Q_{\text{ind}} = -\frac{\lambda}{2\pi}(R^2 - b^2)\frac{2\pi}{R^2 - b^2} = -\lambda.$$

So

$$\boxed{Q_{\text{ind}} = -\lambda}$$

per unit length, as expected.

(d) Force per unit length on the line charge

The force per unit length on the real line charge is the Coulomb force per unit length due to the image line charge.

The image line charge has density

$$-\lambda$$

and is located at distance

$$r' = \frac{b^2}{R}$$

from the origin on the same axis. Therefore the separation between the real line and the image line is

$$s = R - \frac{b^2}{R} = \frac{R^2 - b^2}{R}.$$

The electric field of a line charge of density λ' at distance s is

$$E = \frac{|\lambda'|}{2\pi\varepsilon_0 s}.$$

Hence the force per unit length on the real line charge is

$$f = \lambda E = \frac{\lambda^2}{2\pi\varepsilon_0 s}.$$

Substitute $s = (R^2 - b^2)/R$:

$$f = \frac{\lambda^2}{2\pi\varepsilon_0}\frac{R}{R^2 - b^2}.$$

Because the image charge is opposite in sign, the force is attractive, directed toward the cylinder axis. Thus

$$\boxed{\frac{F}{L} = -\hat{\mathbf{x}}\frac{\lambda^2}{2\pi\varepsilon_0}\frac{R}{R^2 - b^2}.}$$

Its magnitude is

$$\boxed{\left|\frac{F}{L}\right| = \frac{\lambda^2}{2\pi\varepsilon_0}\frac{R}{R^2 - b^2}.}$$

Final results

$$\boxed{\lambda' = -\lambda, \qquad r' = \frac{b^2}{R}}$$

$$\boxed{\Phi(r,\theta) = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{r^2 + \left(\frac{b^2}{R}\right)^2 - 2r\frac{b^2}{R}\cos\theta}{r^2 + R^2 - 2rR\cos\theta}\right]}$$

$$\boxed{V_{\text{cyl}} = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{b}{R}}$$

$$\boxed{\Phi(r,\theta) \sim \frac{\lambda}{2\pi\varepsilon_0}\frac{R^2 - b^2}{R}\frac{\cos\theta}{r} \qquad (r \gg R)}$$

$$\boxed{\sigma(\theta) = -\frac{\lambda}{2\pi b}\,\frac{R^2 - b^2}{R^2 - 2Rb\cos\theta + b^2}}$$

$$\boxed{\frac{\sigma(\theta)}{\lambda/2\pi b} = -\frac{R^2 - b^2}{R^2 - 2Rb\cos\theta + b^2}}$$

$$\boxed{Q_{\text{ind}} = -\lambda}$$

$$\boxed{\left|\frac{F}{L}\right| = \frac{\lambda^2}{2\pi\varepsilon_0}\,\frac{R}{R^2 - b^2}}$$

with the force directed toward the cylinder.

Problem 2.12

16.1 Problem Statement

Starting with the series solution (2.71) for the two-dimensional potential problem with the potential specified on the surface of a cylinder of radius b, evaluate the coefficients formally, substitute them into the series, and sum it to obtain the potential inside the cylinder in the form of Poisson's integral:

$$\Phi(\rho, \phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b, \phi')\frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}\,d\phi'.$$

What modification is necessary if the potential is desired in the region of space bounded by the cylinder and infinity?

16.2 Solution

We consider Laplace's equation in two-dimensional polar coordinates:

$$\nabla^2\Phi = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\Phi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\Phi}{\partial\phi^2} = 0.$$

The boundary is the circle

$$\rho = b,$$

and the boundary value is specified:

$$\Phi(b, \phi) = f(\phi).$$

We first solve for the region $\rho < b$, where the potential must remain finite at the origin.

1. Series solution inside the cylinder

The general separated solution of Laplace's equation in polar coordinates is

$$\Phi(\rho, \phi) = A_0 + B_0\ln\rho + \sum_{n=1}^{\infty}\left[\rho^n(A_n\cos n\phi + B_n\sin n\phi) + \rho^{-n}(C_n\cos n\phi + D_n\sin n\phi)\right].$$

For the interior region $\rho < b$, regularity at $\rho = 0$ requires

- $B_0 = 0$, because $\ln\rho$ diverges at the origin;
- $C_n = D_n = 0$, because ρ^{-n} diverges at the origin.

Hence the interior solution becomes

$$\Phi(\rho, \phi) = A_0 + \sum_{n=1}^{\infty}\rho^n\left(A_n\cos n\phi + B_n\sin n\phi\right).$$

It is convenient to rewrite the coefficients in terms of the boundary radius b:

$$\Phi(\rho, \phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(\frac{\rho}{b}\right)^n\left(a_n\cos n\phi + b_n\sin n\phi\right),$$

where the constants a_n, b_n are the Fourier coefficients of the boundary function $f(\phi) = \Phi(b, \phi)$.

On the boundary $\rho = b$,

$$\Phi(b, \phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\phi + b_n \sin n\phi) .$$

So indeed

$$f(\phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\phi + b_n \sin n\phi) ,$$

with the standard Fourier coefficients

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(\phi')\, d\phi',$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(\phi') \cos n\phi'\, d\phi', \qquad b_n = \frac{1}{\pi} \int_0^{2\pi} f(\phi') \sin n\phi'\, d\phi'.$$

2. Substitute the coefficients into the series

Substitute these expressions into the interior solution:

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi')\, d\phi' + \sum_{n=1}^{\infty} \left(\frac{\rho}{b}\right)^n \left[\frac{1}{\pi} \int_0^{2\pi} f(\phi') \cos n\phi'\, d\phi' \cos n\phi \right.$$

$$\left. + \frac{1}{\pi} \int_0^{2\pi} f(\phi') \sin n\phi'\, d\phi' \sin n\phi \right].$$

Interchange summation and integration:

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi')\, d\phi' + \frac{1}{\pi} \int_0^{2\pi} f(\phi') \sum_{n=1}^{\infty} \left(\frac{\rho}{b}\right)^n [\cos n\phi' \cos n\phi + \sin n\phi' \sin n\phi]\, d\phi'.$$

Now use the trigonometric identity

$$\cos n\phi' \cos n\phi + \sin n\phi' \sin n\phi = \cos n(\phi' - \phi).$$

Therefore

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi')\, d\phi' + \frac{1}{\pi} \int_0^{2\pi} f(\phi') \sum_{n=1}^{\infty} \left(\frac{\rho}{b}\right)^n \cos n(\phi' - \phi)\, d\phi'.$$

Factor out $1/(2\pi)$:

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi') \left[1 + 2 \sum_{n=1}^{\infty} \left(\frac{\rho}{b}\right)^n \cos n(\phi' - \phi)\right] d\phi'.$$

Thus everything reduces to summing the series

$$1 + 2 \sum_{n=1}^{\infty} t^n \cos n\alpha, \qquad t = \frac{\rho}{b}, \qquad \alpha = \phi' - \phi,$$

with $|t| < 1$ since $\rho < b$.

3. Sum the trigonometric series

We use the standard identity

$$1 + 2 \sum_{n=1}^{\infty} t^n \cos n\alpha = \frac{1 - t^2}{1 - 2t \cos \alpha + t^2}, \qquad |t| < 1.$$

Therefore

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi') \frac{1 - (\rho/b)^2}{1 - 2(\rho/b) \cos(\phi' - \phi) + (\rho/b)^2}\, d\phi'.$$

Multiply numerator and denominator by b^2:

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} f(\phi') \frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho \cos(\phi' - \phi)}\, d\phi'.$$

Recalling that $f(\phi') = \Phi(b, \phi')$, we obtain Poisson's integral formula:

$$\boxed{\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} \Phi(b, \phi') \frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}\, d\phi', \qquad \rho < b.}$$

This is exactly the required result.

4. Check of the boundary limit

It is useful to verify that this formula reproduces the prescribed boundary value.

As $\rho \to b^-$, the kernel

$$P(\rho, \phi; \phi') = \frac{1}{2\pi} \frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}$$

becomes sharply peaked at $\phi' = \phi$, and behaves as a delta-function on the circle. Thus

$$\Phi(\rho, \phi) \to \Phi(b, \phi) \qquad \text{as } \rho \to b^-.$$

So the integral formula has the correct boundary behavior.

5. Exterior region: modification needed

Now suppose the potential is desired in the region outside the cylinder,

$$\rho > b,$$

and also vanishes at infinity.

In that case, the solution must be regular at infinity rather than at the origin. Therefore the allowed radial dependence is ρ^{-n}, not ρ^n. The logarithmic term must also be absent if the potential is to vanish at infinity.

So the exterior solution has the form

$$\Phi(\rho, \phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{b}{\rho}\right)^n (a_n \cos n\phi + b_n \sin n\phi)\,.$$

If $\Phi \to 0$ at infinity, then necessarily

$$a_0 = 0.$$

Repeating the same derivation as above, but with

$$t = \frac{b}{\rho}, \qquad |t| < 1,$$

one obtains

$$\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} \Phi(b, \phi') \left[2 \sum_{n=1}^{\infty} \left(\frac{b}{\rho}\right)^n \cos n(\phi' - \phi) \right] d\phi'.$$

Using again the same summation formula, but now without the $n = 0$ term because $a_0 = 0$, the correct exterior Poisson kernel is

$$\boxed{\Phi(\rho, \phi) = \frac{1}{2\pi} \int_0^{2\pi} \Phi(b, \phi') \frac{\rho^2 - b^2}{\rho^2 + b^2 - 2b\rho\cos(\phi' - \phi)}\, d\phi', \qquad \rho > b,}$$

provided the potential vanishes at infinity.

This is the required modification.

6. Physical interpretation of the two kernels

The interior kernel

$$\frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}$$

is appropriate because the solution must be regular at the center and is built out of powers

$$\left(\frac{\rho}{b}\right)^n.$$

The exterior kernel

$$\frac{\rho^2 - b^2}{\rho^2 + b^2 - 2b\rho\cos(\phi' - \phi)}$$

is appropriate because the solution must be regular at infinity and is built out of powers

$$\left(\frac{b}{\rho}\right)^n.$$

So the two formulas are exact analogues of one another, with ρ/b replaced by b/ρ.

Final results

For the interior of the cylinder ($\rho < b$),

$$\boxed{\Phi(\rho, \phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b, \phi') \frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}\, d\phi'.}$$

For the exterior region ($\rho > b$), with $\Phi \to 0$ at infinity, the required modification is

$$\boxed{\Phi(\rho, \phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b, \phi') \frac{\rho^2 - b^2}{\rho^2 + b^2 - 2b\rho\cos(\phi' - \phi)}\, d\phi'.}$$

So the modification is simply that the Poisson kernel changes from

$$\frac{b^2 - \rho^2}{b^2 + \rho^2 - 2b\rho\cos(\phi' - \phi)}$$

to

$$\frac{\rho^2 - b^2}{\rho^2 + b^2 - 2b\rho\cos(\phi' - \phi)},$$

corresponding to replacing interior powers $(\rho/b)^n$ by exterior powers $(b/\rho)^n$.

Problem 2.13

17.1 Problem Statement

(a) Two halves of a long hollow conducting cylinder of inner radius b are separated by small lengthwise gaps on each side, and are kept at different potentials V_1 and V_2. Show that the potential inside is given by

$$\Phi(\rho, \phi) = \frac{V_1 + V_2}{2} + \frac{V_1 - V_2}{\pi} \tan^{-1}\left(\frac{2b\rho\cos\phi}{b^2 - \rho^2}\right),$$

where ϕ is measured from a plane perpendicular to the plane through the gap.

(b) Calculate the surface-charge density on each half of the cylinder.

17.2 Solution

We now *derive* the given potential formula and then compute the surface-charge density.

The geometry is a long hollow cylinder of radius b, split into two longitudinal halves by two narrow gaps. Since the cylinder is long, the problem is two-dimensional in the cross section. Inside the cylinder ($\rho < b$), the potential satisfies Laplace's equation

$$\nabla^2\Phi = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\Phi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\Phi}{\partial\phi^2} = 0.$$

The boundary condition on $\rho = b$ is piecewise constant:

$$\Phi(b, \phi) = \begin{cases} V_1, & \cos\phi > 0, \\ V_2, & \cos\phi < 0. \end{cases}$$

That is, one half-cylinder is at V_1, the opposite half-cylinder is at V_2.

1. General interior solution of Laplace's equation

For the interior of a circle, the regular separated solution of Laplace's equation is

$$\Phi(\rho,\phi) = A_0 + \sum_{n=1}^{\infty} \left(\frac{\rho}{b}\right)^n (A_n \cos n\phi + B_n \sin n\phi).$$

The boundary condition is even in ϕ, so only cosine terms are needed:

$$\Phi(\rho,\phi) = A_0 + \sum_{n=1}^{\infty} A_n \left(\frac{\rho}{b}\right)^n \cos n\phi.$$

Thus the problem reduces to finding the Fourier cosine series of the boundary value

$$f(\phi) \equiv \Phi(b,\phi).$$

2. Fourier series of the boundary function

Write the boundary function as

$$f(\phi) = \frac{V_1+V_2}{2} + \frac{V_1-V_2}{2}\,\mathrm{sgn}(\cos\phi).$$

The square wave $\mathrm{sgn}(\cos\phi)$ has the Fourier cosine series

$$\mathrm{sgn}(\cos\phi) = \frac{4}{\pi}\left(\cos\phi - \frac{1}{3}\cos 3\phi + \frac{1}{5}\cos 5\phi - \cdots\right).$$

Therefore

$$f(\phi) = \frac{V_1+V_2}{2} + \frac{2(V_1-V_2)}{\pi}\left(\cos\phi - \frac{1}{3}\cos 3\phi + \frac{1}{5}\cos 5\phi - \cdots\right).$$

Hence the interior solution is

$$\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{2(V_1-V_2)}{\pi}\sum_{m=0}^{\infty}(-1)^m \frac{1}{2m+1}\left(\frac{\rho}{b}\right)^{2m+1} \cos\big((2m+1)\phi\big).$$

So we have

$$\boxed{\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{2(V_1-V_2)}{\pi}\sum_{m=0}^{\infty}(-1)^m \frac{(\rho/b)^{2m+1}\cos\big((2m+1)\phi\big)}{2m+1}.}$$

3. Sum the series

Let

$$t = \frac{\rho}{b}, \qquad |t| < 1.$$

Define

$$S(t,\phi) = \sum_{m=0}^{\infty}(-1)^m \frac{t^{2m+1}\cos\big((2m+1)\phi\big)}{2m+1}.$$

Introduce the complex variable

$$z = te^{i\phi}.$$

Then

$$S(t,\phi) = \mathrm{Re}\left[\sum_{m=0}^{\infty}(-1)^m \frac{z^{2m+1}}{2m+1}\right].$$

But for $|z| < 1$,

$$\sum_{m=0}^{\infty}(-1)^m \frac{z^{2m+1}}{2m+1} = \arctan z.$$

Therefore

$$S(t,\phi) = \mathrm{Re}\big(\arctan z\big).$$

Now use the standard identity

$$\mathrm{Re}\big(\arctan(te^{i\phi})\big) = \frac{1}{2}\tan^{-1}\left(\frac{2t\cos\phi}{1-t^2}\right).$$

Hence

$$S(t,\phi) = \frac{1}{2}\tan^{-1}\left(\frac{2t\cos\phi}{1-t^2}\right).$$

Substitute this into the potential:

$$\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{2(V_1-V_2)}{\pi}\cdot\frac{1}{2}\tan^{-1}\left(\frac{2(\rho/b)\cos\phi}{1-(\rho/b)^2}\right).$$

Simplify:

$$\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{V_1-V_2}{\pi}\tan^{-1}\left(\frac{2(\rho/b)\cos\phi}{1-\rho^2/b^2}\right).$$

Multiply numerator and denominator inside the arctangent by b^2:

$$\boxed{\Phi(\rho,\phi) = \frac{V_1+V_2}{2} + \frac{V_1-V_2}{\pi}\tan^{-1}\left(\frac{2b\rho\cos\phi}{b^2-\rho^2}\right).}$$

This proves part **(a)**.

4. Check of the boundary values

As $\rho \to b^-$:

- if $\cos\phi > 0$, then

$$\frac{2b\rho\cos\phi}{b^2-\rho^2} \to +\infty, \qquad \tan^{-1}(+\infty) = \frac{\pi}{2},$$

  and therefore

$$\Phi(b,\phi) = \frac{V_1+V_2}{2} + \frac{V_1-V_2}{2} = V_1;$$

- if $\cos\phi < 0$, then

$$\frac{2b\rho\cos\phi}{b^2-\rho^2} \to -\infty, \qquad \tan^{-1}(-\infty) = -\frac{\pi}{2},$$

  and therefore

$$\Phi(b,\phi) = \frac{V_1+V_2}{2} - \frac{V_1-V_2}{2} = V_2.$$

So the formula has exactly the required boundary values on the two halves.

5. Surface-charge density

We now compute the surface-charge density on the inner surface of the cylinder.

For a conductor,

$$\sigma = \varepsilon_0 E_n,$$

where E_n is the field just outside the conductor in the normal direction from the conductor into the vacuum region.

Here the vacuum region is the interior of the cylinder, so at $\rho = b$ the normal from the conductor into the vacuum points in the $-\hat{\boldsymbol{\rho}}$ direction. Therefore

$$E_n = -E_\rho.$$

Since

$$E_\rho = -\frac{\partial\Phi}{\partial\rho},$$

we get

$$E_n = \frac{\partial\Phi}{\partial\rho}.$$

Hence

$$\boxed{\sigma(\phi) = \varepsilon_0 \left.\frac{\partial\Phi}{\partial\rho}\right|_{\rho=b^-}.}$$

6. Differentiate the potential

Let

$$A = \frac{V_1 + V_2}{2}, \qquad B = \frac{V_1 - V_2}{\pi}, \qquad u(\rho, \phi) = \frac{2b\rho\cos\phi}{b^2 - \rho^2}.$$

Then

$$\Phi = A + B\tan^{-1} u.$$

So

$$\frac{\partial \Phi}{\partial \rho} = B\frac{1}{1+u^2}\frac{\partial u}{\partial \rho}.$$

Now

$$u(\rho, \phi) = \frac{2b\rho\cos\phi}{b^2 - \rho^2},$$

so by the quotient rule

$$\frac{\partial u}{\partial \rho} = \frac{2b\cos\phi\,(b^2 - \rho^2) - 2b\rho\cos\phi(-2\rho)}{(b^2 - \rho^2)^2}.$$

Simplifying,

$$\frac{\partial u}{\partial \rho} = \frac{2b(b^2 + \rho^2)\cos\phi}{(b^2 - \rho^2)^2}.$$

Also

$$1 + u^2 = 1 + \frac{4b^2\rho^2\cos^2\phi}{(b^2 - \rho^2)^2} = \frac{(b^2 - \rho^2)^2 + 4b^2\rho^2\cos^2\phi}{(b^2 - \rho^2)^2}.$$

Hence

$$\frac{\partial \Phi}{\partial \rho} = B\frac{2b(b^2 + \rho^2)\cos\phi}{(b^2 - \rho^2)^2 + 4b^2\rho^2\cos^2\phi}.$$

Now let $\rho \to b^-$. Then

$$b^2 + \rho^2 \to 2b^2,$$

and

$$(b^2 - \rho^2)^2 + 4b^2\rho^2\cos^2\phi \to 4b^4\cos^2\phi.$$

So

$$\left.\frac{\partial \Phi}{\partial \rho}\right|_{\rho = b^-} = B\frac{2b(2b^2)\cos\phi}{4b^4\cos^2\phi} = \frac{B}{b\cos\phi}.$$

Thus

$$\boxed{\left.\frac{\partial \Phi}{\partial \rho}\right|_{\rho = b^-} = \frac{V_1 - V_2}{\pi b\cos\phi}.}$$

Therefore the surface-charge density is

$$\boxed{\sigma(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}.}$$

7. Surface-charge density on each half

Now identify the two halves.

- On the half-cylinder at potential V_1, we have $\cos\phi > 0$, i.e.

$$-\frac{\pi}{2} < \phi < \frac{\pi}{2},$$

so

$$\boxed{\sigma_{V_1}(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}, \qquad -\frac{\pi}{2} < \phi < \frac{\pi}{2}.}$$

- On the half-cylinder at potential V_2, we have $\cos\phi < 0$, i.e.

$$\frac{\pi}{2} < \phi < \frac{3\pi}{2},$$

so the same formula applies:

$$\boxed{\sigma_{V_2}(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}, \qquad \frac{\pi}{2} < \phi < \frac{3\pi}{2}.}$$

Since $\cos\phi < 0$ on the second interval, σ_{V_2} automatically has the opposite sign.

8. Physical interpretation

If $V_1 > V_2$, then

- on the V_1 half, $\cos\phi > 0$, so
$$\sigma_{V_1} > 0;$$
- on the V_2 half, $\cos\phi < 0$, so
$$\sigma_{V_2} < 0.$$

So the higher-potential half carries positive charge, and the lower-potential half carries negative charge, exactly as expected.

Also, the charge density diverges as
$$\frac{1}{\cos\phi}$$
near the slit edges at
$$\phi = \frac{\pi}{2}, \qquad \phi = \frac{3\pi}{2}.$$
This is the usual edge singularity for conductor boundaries meeting at a narrow gap.

Final results

The potential inside the cylinder is

$$\boxed{\Phi(\rho,\phi) = \frac{V_1 + V_2}{2} + \frac{V_1 - V_2}{\pi}\tan^{-1}\left(\frac{2b\rho\cos\phi}{b^2 - \rho^2}\right).}$$

The surface-charge density on the cylinder is

$$\boxed{\sigma(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}.}$$

More explicitly,

$$\boxed{\sigma_{V_1}(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}, \qquad -\frac{\pi}{2} < \phi < \frac{\pi}{2},}$$

$$\boxed{\sigma_{V_2}(\phi) = \frac{\varepsilon_0(V_1 - V_2)}{\pi b\cos\phi}, \qquad \frac{\pi}{2} < \phi < \frac{3\pi}{2}.}$$

If $V_1 > V_2$, the V_1 half is positively charged, the V_2 half is negatively charged, and the charge density diverges at the gap edges.